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EPIDEMIOLOGIA MOLECULAR DE LEPTOSPIRAS PATOGÊNICAS NO
BRASIL: UMA ABORDAGEM INTEGRATIVA MULTI-HOSPEDEIROS E
MULTIRREGIONAL

Niterói, RJ

2025

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UMA ABORDAGEM INTEGRATIVA MULTI-HOSPEDEIROS E MULTIRREGIONAL**

Tese apresentada ao Programa de Pós-Graduação em Microbiologia e Parasitologia Aplicadas do Instituto Biomédico da Universidade Federal Fluminense como requisito parcial à obtenção do título de Doutora.

Área de Concentração: Aspectos clínicos, epidemiológicos e evolutivos

Orientador: Prof. Dr. Walter Lilenbaum

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Dedico este trabalho a minha avó Miriam, cuja fé na educação acendeu minha própria vocação. Pedagoga, com sua mente paciente, alfabetizou-me. Com seu olhar além do tempo, mostrou-me o caminho da pesquisa e da docência.

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*“Devemos acreditar que somos talentosas para
alguma coisa, e que essa coisa, a qualquer
custo, deve ser alcançada”*

Marie Curie

RESUMO

A leptospirose é uma doença negligenciada de ampla distribuição no Brasil, com impacto significativo na saúde pública. Diante do aumento das zoonoses associadas às mudanças ambientais e climáticas, a doença permanece como um desafio emergente, exigindo uma compreensão mais profunda de seus padrões de transmissão e da diversidade genética dos agentes envolvidos. Este estudo teve como objetivo investigar a diversidade genética, as relações evolutivas e os padrões de circulação de *Leptospira* spp. patogênicas no Brasil, a partir de diferentes hospedeiros, regiões geográficas e fontes ambientais. Para isso, foram integradas abordagens moleculares *in vitro* e *in silico*, incluindo diagnóstico por PCR, tipificação baseada no gene *secY*, análises filogenéticas e de rede haplotípica, além da curadoria e compilação de extensos bancos de dados genéticos e metadados ecológicos. As análises integradas evidenciaram ampla diversidade ecológica de *Leptospira* spp. patogênicas, geneticamente identificadas principalmente em ruminantes, cães, humanos e pequenos mamíferos silvestres. De forma importante, também foram identificadas em amostras ambientais. Mata Atlântica e Amazônia foram os biomas com maior número de ocorrências. Essa elevada diversidade detectada entre múltiplos hospedeiros e biomas evidencia a complexa estrutura ecoepidemiológica da leptospirose no país. A espécie mais frequente foi *L. interrogans*, seguida por *L. santarosai*, *L. borgpetersenii*, *L. kirschneri* e *L. noguchii*. Observou-se ampla sobreposição genética entre estirpes de diferentes origens e a predominância de um haplótipo clonal de *L. interrogans* do sorogrupo Icterohaemorrhagiae, amplamente disseminado. Essa homogeneidade genética indica a consolidação de um ciclo de transmissão sustentado por reservatórios sinantrópicos e pela intensa interação entre espécies e ambientes. Tais evidências revelam a forte interconexão entre os ciclos urbano, rural e silvestre e confirmam o papel de animais domésticos, silvestres e do ambiente como fontes potenciais de infecção humana. De forma integrada, os resultados destacam a importância de incorporar ferramentas de epidemiologia molecular à vigilância e ao controle, fortalecendo uma abordagem efetivamente *One Health* (Uma só saúde) para o enfrentamento da leptospirose no Brasil.

ABSTRACT

Leptospirosis is a neglected zoonosis with a wide distribution in Brazil, having a significant impact on public health. Given the increase in zoonoses associated with environmental and climatic changes, the disease remains an emerging challenge, requiring a deeper understanding of its transmission patterns and the genetic diversity of the agents involved. This study aimed to investigate the genetic diversity, evolutionary relationships, and circulation patterns of pathogenic *Leptospira* spp. in Brazil, from different hosts, geographic regions, and environmental sources. To this end, *in vitro* and *in silico* molecular approaches were integrated, including PCR diagnosis, *secY* gene-based typing, phylogenetic and haplotype network analyses, as well as the curation and compilation of extensive genetic databases and ecological metadata. The integrated analyses revealed a wide ecological diversity of pathogenic *Leptospira* spp., genetically identified mainly in ruminants, dogs, humans, and small wild mammals. Importantly, they were also identified in environmental samples. The Atlantic Forest and Amazon biomes had the highest number of occurrences. This high diversity detected among multiple hosts and biomes highlights the complex eco-epidemiological structure of leptospirosis in the country. The most frequent species was *L. interrogans*, followed by *L. santarosai*, *L. borgpetersenii*, *L. kirschneri*, and *L. noguchii*. Extensive genetic overlap was observed between strains of different origins, and a clonal haplotype of *L. interrogans* from the widely disseminated serogroup Icterohaemorrhagiae predominated. This genetic homogeneity indicates the consolidation of a transmission cycle sustained by synanthropic reservoirs and the intense interaction among species and environments. Such evidence reveals the strong interconnection between urban, rural, and sylvatic cycles and confirms the role of domestic, wild, and domestic animals as potential sources of human infection. Taken together, the results highlight the importance of incorporating molecular epidemiology tools into surveillance and control, strengthening an effectively One Health approach to tackling leptospirosis in Brazil.

LISTA DE FIGURAS

Figura 1. Ciclo epidemiológico da leptospirose sob a perspectiva “Uma só Saúde”...5

Figura 2. Representação esquemática dos conceitos de SNP, assinatura genética, haplótipo e rede haplotípica.....18

LISTA DE ABREVIATURAS E SIGLAS

BGL: *Bovine Genital Leptospirosis*

c-di-GMP: *Cyclic diguanylate guanosine monophosphate*

DNA: *Deoxyribonucleic acid*

EUA: Estados Unidos da América

Ig-like: *Immunoglobulin-like domain*

LPS: Lipopolissacarídeo

MAT: *Microscopic Agglutination Test*

MIC: *Minimum Inhibitory Concentration*

MLST: *Multi Locus Sequence Typing*

OIE: Organização Mundial da Saúde Animal

OMS: Organização Mundial da Saúde

PCR: *Polymerase Chain Reaction*

pH: Potencial hidrogeniônico

qPCR: *Quantitative Polymerase Chain Reaction*

SLST: *Single Locus Sequence Typing*

SNP: *Single Nucleotide Polymorphism*

SPHL: Síndrome Pulmonar Hemorrágica da Leptospirose

ST: *Sequence Type*

SUMÁRIO

INTRODUÇÃO.....	1
1. O gênero <i>Leptospira</i>	1
2. Hospedeiros e aspectos ecológicos.....	3
3. Leptospirose humana.....	8
4. Leptospirose animal.....	11
5. Diagnóstico laboratorial.....	13
6. Epidemiologia molecular.....	15
7. Justificativa.....	19
OBJETIVOS.....	21
Objetivo geral.....	21
Objetivos específicos.....	21
ARTIGO 1.....	23
Molecular epidemiology of pathogenic <i>Leptospira</i> spp. infecting dogs in Latin America. (<i>Animals</i>)	
ARTIGO 2.....	36
Molecular epidemiology of <i>Leptospira</i> spp. serogroup Sejroe associated with chronic bovine leptospirosis. (<i>Veterinary Microbiology</i>)	
ARTIGO 3.....	46
Genetic characterization and zoonotic potential of <i>Leptospira interrogans</i> identified in small non-flying mammals from Southeastern Atlantic Forest, Brazil. (<i>Tropical Medicine and Infectious Diseases</i>)	
ARTIGO 4.....	58
Ecological range and host–biome associations of pathogenic <i>Leptospira</i> in Brazil: A One Health perspective from a tropical area. (<i>Acta Tropica</i>)	
ARTIGO 5.....	73
Genetic diversity and clonal expansion of pathogenic <i>Leptospira</i> in Brazil: a multi-host and multi-regional panorama. (<i>Microorganisms</i>)	
DISCUSSÃO.....	87
CONCLUSÕES.....	92
REFERÊNCIAS.....	93

INTRODUÇÃO

1. O gênero *Leptospira*

O gênero *Leptospira* pertence ao filo Spirochaetes, classe Spirochaetia, ordem Leptospirales, família Leptospiraceae (Levett, 2015). Dois esquemas de classificação são utilizados para esta bactéria. O primeiro é sorológico, baseado principalmente nas diferenças antigênicas dos lipopolissacarídeos (LPSs) presentes na superfície bacteriana, o que resulta em mais de 30 sorogrupos e cerca de 300 sorovares distintos (Mejía et al., 2022). Historicamente, utiliza-se para esta definição o teste de aglutinação microscópica (MAT), que continua sendo o padrão-ouro para identificação de sorogrupos em laboratórios de referência. Este esquema sorológico permanece fundamental para a epidemiologia, diagnóstico e desenvolvimento de vacinas contra leptospirose (Guedes et al., 2021a). O outro esquema é baseado na taxonomia molecular para definição de espécies. Análises filogenômicas reconhecem 64 espécies de *Leptospira*, que podem ser divididas em três linhagens que refletem seu grau de patogenicidade: saprofíticas ($n=26$), intermediárias ($n=21$) e patogênicas ($n=17$) (Vincent et al., 2019). É importante ressaltar que as classificações sorológica e molecular não estão associadas, e diferentes sorogrupos/sorovares podem pertencer a mesma espécie, sendo seu uso integrativo encorajado.

Leptospira spp. são bactérias aeróbicas de crescimento lento, com características morfológicas marcantes, como seu formato em hélice e um endoflagelo bem desenvolvido, o que lhes confere alta motilidade, um fator crítico de virulência para leptospirosas patogênicas (Nakamura, 2020). Sua parede celular é formada por dupla membrana e sua camada externa é rica em biomoléculas, com destaque para LPS, presente abundantemente em espécies patogênicas. Apesar da estrutura do LPS ser semelhante à de bactérias Gram-negativas, leptospirosas não se coram bem pela técnica de Gram, e apresentam menor toxicidade (Hsu e Yang, 2022). O método clássico e ainda amplamente utilizado para a visualização da bactéria é a microscopia de campo escuro, que permite observar diretamente a morfologia espiralada e a motilidade característica dessas espiroquetas em amostras frescas (Fontana e Reyes, 2020). Alternativamente, a impregnação por sais de prata, como nos métodos de Warthin–Starry ou Fontana–Tribon, permite a observação morfológica de

leptospiras em cortes histológicos, destacando seus filamentos delgados e helicoidais contra o fundo tecidual (Adler e De la Peña Moctezuma, 2010).

Além do LPS, a superfície de *Leptospira* spp. é rica em proteínas de membrana externa, muitas das quais são lipoproteínas ancoradas por ácidos graxos na membrana externa. Entre as mais abundantes e estudadas estão LipL32, LipL41, LipL21, LipL46, Loa22, OmpL36 e LipL71 (Daroz et al., 2021). LipL32, em particular, é a proteína de superfície mais abundante nas espécies patogênicas, apresenta estrutura globular com um motivo “jelly-roll” e regiões ácidas que interagem com componentes da matriz extracelular, como laminina, sugerindo papel na adesão e evasão imune (Hsu et al., 2024). Outras proteínas, como aquelas da família Lig (LigA, LigB), possuem domínios imunoglobulina-like (Ig-like) e são fundamentais para a ligação à matriz extracelular e fatores do complemento, contribuindo para a virulência e evasão do sistema imunológico (Kumar et al., 2023).

Leptospira spp. possuem, ainda, a capacidade de formar biofilmes em ambientes naturais e em hospedeiros animais e humanos, incluindo ambientes aquáticos, solos, superfícies artificiais e, de forma importante, nos túbulos renais de roedores reservatórios, como *Rattus norvegicus*, onde favorecem a persistência e a transmissão da bactéria (Santos et al., 2021; Dos Santos Ribeiro et al., 2023). Tanto espécies patogênicas quanto saprofíticas exibem essa habilidade, com biofilmes compostos por matriz polimérica que protege as células contra estresses ambientais e sustenta sua sobrevivência fora do hospedeiro (Dias; Pinna, 2025). A formação do biofilme é regulada por sistemas intracelulares, como o c-di-GMP, que controla a expressão de fatores de adesão e componentes estruturais, estando associada a maior colonização renal em infecções crônicas e à persistência bacteriana em infecções agudas. Além disso, biofilmes de leptospiras apresentam redução da suscetibilidade a antibióticos, com aumento das concentrações inibitórias mínimas (MICs) e indução de sua formação por exposição a doses subinibitórias de antimicrobianos (Vinod Kumar et al., 2018). Em conjunto, esses achados indicam que a formação de biofilme constitui um mecanismo-chave de sobrevivência, persistência ambiental, resistência antimicrobiana e transmissão de leptospiras.

2. Hospedeiros e aspectos ecológicos

Leptospirose é uma zoonose clássica. Mais de 160 espécies de mamíferos e algumas poucas espécies de répteis e anfíbios já foram identificadas como hospedeiros e/ou reservatórios da bactéria, que disseminam o patógeno por meio de seus fluidos corporais (principalmente urina), contaminando cursos d'água e solo (Cilia et al., 2021). Historicamente, a etiologia das infecções por leptospirosas em seus hospedeiros é dividida naquelas determinadas por sorogrupos incidentais e sorogrupos adaptados. Estirpes incidentais ocorrem em hospedeiros atípicos, e levam à um quadro agudo com sintomas aparentes. Estirpes adaptadas são aquelas observadas no hospedeiro animal reservatório, fruto de um longo processo de coevolução patógeno-hospedeiro, levando à doença assintomática ou inaparente, ou sintomas leves e subclínicos de natureza crônica. Hospedeiros de manutenção indicam os animais infectados que são assintomáticos e liberam leptospirosas viáveis em seus fluidos, atuando simultaneamente como hospedeiros e reservatórios (Putz e Nally, 2020). Esses hospedeiros desempenham um papel importante na epidemiologia da leptospirose, pois abrigam e disseminam a bactéria no ambiente por longos períodos, tornando-se importantes fontes de infecção para hospedeiros não adaptados (Adler e De la Peña Moctezuma, 2010). Exemplos de sorovares adaptados são Bratislava em equinos, Pomona em suínos, Hardjo em ruminantes, Canicola em cães e Icterohaemorrhagiae em roedores (Hagedoorn et al., 2024). De maneira importante, espécies do sorogrupo Icterohaemorrhagiae são as mais associadas à doença aguda humana (Browne et al., 2023), evidenciando o caráter incidental da infecção no homem.

Sabe-se também que, como leptospirosas sobrevivem por um período prolongado no solo e se multiplicam em terrenos úmidos, e como a leptospirose é mais frequente após inundações, deve-se considerar também a transmissão ambiental (Yanagihara et al., 2022). Fatores ambientais, como pH, temperatura, presença de matéria orgânica e características do solo e da água influenciam a persistência das leptospirosas no ambiente, sendo que a bactéria pode sobreviver por semanas a meses em condições favoráveis (Bierque et al., 2020). Estudos recentes demonstram que a abundância de reservatórios (especialmente roedores), a taxa de eliminação de leptospirosas e fatores hidrológicos (como enchentes e entrada de água contaminada em residências) são determinantes

críticos para o risco de infecção humana, superando até mesmo a influência da carga bacteriana individual dos animais (Soni et al., 2024). Sendo assim, a leptospirose é um clássico exemplo a ser estudo no contexto de “Saúde Única” (do inglês, *One Health*) (Sykes et al., 2022b) (Figura 1). O Ministério do Meio Ambiente do Brasil passou a adotar oficialmente a nomenclatura “Uma Só Saúde” (Brasil, 2024), a qual será empregada ao longo deste documento.

As vias de transmissão das leptospirosas no ambiente são complexas e têm sido relatadas em uma variedade de cenários, incluindo grandes centros urbanos, áreas rurais e ambientes silvestres (Harran et al., 2022) (Figura 1). Nos centros urbanos, o rato de esgoto/ratazana (*Rattus norvegicus*) é o principal reservatório e disseminador da bactéria (Boey et al., 2019), enquanto ruminantes e pequenos mamíferos (roedores e marsupiais) exercem papel importante nos meios rural e silvestre, respectivamente (Martins e Lilenbaum, 2017; Rosli et al., 2023). Estudos recentes vêm sugerindo o papel importante dos cães na transmissão urbana e periurbana (Sant’Anna et al., 2021; Azócar-Aedo e Monti, 2022; Balboni et al., 2022; Guzmán et al., 2023; Ricardo et al., 2024). De forma relevante, bovinos podem excretar leptospirosas de maneira intermitente tanto na urina como em fluidos genitais, com períodos que variam de 79 a 259 dias, e em quantidades significativas, o que contribui para a manutenção da bactéria no ambiente rural (Monti et al., 2023).

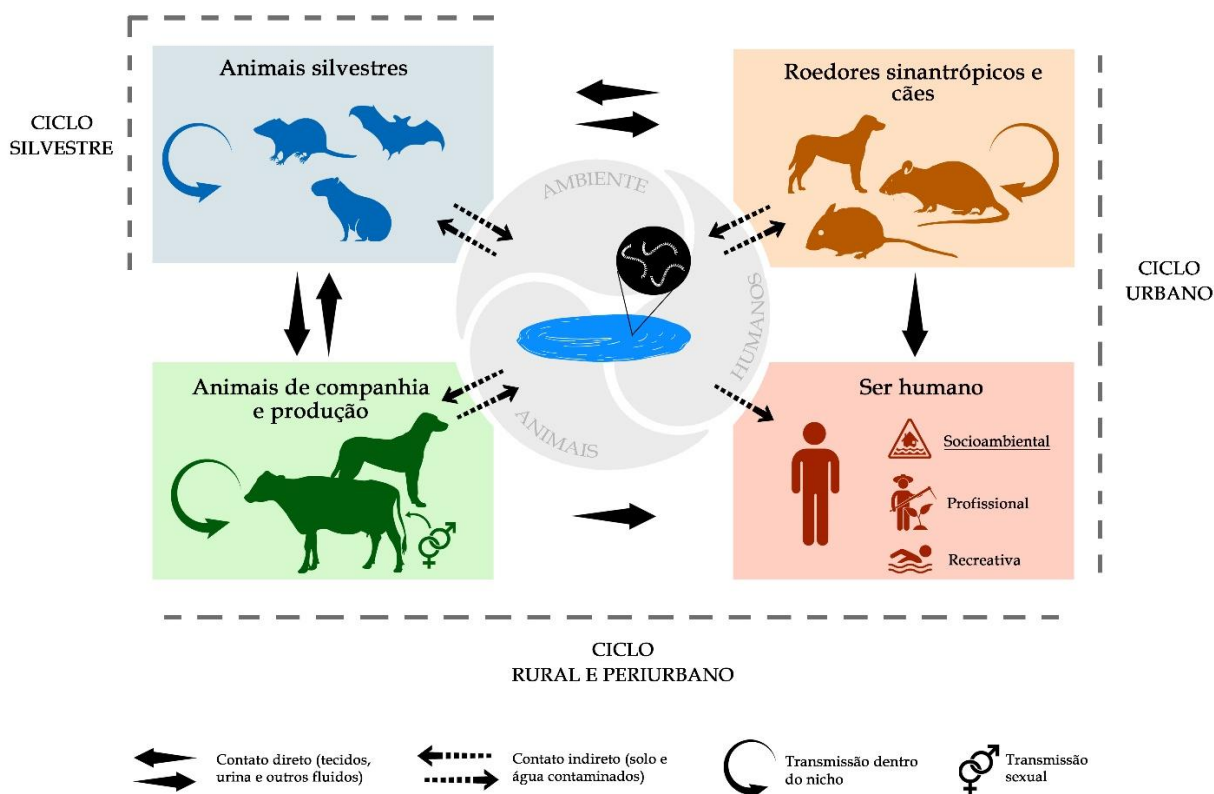


Figura 1. Ciclo epidemiológico da leptospirose sob a perspectiva “Uma só Saúde”. Representação esquemática evidenciando a interação entre animais silvestres, roedores sinantrópicos, animais de companhia e produção, seres humanos e o ambiente. Os principais componentes dos ciclos silvestre, rural/periurbano e urbano são mostrados. As setas indicam as principais rotas de transmissão: direta (contato com tecidos, urina e outros fluidos), indireta (contato com solo e água contaminados) e sexual (bovinos). Os contextos de exposição humana são mostrados: socioambiental (principal), ocupacional e recreativa. É importante observar que o homem é hospedeiro acidental e não apresenta importância epidemiológica no ciclo da bactéria. (Elaborado pela autora).

A principal via de transmissão para o hospedeiro humano é indireta, por meio do contato de mucosas (ocular, oral, nasal) ou pele lesada com água, solo ou alimentos contaminados pela urina de animais infectados. A exposição a ambientes úmidos, inundações, enchentes e solos encharcados aumenta o risco, pois favorece a sobrevivência e dispersão das leptospiros no ambiente,

especialmente em áreas urbanas com saneamento precário e alta densidade de roedores (CDC, 2025). Eventos de chuva intensa podem ressuspender leptospiros presentes no solo, carreando-as para corpos d'água superficiais e ampliando a exposição (Bierque et al., 2020). A transmissão direta é comum entre animais e ocorre principalmente por contato com urina e tecidos contaminados, mas também pode envolver secreções genitais, placentárias ou leite, especialmente em bovinos, onde a transmissão sexual é reconhecida e contribui para a disseminação intraespecífica (Haji Hajikolaei et al., 2022; Regassa e Obsu, 2024). Em humanos, a infecção direta é rara, mas pode ocorrer por contato ocupacional (trabalhadores rurais, médicos veterinários, coletores de lixo) ou recreacional (esportes aquáticos em águas contaminadas) (Barragan et al., 2017) (Figura 1). A transmissão por mordedura de animal é também rara, e a transmissão inter-humana é insignificante (Rajapakse et al., 2025).

No Brasil, a leptospirose humana é uma doença endêmica, tornando-se epidêmica em períodos chuvosos, principalmente nas capitais e áreas metropolitanas, devido às enchentes associadas à aglomeração populacional de baixa renda, às condições inadequadas de saneamento e à alta infestação de roedores infectados, sendo significativamente relacionada à vulnerabilidade da população (Teles et al., 2023). A incidência urbana é mais elevada nas regiões Norte e Nordeste, e rural mais alta no Sudeste e Sul (Linhares et al., 2024). A maioria dos surtos humanos no Brasil é causada por *L. interrogans* do sorovar Copenhageni (sorogrupo Icterohaemorrhagiae), tradicionalmente associado a roedores urbanos (Pereira et al., 2000). As espécies *L. noguchii* e *L. kirschneri* também já foram associadas a casos humanos (Silva et al., 2009; Balassiano et al., 2017). Estudos sorológicos em diferentes regiões do país demonstram, em menor grau, a circulação de outros sorovares, como Pomona, Tarassovi, Hardjo, Australis, Grippotyphosa e Canicola, refletindo a diversidade de fontes de infecção (Oliveira et al., 2017; De Oliveira et al., 2024).

A leptospirose animal no Brasil permanece altamente prevalente e representa um desafio significativo para a saúde animal, saúde pública e produção agropecuária. Estudos demonstram que a infecção por *Leptospira* spp. é endêmica em múltiplas espécies domésticas e silvestres, com variações regionais importantes conforme o hospedeiro e o bioma. Em cães, a soroprevalência nacional é estimada em aproximadamente 20% (Esteves et al.,

2023), a maioria do sorogrupo *Icterohaemorrhagiae* (Jaeger et al., 2018; Sant'Anna et al., 2021; Di Azevedo et al., 2022; Esteves et al., 2023).

Já em bovinos, a soroprevalência pode ultrapassar 50% em determinadas regiões, com destaque para infecções causadas pelo sorovar Hardjo (sorogrupo Sejroe), associadas a uma forma crônica reprodutiva subclínica e silenciosa, muitas vezes negligenciada (Ellis, 2015). Historicamente, os rins têm sido o foco principal dos estudos de colonização por *Leptospira* spp. (Hernández-Rodríguez et al., 2011; Guedes et al., 2021b; Saglam et al., 2022). No entanto, pesquisas recentes têm elucidado infecções do trato genital por estirpes do sorogrupo Sejroe no útero (Aymée et al., 2022b; Di Azevedo et al., 2020), fluido folicular (Di Azevedo et al., 2021; Pereira et al., 2022), muco cervicovaginal (Loureiro et al., 2016; Pimenta et al., 2019; Gonçalves et al., 2021), sêmen (Vinodh et al., 2008), testículo e epidídimo (Pedrosa et al., 2024). De forma importante, leptospirosas do sorogrupo Sejroe já foram identificadas em infecções humanas (Romero e Yasuda, 2006; Kakita et al., 2020).

Apesar da diversidade e abundância da fauna silvestre brasileira, pouco se sabe sobre o papel de espécies silvestres na epidemiologia da leptospirose no Brasil. No bioma Pantanal, 40,6% das amostras foram positivas molecularmente (PCR), incluindo cachorro-do-mato (*Cerdocyon thous*), quati (*Nasua nasua*), veado-campeiro (*Ozotoceros bezoarticus*) e javali (*Sus scrofa*), sendo os sorogrupos mais prevalentes Javanica e Djasiman (Vieira et al., 2016). Na Mata Atlântica, estudos com pequenos mamíferos, especialmente roedores silvestres das ordens Rodentia e Didelphimorphia, revelam altas taxas de infecção renal por leptospirosas patogênicas, com positividade em até 28% dos animais testados. Foram geneticamente identificadas quatro espécies patogênicas circulantes: *L. interrogans*, *L. borgpetersenii*, *L. santarosai* e *L. noguchii* (Vieira et al., 2019a). Adicionalmente, roedores neotropicais dos gêneros *Akodon* e *Oligoryzomys* apresentam taxas positividade superiores a 30%, com confirmação molecular de espécies patogênicas, reforçando a importância desses hospedeiros na manutenção do ciclo silvestre da leptospirose na Mata Atlântica (Vieira et al., 2019b). Uma prevalência de 16% foi encontrada em gambás (*Didelphis albiventris*) e quatis (*Nasua nasua*) no estado de São Paulo. Os sorogrupos mais comuns foram Djasiman e Australis, enquanto *L. santarosai* foi a espécie mais prevalente (Fornazari et al., 2018).

A fauna silvestre amazônica também é reservatório de leptospiros. Capivaras em áreas periurbanas e rurais do oeste da Amazônia apresentam altas taxas de infecção crônica, com isolamento de leptospiros dos sorogrupos Icterohaemorrhagiae, Grippotyphosa e Shermani, com destaque para Icterohaemorrhagiae, sugerindo papel análogo ao de roedores urbanos como reservatórios (Albuquerque et al., 2017). *Leptospira interrogans*, *L. noguchii* e *L. santarosai* foram identificadas em didelfimorfos e roedores da Amazônia Ocidental brasileira, com 44,7% de positividade (Medeiros et al., 2020). Na mesma região, morcegos foram reconhecidos como reservatórios de *L. interrogans* (21% de positividade) e de uma potencial espécie nova (Di Azevedo et al., 2025), com elevada ocorrência independentemente do tipo de paisagem (urbana, rural ou de floresta), com sanguinívoros apresentando maior positividade (Verde et al., 2024).

3. Leptospirose humana

A leptospirose humana permanece um importante problema de saúde pública global, com incidência significativa em regiões tropicais e subtropicais (Costa et al., 2015), mas também com casos esporádicos em áreas consideradas de baixo risco, onde surtos em animais podem preceder casos humanos (Waranius et al., 2024). Estima-se que ocorram cerca de um milhão de casos e 60.000 mortes por ano no mundo (Costa et al., 2015). No Brasil, existem registros de leptospirose em todas as unidades da federação, com um maior número de casos nas regiões sul e sudeste. A média anual é de 3.600 casos e 375 óbitos, sendo indivíduos do sexo masculino com faixa etária entre 20 e 49 anos os mais atingidos (Brasil, 2024). Em relação a eventos recentes, o estado do Rio Grande do Sul confirmou 25 mortes por leptospirose até julho de 2024, muito provavelmente relacionadas às enchentes que atingiram o estado (Michelon et al., 2025). É importante destacar que esses dados provavelmente são subestimados, sobretudo devido à sua ocorrência em áreas co-endêmicas com outras doenças agudas febris infecciosas, como dengue, chikungunya e malária. Nesses contextos, o diagnóstico clínico, baseado apenas na apresentação inespecífica de sintomas tende a direcionar o diagnóstico para outras etiologias mais comuns ou com maior visibilidade epidemiológica (Esteves et al., 2025). Como resultado, casos de leptospirose podem passar despercebidos, especialmente na ausência de

manifestações graves. Além disso, o diagnóstico laboratorial definitivo é limitado: o MAT, teste mais empregado, exige infraestrutura laboratorial especializada, acesso a painéis vivos de sorovares e capacitação técnica, estando disponível apenas em laboratórios de referência (Ciurariu et al., 2025), o que contribui ainda mais para a subnotificação da doença e dificulta a compreensão de sua real magnitude no país.

A leptospirose em humanos apresenta manifestações clínicas muito diversas, podendo variar desde uma doença febril aguda leve e autolimitada até uma condição grave e potencialmente fatal com disfunção de múltiplos órgãos (Haake e Levett, 2015). As características clínicas são similares a outras doenças febris, especialmente aquelas vistas nos trópicos, como dengue, febres hemorrágicas, infecções por rickettsias, malária e sepse bacteriana (Rajapakse, 2022). Embora a maioria dos pacientes apresente febre não complicada, aproximadamente 10% desenvolvem doença grave (Adler e De la Peña Moctezuma, 2010).

O período de incubação varia amplamente, de 2 a 20 dias, geralmente entre 7 a 12 dias. O curso clínico é classicamente dividido em duas fases: a "fase leptospirêmica" ou fase aguda (que dura de 3 a 9 dias) e a "fase imune". A fase inicial apresenta-se como uma doença febril aguda inespecífica com febre, calafrios, mialgia e cefaleia. A sufusão conjuntival é um achado característico que se desenvolve no terceiro ou quarto dia. A mialgia pode ser severa, afetando principalmente panturrilhas, abdômen e músculos paraespinhais (Crecelius e Burnett, 2020).

A apresentação clássica grave é conhecida como síndrome de Weil, caracterizada por hemorragia, icterícia e lesão renal aguda. A Síndrome Pulmonar Hemorrágica da Leptospirose (SPHL) tem sido reconhecida como uma importante causa de mortalidade (Nicodemo e Duarte-Neto, 2021). Outras manifestações graves incluem envolvimento miocárdico, disfunção endotelial e vasculite. Diversas manifestações atípicas podem ocorrer, incluindo complicações neurológicas (encefalomielite, hidrocefalia, eventos vasculares intracranianos), oculares (uveíte, neurite óptica), hematológicas (pancitopenia, anemia hemolítica) e gastrointestinais (pancreatite, colecistite) (Rajapakse et al., 2015).

Com relação ao controle, medidas de prevenção importantes incluem: i) evitar o contato com roedores e animais domésticos; ii) evitar contato direto com solo

e bolsões de água em regiões endêmicas; iii) uso de equipamentos de proteção adequados no caso de pessoas que estão em contato constante com água doce, como agricultores e esportistas aquáticos; iv) implementação de programas de controle de enchentes, prevenindo inundações; v) evitar contato com a água ou utilizar os devidos equipamentos de proteção em casos de enchentes. As medidas clínicas preventivas incluem quimioprevenção e vacinação (Rajapakse et al., 2025).

O tratamento deve ser iniciado precocemente, idealmente na fase inicial da doença. Para casos leves, recomenda-se doxiciclina como primeira escolha, sendo alternativas ampicilina, amoxicilina ou azitromicina, especialmente em situações de contraindicação. Em casos graves, como na presença de disfunção orgânica (icterícia, insuficiência renal, hemorragia pulmonar), a penicilina intravenosa é tradicionalmente o antibiótico de escolha, com ceftriaxona e cefotaxima como alternativas igualmente eficazes (Mendu et al., 2025).

O desenvolvimento de uma vacina universal é dificultado pela grande diversidade antigênica dos sorovares patogênicos e pela ausência de correlatos imunológicos de proteção robustos (Azevedo e Amamura; Isaac, 2023). Vacinas comerciais para uso em humanos estão disponíveis apenas em alguns países (Japão, China, Cuba e França), sendo compostas por bacterinas (leptospiras inativadas). Essas vacinas induzem resposta imune de curta duração e limitada ao(s) sorovar(es) incluído(s) na formulação, sem proteção cruzada significativa. Além disso, são associadas a efeitos adversos relevantes, o que limita sua aceitação e uso em larga escala (Barazzzone et al., 2021). Não há vacinas recombinantes aprovadas para uso humano, apesar de décadas de pesquisa e múltiplos candidatos promissores em modelos animais (Oliveira et al., 2022). Estratégias inovadoras, como vacinas de DNA, proteínas recombinantes, abordagens de vacinologia reversa e vacinas multiepitópicas têm sido investigadas, mas nenhuma avançou para uso clínico amplo (Felix et al., 2020). Uma vacina universal, capaz de induzir imunidade duradoura e proteção cruzada, ainda está distante da implementação clínica.

4. Leptospirose animal

A leptospirose em animais é onipresente, sendo relatada em diversas espécies em todas as regiões do mundo (Ellis, 2015). Parece haver uma gama de suscetibilidade à infecção, com cavalos sendo amplamente suscetíveis (Díaz et al., 2023), enquanto a infecção em gatos é rara (Miotto et al., 2024). Em cães, a leptospirose apresenta-se predominantemente como doença aguda hepatorenal, com sinais clínicos e laboratoriais inespecíficos, podendo evoluir para falência multiorgânica e morte, especialmente quando há envolvimento pulmonar ou renal grave (Sykes et al., 2023). Além da apresentação aguda, doença renal crônica e quadros assintomáticos também são comuns, e parece não existir uma correlação clara entre desfecho clínico e sorovares e/ou perfil genético da bactéria (Di Azevedo et al., 2022).

Em bovinos, assim como em outros ruminantes, a forma aguda e severa da leptospirose é incomum e frequentemente associada a surtos esporádicos em bezerros causados por estirpes incidentais (Ellis, 2015). A apresentação mais comum da leptospirose animal é a forma crônica subclínica e silenciosa, muitas vezes negligenciada, onde sinais reprodutivos são predominantes. Como consequência da colonização do trato reprodutivo por *Leptospira* spp., essa forma crônica determina importantes disfunções reprodutivas, como subfertilidade, repetição de estro, abortos e natimortos, levando a sofrimento animal e perdas econômicas impactantes. Essa forma reprodutiva silenciosa da leptospirose hoje é considerada como uma síndrome distinta e específica, chamada Leptospirose Genital Bovina (BGL) (Loureiro e Lilenbaum, 2020).

A infecção por *Leptospira* spp. em pequenos mamíferos, como roedores e marsupiais, é predominantemente assintomática, com colonização renal crônica e excreção urinária persistente do agente (Mohd-Taib et al., 2020). Em modelos experimentais, a gravidade das manifestações clínicas varia conforme a espécie e a estirpe, sendo hamsters particularmente suscetíveis a formas agudas e fatais, enquanto camundongos e roedores silvestres geralmente permanecem assintomáticos (Philip et al., 2021; Beato-Benítez et al., 2024).

O controle da leptospirose animal é complexo, uma vez que envolve um cenário muito propenso para transmissão e de difícil monitoramento. Entretanto, apesar das dificuldades, o controle é possível e deve ser incentivado e envolve a implementação da tríade antibioticoterapia - vacinação – manejo ambiental

(Martins e Lilenbaum, 2017). O tratamento da leptospirose em cães baseia-se principalmente no uso de antibióticos, sendo doxiciclina a droga de escolha por sua eficácia na eliminação da bactéria dos túbulos renais, interrupção da leptospiúria e consequente transmissão zoonótica (Sykes et al., 2023). No entanto, a persistência da leptospiúria mesmo após tratamento adequado com doxiciclina ou enrofloxacin tem sido relatada, recomendando-se acompanhamento com PCR urinário e uso de claritromicina, embora com evidências limitadas (Hetrick et al., 2022). A vacinação anual é uma medida preventiva essencial, com eficácia de aproximadamente 84% contra a doença clínica e 88% contra o estado de portador renal, variando conforme o sorovar infectante e a vacina utilizada (Esteves et al., 2022).

Em bovinos, a leptospirose renal é tradicionalmente tratada com dose única de estreptomicina (25 mg/kg), geralmente eficaz (Ellis, 2015). No entanto, infecções genitais, especialmente por estirpes do sorogrupo Sejroe, exigem protocolos prolongados, com três doses sendo mais eficazes, com taxa de sucesso superior a 90% (Aymée et al., 2023). Há relatos de suscetibilidade reduzida de estirpes bovinas a estreptomicina, tetraciclina e ceftiofur, destacando a importância do monitoramento da resposta terapêutica e da resistência antimicrobiana (Correia et al., 2018). Atualmente, todas as vacinas comerciais contra leptospirose bovina são bacterinas. Embora a proteção seja mais eficaz entre sorovares do mesmo sorogrupo, alguma proteção cruzada entre diferentes sorogrupos pode ocorrer. A inclusão de estirpes representantes dos sorogrupos mais prevalentes nas vacinas é fundamental, sendo a vigilância sorológica essencial para orientar essa escolha (Aymée et al., 2024).

Devido à sobrevivência da bactéria no ambiente e à alta diversidade de animais portadores, tanto domésticos quanto selvagens, o controle ambiental é particularmente desafiador, especialmente em regiões tropicais. Estratégias ambientais eficazes incluem a redução da exposição dos animais a fontes de água potencialmente contaminadas, o manejo adequado de resíduos e dejetos, a drenagem de áreas alagadas e o controle rigoroso de roedores (Soo et al., 2020). Revisões sistemáticas apontam que intervenções ambientais, como o controle de enchentes, melhoria do saneamento e redução do contato dos animais com águas contaminadas são fundamentais para interromper os ciclos

de transmissão, tanto entre animais quanto na interface animal-humano (Mwachui et al., 2015; Naing et al., 2019).

Embora seja amplamente conhecido que as estações chuvosas facilitam a ocorrência de surtos causados por estirpes incidentais, principalmente no ambiente urbano (Cunha et al., 2022), isso não parece ser marcante em infecções por estirpes adaptadas, como aquelas associadas à BGL, que podem ocorrer independentemente das condições ambientais (Correia et al., 2017). Nesses casos, o controle da doença depende principalmente de medidas sanitárias específicas voltadas ao manejo reprodutivo dos animais (Aymée et al., 2024).

5. Diagnóstico laboratorial

O diagnóstico correto da leptospirose depende de alguns fatores importantes, como a coleta de espécimes adequados e, principalmente, o estágio da infecção. Atualmente, o diagnóstico pode ser realizado de forma direta pela detecção da bactéria por meio da observação a fresco de leptospiras, cultura ou detecção de seu DNA; e indiretamente pela detecção de anticorpos específicos (sorologia) (Koizumi, 2020).

A visualização direta de leptospiras no sangue, urina e leite de animais infectados pode ser realizada usando microscopia de campo escuro, mas a sensibilidade e a especificidade são baixas. Portanto, este método não é usado rotineiramente no diagnóstico de leptospirose (Sykes et al., 2022a). Uma cultura positiva de amostras biológicas (sangue, urina, tecido) é a prova definitiva de infecção, mas cultivar leptospiras é trabalhoso e fastidioso. As bactérias requerem meios de crescimento especiais e a incubação pode durar meses. Embora altamente específica, a sensibilidade da cultura é baixa, o que pode estar relacionado à baixa bacteremia, administração de antibióticos antes da coleta da amostra ou dificuldades técnicas (Goarant et al., 2020).

No que se refere ao diagnóstico indireto, a sorologia, principalmente o MAT, é adequada para o diagnóstico de leptospirose aguda em humanos e cães, onde pacientes infectados apresentam altos títulos de IgM e IgG, facilmente detectáveis, e seu uso é indicado pela OIE e OMS para fins diagnósticos (Rodríguez-Rodríguez et al., 2024). Seu valor epidemiológico reside em sua capacidade de indicar o sorogrupo circulante. No entanto, este ensaio requer

experiência significativa para ser executado e manutenção de culturas vivas, o que limita sua aplicabilidade em larga escala, principalmente em áreas rurais ou de baixa renda (Yang et al., 2019). Além disso, embora útil para o diagnóstico coletivo, a sorologia apresenta limitações importantes para a leptospirose animal crônica, não sendo adequada para o diagnóstico individual (Di Azevedo e Lilenbaum, 2021).

A introdução de métodos moleculares, especialmente a PCR, representa um avanço importante, permitindo a detecção direta do DNA leptospírico em amostras clínicas, principalmente sangue e urina, durante a fase aguda da infecção (Verma et al., 2020). A PCR apresenta alta sensibilidade e especificidade nos primeiros dias da doença, mesmo antes da soroconversão, e pode ser utilizada em combinação com testes sorológicos para aumentar a acurácia diagnóstica ao longo das diferentes fases clínicas (Ciurariu et al., 2025). No entanto, a padronização dos protocolos de PCR e a disponibilidade desses testes ainda são desafios, especialmente em ambientes com recursos limitados.

Embora o diagnóstico da doença aguda seja razoavelmente padronizado tanto pelo método direto quanto pelo indireto, o diagnóstico da forma crônica silenciosa permanece um desafio. A sintomatologia leve, frequentemente inaparente, torna a suspeita clínica muito rara. Na maioria dos animais domésticos infectados, a eliminação de leptospiras é intermitente e não muito intensa; além disso, apresentam baixos títulos de anticorpos (Nally et al., 2018), dificultando tanto o diagnóstico direto quanto o indireto. Além disso, apesar dos problemas reprodutivos serem amplamente relatados, a maioria dos estudos enfoca a investigação bacteriana do trato urinário, e as amostras genitais permanecem subutilizadas, quando, na verdade, deveriam ser a amostra primária de escolha para o diagnóstico da forma crônica da leptospirose animal (Loureiro e Lilenbaum, 2020).

O gene *lipL32*, que codifica uma lipoproteína de 32kDa localizada na membrana externa de *Leptospira* spp. patogênicas, tem sido amplamente utilizado como alvo em ensaios moleculares devido à sua alta especificidade e expressão abundante durante a infecção em hospedeiros mamíferos (Stoddard et al., 2009). Esta lipoproteína está ausente na maioria das espécies saprofíticas e intermediárias, o que confere ao marcador uma excelente capacidade discriminatória entre estirpes patogênicas e não patogênicas (Ahmed et al.,

2009). Métodos baseados em PCR convencional ou em tempo real (qPCR) que amplificam esse gene demonstram alta sensibilidade e especificidade, sendo considerados ferramentas valiosas para o diagnóstico direto da leptospirose em humanos e animais e já são uma realidade em laboratórios de referência (Khaki et al., 2023).

6. Epidemiologia molecular

A epidemiologia molecular é o ramo da epidemiologia que estuda, numa abordagem molecular, o papel dos fatores genéticos e suas interações com fatores ambientais na ocorrência e distribuição de doenças em populações. Dentre seus objetivos estão identificar os microrganismos causadores de doenças e determinar suas fontes (hospedeiros), suas relações biológicas (filogenia) e seus modos de transmissão. Um dos resultados do emprego desta abordagem é a elucidação da etiologia da doença e seus padrões de distribuição, o que constituiu peça fundamental para mitigação de impactos, prevenção e controle de doenças (Honardoost et al., 2018).

Um conjunto vasto e integrado de ferramentas laboratoriais e computacionais pode ser empregado para estudos de epidemiologia molecular. As abordagens laboratoriais são essenciais para a detecção direta do patógeno, identificação precisa em nível de espécie e tipificação genotípica (Tümmler, 2020). Em um segundo momento, ferramentas de bioinformática desempenham papel crucial na triagem, curadoria, alinhamento de sequências e análise integrativa dos metadados, incorporando informações epidemiológicas complementares (ex: hospedeiro, origem geográfica, ano de coleta, quadro clínico, etc) (Elrashedy et al., 2025). A partir dessa integração, é possível gerar inferências que permitem compreender padrões de disseminação, fontes de infecção e relações evolutivas entre as sequências (Riley, 2018).

Para fins de epidemiologia molecular, a tipificação bacteriana é muito importante. A análise genética de espécies patogênicas de *Leptospira* spp. é mais eficiente quando realizada em estirpes bacterianas obtidas de animais infectados (uma vez que o DNA está mais concentrado, íntegro e livre de contaminantes), mas, como exposto anteriormente, o cultivo de leptospiros não é trivial. Por outro lado, a identificação e, principalmente, a tipificação de leptospiros diretamente de

amostras clínicas sem cultivo é um desafio. O método clássico, *Multi Locus Sequence Typing* (MLST), que utiliza polimorfismos de sequências de múltiplos genes *housekeeping* para caracterização e investigação de relações evolutivas entre bactérias intimamente relacionadas, na grande maioria das vezes só pode ser realizado com total eficiência após o cultivo (Varni et al., 2014). Assim, esforços têm sido feitos para realizar investigações genéticas diretamente de amostras clínicas usando regiões curtas do DNA (Guernier et al., 2018).

Uma estratégia eficiente e amplamente empregada é o sequenciamento de genes individuais do genoma bacteriano, ou *Single Locus Sequence Typing* (SLST). Essa metodologia baseia-se no sequenciamento de um único gene informativo, que apresenta regiões conservadas suficientes para amplificação universal, mas também variabilidade suficiente para discriminar espécies patogênicas e, ainda, estabelecer relações intraespecíficas (Ederveen et al., 2019). Dentre os alvos utilizados para caracterização de *Leptospira* spp., o gene *secY* é amplamente utilizado por sua alta capacidade de diferenciação interespecie e boa performance mesmo em amostras clínicas com baixa carga bacteriana (Di Azevedo e Lilenbaum, 2021). Este gene codifica uma subunidade essencial do sistema de translocação proteica bacteriano, e tem se destacado como um marcador molecular robusto para a detecção e tipificação de *Leptospira* spp. patogênicas (Lam et al., 2020). Ensaio baseados na amplificação parcial (~500pb) e sequenciamento deste gene têm sido utilizados com sucesso em estudos epidemiológicos, especialmente em contextos onde o isolamento bacteriano não é possível (Dogonyaro et al., 2023; Sarangi et al., 2025; Silva-Ramos et al., 2025). Além disso, a tipificação molecular baseada neste marcador tem se mostrado uma ferramenta valiosa para a investigação de surtos e para a compreensão da distribuição das espécies de *Leptospira* em diversos hospedeiros e regiões geográficas, contribuindo significativamente para a vigilância da leptospirose (Mayer-Scholl et al., 2019; Aymée et al., 2022; Javati et al., 2022). Apesar de apresentar resolução inferior em comparação com métodos multigênicos como o MLST ou análises genômicas completas, o SLST tem se mostrado valioso em estudos taxonômicos e epidemiológicos, fornecendo dados filogenéticos relevantes de maneira rápida e com custo reduzido.

Enquanto análises MLST geram sequências-tipo (STs), o SLST produz haplótipos, definidos como combinações específicas de variantes genéticas herdadas em conjunto que permitem distinguir linhagens ou populações de um mesmo organismo (Scholz; Jensen, 2017). Essas variantes geralmente correspondem a polimorfismos de nucleotídeo único (SNPs): substituições pontuais em uma única base do DNA que representam a forma mais simples e comum de variação genômica entre indivíduos. Quando determinados conjuntos de SNPs ocorrem de maneira consistente em um grupo específico de isolados, formam o que se denomina assinatura genética, isto é, um padrão característico de mutações que identifica uma linhagem, espécie ou população. Assim, sequências idênticas provenientes de diferentes origens correspondem a um mesmo haplótipo, o que confere à análise haplotípica grande valor para inferir relações evolutivas, rotas de dispersão e padrões de transmissão. A construção de redes haplotípicas, por sua vez, oferece uma visualização clara das conexões genéticas entre isolados, permitindo estimar ancestralidade e identificar eventos de diversificação ou expansão clonal, fornecendo bases sólidas para compreender a estrutura populacional e a dinâmica epidemiológica de patógenos (Paradis et al., 2018) (Figura 2).

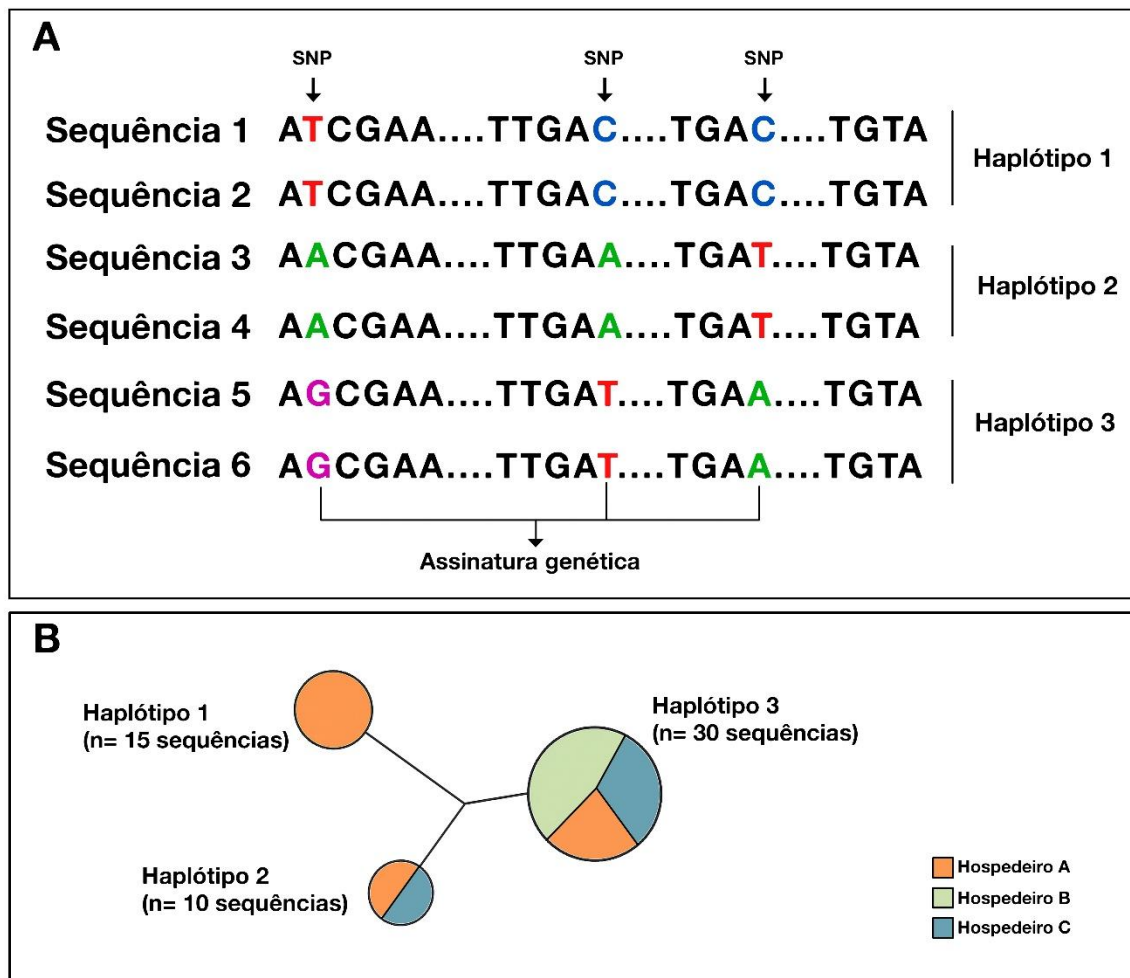


Figura 2. Representação esquemática dos conceitos de SNP, assinatura genética, haplótipo e rede haplotípica. (A) Alinhamento ilustrativo de seis sequências de DNA provenientes de diferentes origens, destacando polimorfismos de nucleotídeo único (SNPs). Combinações específicas de SNPs definem assinaturas genéticas características, que permitem a discriminação entre haplótipos (Haplótipos 1, 2 e 3). (B) Rede haplotípica construída a partir de sequências nucleotídicas, em que cada círculo representa um haplótipo, proporcionalmente ao número de sequências associadas. As cores internas indicam a origem das sequências (hospedeiros A, B e C), evidenciando a diversidade e compartilhamento genético entre diferentes hospedeiros. (Elaborado pela autora).

7. Justificativa

As últimas décadas trouxeram preocupações crescentes quanto à expansão e ao recrudescimento de doenças infecciosas emergentes e reemergentes, responsáveis por importantes epidemias e/ou pandemias. A maioria dessas doenças são zoonoses, cuja emergência está intimamente associada a processos antrópicos, incluindo alterações ambientais, mudanças nos ecossistemas e uso da terra, intensificação da produção animal e agrícola, urbanização e aumento da mobilidade humana e do comércio internacional (McArthur, 2019). Um exemplo recente são as enchentes de grande magnitude ocorridas no estado do Rio Grande do Sul em 2024, que, além das incalculáveis perdas materiais, sociais e econômicas, evidenciaram o risco de surtos de doenças infecciosas relacionadas a desastres naturais, com destaque para a leptospirose (Martins-Filho et al., 2024). Essa triste catástrofe no Sul do Brasil revela de forma contundente a vulnerabilidade das populações humanas diante das mudanças climáticas e sublinha a necessidade urgente de fortalecer a capacidade de vigilância, preparação e resposta a eventos extremos, por meio de estratégias integradas de mitigação e prevenção.

No Brasil, a grande maioria dos estudos epidemiológicos sobre leptospirose ainda se baseia predominantemente em achados sorológicos, limitando-se à detecção indireta da infecção (Nardoni Marteli et al., 2022). Além disso, a maioria das investigações tem enfoque urbano e privilegia variáveis ambientais (Schneider et al., 2015; Santos et al., 2018; Magalhães e Acosta, 2019; Diz; Conceição, 2021; Galan et al., 2021; Costa et al., 2022), negligenciando a interface rural e silvestre da doença, onde se estabelecem importantes ciclos de manutenção. Mesmo os estudos que integraram dados de origem animal o fizeram em escalas espaço-temporais restritas (Chapola et al., 2005; Silva et al., 2009; Jorge et al., 2017; Cerveira et al., 2020), o que limita a compreensão da dinâmica ecológica e evolutiva do agente.

Os progressos recentes e a democratização das tecnologias de sequenciamento de DNA impulsionaram a epidemiologia molecular, tornando suas análises mais rápidas, acessíveis e abrangentes, e viabilizando o uso de grandes volumes de dados genéticos e genômicos provenientes de bases públicas de livre acesso. Atualmente, embora exista um acervo expressivo de sequências de *Leptospira* spp. provenientes do Brasil, nenhum estudo genético

abrangente *in silico* foi conduzido para explorar inferências epidemiológicas e evolutivas integrando todos os hospedeiros, biomas e contextos ecológicos.

A escolha dos hospedeiros investigados (cães, bovinos e pequenos mamíferos silvestres) fundamenta-se em seus papéis complementares na ecoepidemiologia da leptospirose. Cães atuam como importantes sentinelas urbanos e potenciais amplificadores da transmissão em áreas periurbanas, refletindo diretamente a exposição humana (Guzmán et al., 2023). Bovinos representam um dos principais reservatórios de leptospirosas patogênicas no ambiente rural, especialmente na forma crônica genital, contribuindo para perdas reprodutivas e manutenção silenciosa do agente nos rebanhos (Aymée et al., 2024). Pequenos mamíferos, por sua vez, constituem hospedeiros-chave na manutenção natural das espécies de *Leptospira*, funcionando como pontes entre ambientes silvestres, rurais e urbanos (Rosli et al., 2023). Assim, a análise integrada desses três grupos pode oferecer uma visão abrangente e comparativa da diversidade genética, circulação e estrutura populacional das leptospirosas no Brasil.

O presente projeto insere-se no contexto da abordagem Uma Só Saúde, que é especialmente relevante em um país megadiverso como o Brasil, onde a interação entre seres humanos, animais e ambiente favorece a emergência e persistência de patógenos zoonóticos (Silva et al., 2020). Além disso, a abordagem proposta é inovadora e estratégica, pois integra dados moleculares, ambientais e ecológicos para subsidiar ações de vigilância inteligente, contribuindo para o fortalecimento das políticas públicas de saúde e para a consolidação de uma vigilância da leptospirose baseada em evidências e inteligência territorial no Brasil.

Com base nessas premissas, parte-se da hipótese de que leptospirosas patogênicas que circulam no Brasil apresentam uma estrutura genética conectada entre diferentes hospedeiros e ecossistemas, refletindo uma dinâmica de transmissão integrada e sustentada por poucos haplogrupos amplamente distribuídos e adaptados às condições ecológicas locais.

OBJETIVOS

Geral

Investigar a diversidade genética, as relações evolutivas e os padrões de circulação de *Leptospira* spp. patogênicas no Brasil, integrando dados moleculares de diferentes hospedeiros (humanos e animais) e regiões geográficas, com ênfase na avaliação do potencial zoonótico e da estrutura ecoepidemiológica das estirpes circulantes.

Específicos

1. Realizar análises *in silico* de sequências de *Leptospira* spp. identificadas em cães da América Latina, visando caracterizar a diversidade genética, a distribuição geográfica e a estrutura populacional das linhagens associadas à infecção canina.
2. Realizar a detecção molecular de *Leptospira* spp. patogênicas em amostras reprodutivas de bovinos do Sudeste brasileiro e caracterizar geneticamente as amostras positivas, avaliando sua circulação e possíveis associações com a forma crônica da leptospirose bovina.
3. Diagnosticar e caracterizar geneticamente *Leptospira* spp. em pequenos mamíferos da Mata Atlântica, analisando suas relações filogenéticas e haplotípicas para inferir o potencial zoonótico e o papel ecológico desses reservatórios.
4. Integrar e analisar sequências de *Leptospira* spp. patogênicas reportadas no Brasil, provenientes de diferentes genes, hospedeiros e biomas, para caracterizar suas associações ecológicas e padrões ecoepidemiológicos sob a perspectiva Uma Só Saúde.
5. Realizar análises filogenéticas e de diversidade genética de *Leptospira* spp. identificadas em todo território nacional para inferir relações evolutivas intra e interespecíficas, estruturação geográfica e eventos de expansão clonal, visando compreender a dinâmica evolutiva e a distribuição molecular das linhagens circulantes no país.

A presente tese foi escrita no formato compacto, portanto, seguindo as instruções do PPGMPA/UFF, as seções “Material e métodos” e “Resultados” foram substituídas pelos artigos publicados. O resumo dos trabalhos é apresentado em português antes de cada artigo.

ARTIGO 1: Epidemiologia molecular de *Leptospira* spp. patogênicas infectando cães na América Latina

Revista: *Animals* (IF= 2.7)

A leptospirose canina é uma doença bacteriana causada por espiroquetas do gênero *Leptospira*. As infecções podem variar desde infecções assintomáticas e crônicas a doenças clínicas agudas. A doença é endêmica em áreas tropicais, como países da América Latina, mas uma ampla compreensão da dinâmica de circulação de estirpes, baseada em dados moleculares, ainda não foi realizada. Com base em análises *in silico*, o presente estudo teve como objetivo analisar a diversidade genética e os padrões de circulação de haplótipos de leptospiros patogênicas que infectam cães na América Latina. Sequências de DNA foram obtidas da plataforma GenBank, curadas e alinhadas. Distâncias genéticas foram calculadas e uma árvore de máxima verossimilhança e rede de haplótipos foram construídas. De acordo com os critérios de inclusão adotados, um total de 148 sequências foram identificadas. A maioria dos registros era do Brasil, em sua maioria sequências de *L. interrogans* sorogrupo Icterohaemorrhagiae. A análise filogenética mostrou um *cluster* geneticamente relacionado, constituído por um haplótipo maior que inclui a estirpe de referência Fiocruz L1-130, conhecida por ser a principal estirpe circulante em humanos. Além disso, não foram observadas variações genéticas de acordo com o histórico clínico e/ou localização geográfica. Descrevemos a epidemiologia molecular das leptospiros circulantes entre cães na América Latina e demonstramos um grupo geneticamente bastante homogêneo, elucidando seu padrão de circulação ubíquo e chamando a atenção para o importante papel dos cães na dinâmica de transmissão da leptospirose no contexto “Uma só Saúde



Article

Molecular Epidemiology of Pathogenic *Leptospira* spp. Infecting Dogs in Latin America

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Simple Summary: Leptospirosis is a neglected zoonosis caused by a bacterium of the genus *Leptospira*. Dogs are highly susceptible to infection, which can range from chronic and asymptomatic to acute and severe disease. Although serosurveys of canine leptospirosis in Latin America are widely published, the gathering of molecular data of strains identified from dogs has not yet been performed. Based on *Leptospira* spp. DNA sequences collected on digital platforms, we aimed to genetically analyze the strains circulating in Latin America in order to provide an unprecedented overview of the molecular epidemiology of pathogenic leptospires infecting dogs in the region. We found a very high genetic similarity between strains apart serogroup, clinical signs, or geographical localization. The strains were similar to the one that most circulates in humans. In addition to the importance from a zoonotic point of view, this high genetic similarity between strains can, in theory, facilitate the development of an effective broad-spectrum vaccine across the continent, as well as enable the standardization of rapid diagnostic tests.

Abstract: Canine leptospirosis is a bacterial disease caused by spirochetes of the genus *Leptospira*. Infections can vary from asymptomatic and chronic infections to clinical acute diseases. The disease is endemic in tropical areas, such as Latin American countries, but a broad understanding of the dynamics of circulation of strains, based on molecular data, has not yet been performed. Based on in silico analyses, the present study aims to analyze the genetic diversity and circulation patterns of haplotypes from pathogenic leptospires infecting dogs in Latin America. DNA sequences were obtained from GenBank platform, curated, and aligned. Genetic distances were calculated, and a maximum likelihood tree and haplotype network were constructed. According to the inclusion criteria adopted, a total of 148 sequences were identified. Most of the records were from Brazil, including sequences from *L. interrogans* serogroup Icterohaemorrhagiae. Phylogenetic analysis showed a genetically closely related cluster, consisting of a larger haplogroup that includes the reference strain Fiocruz L1-130, known to be the major circulating strain in humans. Moreover, no genetic variations were observed according to clinical history and/or geographical localization. We described the molecular epidemiology of leptospires circulating among dogs in Latin America and demonstrated a very genetically homogeneous group, elucidating its ubiquitous circulation pattern and drawing attention to the important role of dogs in the One Health transmission dynamics of leptospirosis.

Keywords: canine leptospirosis; bioinformatics; *secY*; 16S rRNA; one health



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1. Introduction

Leptospirosis is a neglected zoonosis caused by bacteria of the genus *Leptospira*, and affects domestic and wild animals, as well as humans [1]. Leptospire have a thin and spiral morphology, presenting 6 to 20 µm length and two periplasmic flagella, producing

their unique motility [2,3]. The presence of lipopolysaccharides in their outer membrane classifies the genus as Gram-negative [2,4]. The leptospires can be classified either by their serological or by genetic features. Serological classification is based on the homogeneity of LPS antigen between serovars of a serogroup [5]. Regarding genetic classification, the leptospiral species are currently divided into approximately 68 genomospecies separated in pathogenic (subclade P1), intermediate (subclade P2), and saprophytic (subclades S1 and S2) [6,7]. Based on both molecular and serological characterization, 17 pathogenic species and approximately 300 serovars grouped in 24 serogroups, have been identified [6,8].

The life cycle of leptospires is complex, including the natural environment, asymptomatic reservoirs, and susceptible hosts [9]. Due to the great variety of strains, leptospires can infect a wide range of animal species and lead to a potentially lethal disease [10,11]. Regarding human infection, it infects more than 1 million people and is responsible for about 60,000 deaths per year worldwide [1]. Human leptospirosis has diverse clinical manifestations, ranging from a mild, self-limiting acute febrile illness to a severe, life-threatening condition with multiple organ dysfunction [12]. In urban areas, the transmission is directly associated to the proximity with rodent populations, especially the brown rat (*Rattus norvegicus*), the most important source of human infections [12]. The characteristics of urban poor communities (e.g., open sewers, accumulated trash, and dirt floors) create a habitat ideal for rats, which leads to high infestation rates and frequent contact with residents [13].

Transmission occurs by direct contact with infected animals, or with water or soil contaminated with leptospires (shed in the urine of infected animals) via penetration through cuts or abrasions in the skin and/or external mucous membranes such as oral, conjunctiva, respiratory, and genitourinary [14]. Although leptospires do not replicate outside the host and are easily inactivated under adverse conditions, pathogenic serovars can remain viable in water or soil for weeks [15].

Due to the great variety of strains, leptospires can infect a wide range of animal species and lead to a potentially lethal disease [11,12]. Historically, the etiology of leptospirosis infections is divided into those determined by adapted or incidental strains. Adapted strains are harbored by the usual animal host, with a long process of pathogen-host coevolution, leading to asymptomatic or inapparent disease, or mild and subclinical symptoms of a chronic nature. Maintenance hosts indicate the infected animals which are asymptotically infected and shed viable leptospires in their fluids, acting simultaneously as hosts and reservoirs [16]. Those hosts play an important role in the epidemiology of leptospirosis, as they harbor and excrete leptospires in the environment for long periods, becoming important sources of infection of leptospires of virulent strains to non-adapted to incidental hosts [17,18]. Examples of adapted serovars are Bratislava in horses, Pomona in pigs, Hardjo in ruminants, and Canicola in dogs.

Dogs have been recognized as hosts of pathogenic leptospires since 1931 when Klarenbeek and Schuffner first isolated the bacterium from the urine of a dog affected by nephritis [19]. Currently, canine leptospirosis has been reported worldwide [18,20]. Dogs are highly susceptible to infection by leptospires, which can be maintained in the kidneys and shed through urine and other body fluids, elucidating its important role as source of infection for human infection [21]. Clinical manifestations in dogs range from chronic and asymptomatic infection to acute and severe disease [10,22]. Renal involvement ranges from acute to chronic kidney injury and can be characterized by polyuria, with or without azotemia, or even oliguria and anuria [18,23]. Regarding the hepatic manifestations, leptospirosis can cause mild enzyme alterations to severe failure, with hepatic encephalopathy and jaundice [24,25]. Fever and pain are associated, and can be caused by myositis, meningitis, or other inflammations [15,18]. Cardiac and respiratory manifestations can also occur, such as pulmonary hemorrhage [26]. The incubation is approximately seven days in experimental studies, varying according to the host immune response, virulence, and inoculum [27,28].

Infection by *L. interrogans* serogroup Icterohaemorrhagiae has been assigned to be the major cause of the acute signs, while adapted strains of *L. interrogans* serogroup Canicola are more frequently associated with mild disease [21,22,29]. Nevertheless, not only the virulence of the strain determines the severity of the infection. Recent evidence demonstrated that the same strain can provoke clinically severe features or induce an asymptomatic infection in different dogs [30]. This study analyzed genetic diversity on the *secY* gene of *L. interrogans* serogroup Icterohaemorrhagiae strains obtained from 13 dogs, including animals with signs of acute disease, asymptomatic, and presenting chronic kidney disease. There were no differences between the groups' sequences.

Despite the clinical outcomes, infected animals shed leptospires in the environment, promoting the exposure of other mammals [10,31]. Due to the close proximity to humans, infected dogs play an important role in One Health [32], especially in the cases of asymptomatic harboring [22]. The real impact of canine leptospiral infection and its transmission to humans is poorly understood [32,33]. Nonetheless, reports show that leptospirosis is endemic in tropical areas, such as Latin American countries [21]. Serosurveys of canine leptospirosis in those countries are widely published. In Nicaragua, an epidemiological study from 2007 to 2013 revealed the influence of serogroups Canicola, Pyrogenes, and Louisiana in canine leptospirosis [34]. A serosurvey in Colombia found Autumnalis and Canicola as the most prevalent serovars [35]. Still, seroprevalence studies in Chile reported the presence of Canicola, Pomona, and Autumnalis in both urban and rural areas [36,37]. Isolates were recovered from urine ($n = 6$), blood ($n = 1$), and fetus ($n = 1$) in Argentina. The genetic profile of the strains was identical to *L. interrogans* serogroup Canicola, *L. interrogans* serogroup Icterohaemorrhagiae and *L. interrogans* serogroup Pomona [38]. Recently, a review conducted in Brazil revealed the serogroups Canicola, Icterohaemorrhagiae, and Autumnalis as the most prevalent [29].

As abovementioned, most of studies regarding leptospiral infection in dogs are based on serodiagnosis tests. Serology constitutes an indirect method, and its epidemiological value resides in its ability to predict the circulating serogroups. Even though useful for a collective diagnosis, it presents important limitations for chronic animal leptospirosis, and is not adequate for an individual diagnostic, as serological and molecular results are very often discrepant, and many infected animals may present seronegative. In this context, the complementary use of molecular tools has been encouraged [39]. In the last decades, DNA sequencing has become less costly and more streamlined by automating processes. In addition, data are being increasingly deposited on free-access digital platforms, making it possible to perform advanced studies through bioinformatics tools. Currently, despite a representative sequence bank available, the gathering of molecular data of strains identified from dogs has not yet been performed. As a consequence, Caimi et al. [34] were able to perform a study involving almost 500 leptospiral genomes from worldwide, applying R7L-MLST and obtaining important insights in the phylogenetic and epidemiological scenarios. Therefore, based on in silico analyses, the present study aims to analyze the genetic diversity and circulation patterns of haplotypes from pathogenic leptospires infecting dogs in Latin America.

2. Materials and Methods

2.1. Collection of Genetic Data

In January 2023, DNA sequences were obtained from the GenBank platform (NCBI/nucleotide) using the following search terms: "*Leptospira*" AND "dog" OR "*Canis familiaris*" OR "*Canis lupus familiaris*" OR "canine" AND "Argentina" OR "Bolivia" OR "Brazil" OR "Chile" OR "Colombia" OR "Costa Rica" OR "Cuba" OR "Ecuador" OR "El Salvador" OR "Guatemala" OR "Haiti" OR "Honduras" OR "Mexico" OR "Nicaragua" OR "Panama" OR "Paraguay" OR "Peru" OR "Dominican Republic" OR "Uruguay" OR "Venezuela". No date interval filter was applied. Sequences referring to any genetic marker were included, as well as complete genomes.

The sequences were carefully manually reviewed and, once confirmed as being pathogenic leptospires obtained from dogs in Latin America, information on species, serogroup, country of origin, and clinical history was collected. Geographical distribution of the genetic data on Latin America was performed on R-studio (v3.6.1.) using the *rgdal* package.

2.2. Phylogenetic Analyses

Sequences from 16S rRNA and *secY* genes were selected for phylogenetic analysis due to its more representative number of sequences and a well-defined use for leptospiral genetic characterization [39]. Very short sequences were excluded to avoid bias. After the alignment of sequences on ClustalX v 2.0 software [40], a maximum likelihood (ML) tree was constructed using the Tamura-Nei model (TN92) in MEGA X version 10.0.5 software [41], as it was determined to be the best-fitting model of DNA substitution using the Bayesian information criterion. Genetic distances were calculated using the TN92 model on MEGA X.

For better visualization of haplotype distribution, a haplotype network, a widely used approach for analyzing and visualizing the relationships among DNA sequences within a population or species [42], based on *secY* sequences was constructed through the population genetics software PopART [43], using the media-joining inference method [44]. The reference sequence *L. interrogans* L1-130 was included in all analyses.

3. Results

3.1. Description of Genetic Data Obtained

After a careful search in the GenBank platform, a total of 148 sequences were found according to the inclusion criteria adopted (pathogenic *Leptospira* species infecting dogs from Latin America). Summarized information about the sequences included can be found in Table 1 and detailed information is shown in the supplementary material (Table S1).

Table 1. Summarized information about pathogenic *Leptospira* spp. identified through genetic sequencing in dogs from Latin America.

Country	Species	Strain/Sample ID	Clinical History	Serogroup
Argentina (n = 1)	<i>L. interrogans</i> (n = 1)	28	symptomatic	Icterohaemorrhagiae
Brazil (n = 105)	<i>L. interrogans</i> (n = 99)	18R_CRD, 7R, C1/2012, C20/2013, C25/2013, C29/2013, C3/2012, C41/2015, C51/2015, C52/2015, C53/2015, C56/2015, C7/2012, C72/2017, Dog 10, Dog 16, Dog 2, Dog 9, NEG7, RCA	symptomatic	Icterohaemorrhagiae
		24AP, 34AP, 67AP, SG10, SG13, SG27, SG28, SG29, SG3, SG32, SG33, SG34, SG37, SG54, SG55, SG6, SG69, SG7, SG70, SG73, SG75, SG88, SG98	asymptomatic	
		M10/99, M64/06, UFFB15, UFFG001, UFFG19	unknown	
		C80/2017, Dog 6	symptomatic	
		DU114	asymptomatic	
	<i>L. interrogans</i> (n = 99)	Dog 25, Dog 26, Dog 8	symptomatic	Australis
		Dog 13, Dog 30, Dog 4, Dog 5	symptomatic	
		01/014, 01/017, 01/037, 01/070, 01/072, 01/077, 01/078, 01/089, 01/095, 07/070, 13R, 14F, 15F, 16F, 17F, 18F, 21F, 22F, DUZO, H05, H07, H08, H09, H10, H18, H20, H24, H30, H33, H41, H43, H44, Kito, M12/20, M24/19, M60/19, USPIA USPIB, USPIC, USPID, USPIIA	unknown	
		SG95, SG97	asymptomatic	
		Hook	unknown	
	<i>L. noguchii</i> (n = 3)	DU92, DUPA	asymptomatic	Sejroe
	<i>L. santarosai</i> (n = 3)	DU01/96	asymptomatic	N/D

Table 1. Cont.

Country	Species	Strain/Sample ID	Clinical History	Serogroup
Colombia (n = 17)	<i>L. interrogans</i> (n = 17)	CfCASol-110-Or, CfCASol-112-Or, CfCASol-135-Or, CfCASol-37-Or, CfCASol-40-As, CfCASol-53-As, CfCASol-59-Or, CfCASol-61-A, CfCASol-72-As, RCA	symptomatic	N/D
		CfCASol-23-Or, CfCASol-34-O, CfCASol-36-Or, Dog 133, Dog 146, Dog 167, Dog 196	unknown	N/D
Ecuador (n = 1)	<i>L. santarosai</i>	Calderón-1	unknown	N/D
Mexico (n = 1)	<i>L. interrogans</i>	LOCaS46	asymptomatic	Canicola

N/D: Not defined.

The records included sequences from the genes 16S rRNA ($n = 80/54\%$), *secY* ($n = 52/35.1\%$), *lipL32* ($n = 10/6.8\%$), and *gsp* ($n = 1/0.6\%$) and five whole genomes (3.4%). From these, the majority was from Brazil ($n = 121/81.8\%$), followed by Colombia ($n = 24/16.2\%$), with few reports in Ecuador, Argentina, and Mexico (only one in each, totalizing 3% of the records) (Figure 1).

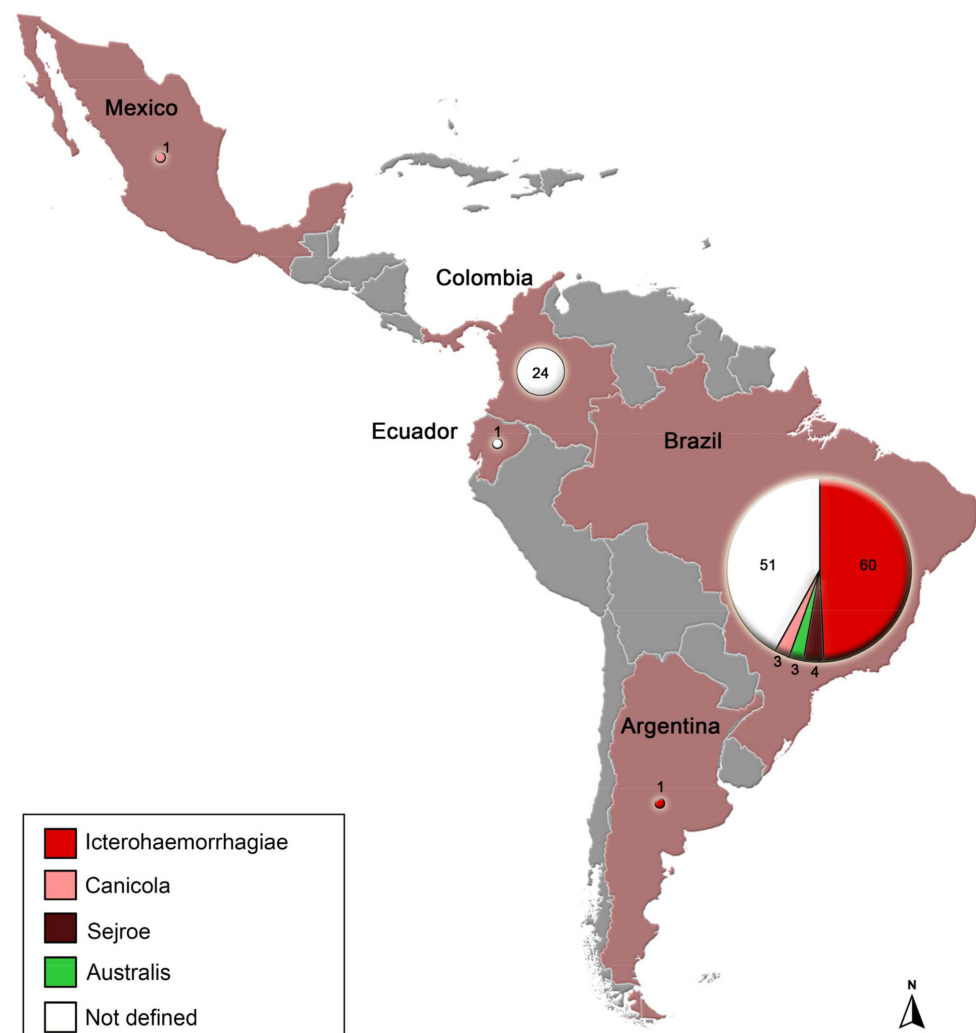


Figure 1. Distribution of pathogenic *Leptospira* spp. From dogs identified by genetic markers according to serogroup in Latin America. Numbers inside circles refer to absolute frequency.

Three species were identified, *L. interrogans*, the most prevalent ($n = 138/93.2\%$), with few reports of *L. santarosai* ($n = 7/4.8\%$) and *L. noguchii* ($n = 3/2\%$). Serogroups were identified in only 72 strains, with Icterohaemorrhagiae being the most prevalent

($n = 61/84.7\%$). Other serogroups identified were Australis ($n = 3/4.2\%$), Canicola ($n = 4/5.5\%$), and Sejroe ($n = 4/5.5\%$). Clinical history of dogs was provided in 83 records, with over half identified as symptomatic ($n = 53/63.9\%$) and 30 (36.1%) as asymptomatic.

3.2. Molecular Epidemiology Based on Phylogenetic Analyses

Sequences from 16S rRNA and *secY* genes were aligned and submitted to genetic analysis, with a cutoff point for the alignments of 258 bp and 393 bp, respectively. A very low genetic distance was observed in both genetic markers ($TN92 < 0.01 \pm 0.00$), reflecting a higher than 99% similarity between sequences.

The ML phylogenetic tree based on *secY* sequences ($n = 55$) confirmed the three circulating species (*L. interrogans*, *L. noguchii*, and *L. santarosai*), and demonstrated a very homogeneous subcluster in *L. interrogans*, including the reference strain Fiocruz L1-130. No specific clusters were observed according to serogroup, clinical history, or geographical localization, as clearly illustrated in Figure 2.

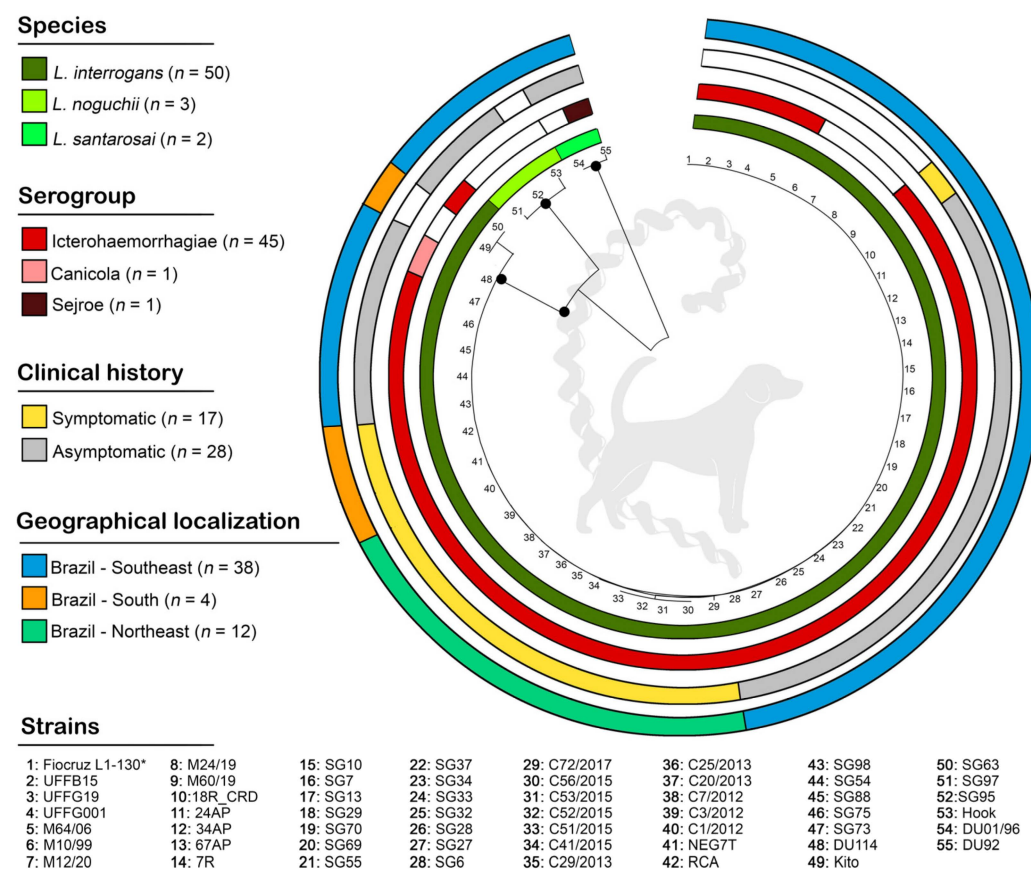


Figure 2. Maximum likelihood phylogenetic tree (1000 bootstrap replicates) based on *secY* gene sequences from Latin American dogs. Bootstrap values are shown with the size of circles in the middle of the branches (bootstraps lower than 50 are not shown). From inner to outer circles, the phylogeny is characterized according to species, serogroups, clinical history, and geographical localization, given by colors (white color indicates that the information was not available). * Reference sequence (host = human).

The haplotype network demonstrated single haplotypes for the species *L. noguchii* ($n = 3$) and *L. santarosai* ($n = 2$), and four haplotypes in *L. interrogans*, including a major comprising 40 sequences, most were identified as Icterohaemorrhagiae, all identical to the strain Fiocruz L1-130 (Figure 3).

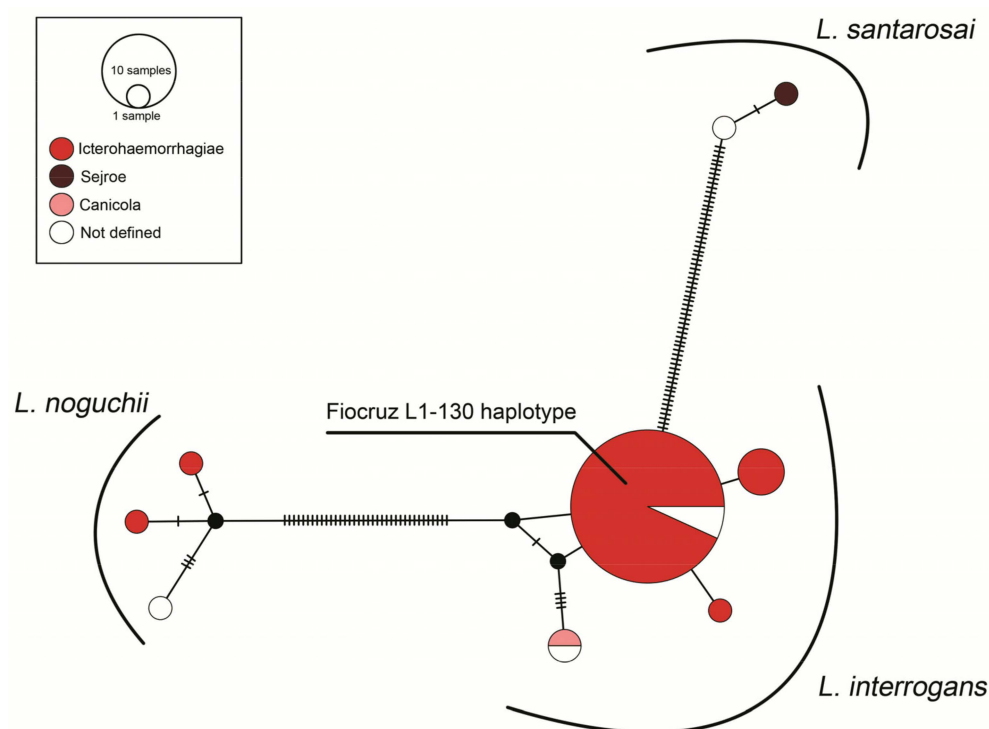


Figure 3. Haplotype network based on *Leptospira* spp. *secY* locus ($n = 56$), distributed according serogroup. Area of the circle is proportional to number of sequences. Hatch marks indicate the number of mutations.

4. Discussion

In the present study, we gathered and genetically analyzed the deposited sequences of *Leptospira* spp. identified in dogs from Latin America. We have searched for sequences of all Latin American countries, most of them belonged to Brazilian studies (81.8%). Nevertheless, this outcome does not reflect that Brazil has a higher occurrence of canine leptospirosis than the others. The high quantity of deposited sequences of Brazilian dogs is most probably a mirror of the research effort in this field. Therefore, the study of canine leptospirosis should be extensively performed in other countries.

The *L. interrogans* prevalence in dogs was drastically higher than other leptospiral species, just as observed in humans and rats [45,46]. *L. noguchii* and *L. santarosai*, although less prevalent than *L. interrogans*, were also found herein. In Brazil, a strain of *L. noguchii* serogroup Australis were recovered from a symptomatic male stray dog kidney tissue [47], and two urine samples of asymptomatic dogs were identified as *L. noguchii* by PCR targeting the gene *secY* [22]. *L. santarosai* serogroup Sejroe was recovered from the urine of an asymptomatic dog [48]. In Colombia, *L. santarosai* was identified by PCR in one unvaccinated dog's urine [49]. Thereby, the presence of pathogenic species of *Leptospira* other than *L. interrogans*, although not common, cannot be neglected, since they may act as emerging pathogens to dogs. In addition, *L. noguchii* and *L. santarosai* have also been reported in other animal hosts in South America, such as ruminants, opossums, weasels, and armadillos [50–52].

In relation to the serogroup distribution, even though only half of the sequences had that information, strains of Icterohaemorrhagiae were by far the major agent of canine infection. In contrast, many serosurveys studies carried out in Latin America frequently reported the presence of strains of Canicola, Autumnalis, and Pomona besides Icterohaemorrhagiae [29,35–38], and their real role in canine leptospirosis remains to be elucidated. On the other hand, the number of sequences recognized as the Canicola serogroup was surprisingly low since dogs are known to be the maintenance host of this serogroup [10]. Although few sequences of Canicola were deposited, this serogroup plays a major role

in serosurveys conducted in South America [49,53,54]. The fact that only 36.1% of the records were identified to be originated from asymptomatic dogs may be responsible for this low rate of Canicola sequences, this number could be higher if more studies were conducted in asymptomatic or chronically infected dogs. Until recently, infections by incidental serogroups, such as Icterohaemorrhagiae, Pomona, and Australis, were linked to severe clinical illness [11,21]. Our outcomes demonstrate that, despite a great number of sequences being recognized as incidental serogroups, almost one-third of the animals were asymptomatic carriers. Those results reinforce previous studies that demonstrated that, besides the classic host-adapted features between dogs and Canicola strains, dogs can also be infected and shed non-adapted strains and remain asymptomatic [22,30,31].

In fact, the high genetic similarity between the sequences of clinically ill and asymptomatic dogs reinforces the hypothesis that the same strain can lead to different clinical outcomes [30]. However, a better understanding of this question with more accurate and definitive answers will only be obtained with advances in genomics, proteomics, and transcriptomics studies, particularly focused on agent–host interactions.

The genetic characterization of the *Leptospira* spp. from dogs is discrete. The absence of a standardized genetic marker for sequencing is an important factor in the gap in the molecular characterization of leptospires. In that context, the use of the *secY* gene as a sequencing target has been encouraged, due to its good taxonomic resolution [39]. Other studies have already performed a wide evaluation of deposited leptospiral *secY* sequences of other animal species and from different countries [55,56], demonstrating the feasibility of this gene as a target in molecular studies. Although *secY* sequencing has been encouraged, in the present study, it represented only 35.1% of the obtained sequences. The target 16S rRNA represented more than half of the sequences; this target was one of the firsts used in *Leptospira* spp. molecular characterization, but its low taxonomic resolution is not useful to differentiate and characterize pathogenic species of the genus [57]. The *lipL32* gene, representing 6.8% of the sequences, is specific to pathogenic species of *Leptospira* and its sequence is very conserved in *Leptospira* spp., that is why its taxonomic resolution is also low [58]. Nevertheless, even with different gene targets being evaluated, the sequences showed high homology (>99%) between them. It is noteworthy that such similarity suggests the presence of a major well-adapted strain circulating in Latin America.

Regarding the haplotype network, the distribution of the clusters was not influenced by the serogroups, clinical aspects, or the geographic origin of the sequences. The larger haplogroup was composed of more than 40 sequences, all genetically similar to the strain Fiocruz L1-130, (*L. interrogans* serogroup Icterohaemorrhagiae) known as the major agent of human leptospirosis in Brazil [30,59–61]. The distribution of those clonal strains of L1-130 by Latin America highlights a ubiquitous circulation pattern and a huge concern to public health.

Concerns about the possibility of infectious disease outbreaks and epidemics have grown during the past few decades [62]. Almost all these illnesses are zoonoses and call for a multidisciplinary approach. These worries led to the development of One Health, a collaborative, multidisciplinary, and cross-sectoral strategy to address the threats from Environment-human relations. WHO defines One Health as “an integrated, unifying approach to balance and optimize the health of people, animals and the environment” [63].

Leptospirosis, as one of the most common zoonotic bacterial infections worldwide, encompasses all aspects of a One Health approach. Its understanding involves in-depth knowledge of the transmission processes, animal reservoir hosts involved, environmental sources of the organism, climate conditions that affect the transmission, and the effects of human occupational activities. Maintaining transmission depends heavily on pathogen characteristics, reservoir host animals, and environmental conditions.

Dogs may act as sentinels of human exposure to leptospirosis [33] and can harbor and shed pathogenic leptospires. Dogs and their owners may share a similar ecological bioaccumulation habitat, and they may serve as sentinels or reservoirs for a variety of zoonotic diseases. Leptospirosis is one of these, and it poses a risk to both humans' and dogs' lives

because it can seriously affect several organ systems and result in death. Particularly in metropolitan areas, the full impact of dogs on the spread of zoonotic illnesses is unknown. In that context, it is known that asymptomatic dogs can persistently shed leptospires in their urine [22,31]. In Latin America, as most countries in the world, licensed vaccines against canine leptospirosis are bivalent, composed of serovars Icterohaemorrhagiae and Canicola [64]. Although vaccination can reduce urinary shedding, even worldwide recommended protocols of immunization do not prevent kidney colonization and vaccinated dogs can act as carriers of leptospires [65]. Similarly, cats infected with *Leptospira* spp. remain asymptomatic with very low antibody titres, even after vaccination, reflecting a short-term adaptative immune response that tolerates the infection and contributes with the maintenance of the bacteria in rural and urban areas [66].

Currently, both human and canine species have close proximity and share the same housing space, which qualifies dogs as an important pillar in the epidemiology of zoonotic diseases. Latin American countries have an alarming dense population of stray or free-roaming dogs [67] that allied to the tropical environment and high endemicity, can play a crucial role in leptospirosis transmission to humans. Still, little is known about the real impact of the closeness of the canines with humans or the high presence of stray dogs with human leptospirosis features. However, the association of asymptomatic shedding and the infection by strains with high zoonotic potential puts canine leptospirosis as a hazard to One Health.

In this context, the findings of the present study, that strains obtained from dogs in Latin America are essentially the same that determine severe disease in humans and circulates in other domestic and wild species deserves attention. Specifically in the case of dogs and humans, both species share the same environment and epidemiological aspects, reinforce the role of the need for a One Health approach in order to reduce both animal and human cases simultaneously. They are interdependent and, particularly in urban settings where they share the same sanitary conditions and are exposed to the same rodent population, it is not possible to consider the study or the understanding and control of leptospirosis without considering a broad approach to it.

5. Conclusions

We described the molecular epidemiology of leptospires circulating among dogs in Latin America and demonstrated a very genetically homogeneous group, with a predominance of a larger haplogroup that includes identical sequences to *L. interrogans* Fio Cruz L1-130, elucidating its ubiquitous circulation pattern and alerting to the important role of dogs in the One Health transmission dynamics of leptospirosis.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/ani13152422/s1>, Table S1: Detailed information about pathogenic *Leptospira* spp. identified through genetic sequencing in dogs from Latin America. Strains are presented in alphabetical order.

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ARTIGO 2: Epidemiologia molecular de *Leptospira* spp. sorogrupo Sejroe associada à leptospirose bovina crônica

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Bactérias patogênicas do gênero *Leptospira* spp. são comumente associadas à leptospirose bovina, caracterizada principalmente por distúrbios reprodutivos crônicos e subclínicos. Estirpes do sorogrupo Sejroe desempenham um papel significativo nessas infecções genitais crônicas conhecidas como Leptospirose Genital Bovina (BGL), que impactam notavelmente a saúde do gado. Este estudo visa aprofundar nossa compreensão da BGL investigando a diversidade genética, a distribuição geográfica e os sítios anatômicos específicos de infecção dos agentes causadores. Inicialmente, fragmentos uterinos e muco cervicovaginal foram coletados de 47 vacas e submetidos à PCR com base no gene *lipL32*. Amostras positivas na PCR-*lipL32* (n=9) foram submetidas à genotipagem com base no gene *secY*. Posteriormente, as sequências foram alinhadas com sequências-referência do *GenBank* (n=108) e analisadas *in silico*. Todas as nove sequências deste estudo foram identificadas como *L. interrogans* com uma identidade >99% com as estirpes de referência do sorogrupo Sejroe (Norma e L53). Na análise mais ampla, a espécie mais prevalente observada foi *L. borgpetersenii*, seguida por *L. interrogans* e *L. santarosai*. A rede de haplótipos de *L. interrogans* revelou que os haplogrupos B e C incluíam exclusivamente estirpes de *L. interrogans* de origem genital, enquanto o haplogrupo A também abrangia estirpes de origem renal. Essas descobertas ressaltam a importância do genótipo Hardjobovis de *L. borgpetersenii* e do genótipo Hardjoprajitno de *L. interrogans* como as estirpes circulantes predominantes e destacam a existência de haplótipos distintos de leptospirosas patogênicas originárias de fontes genitais. Defendemos o uso de *secY* como um marcador genético eficaz para *Leptospira* spp. e enfatizamos a necessidade de pesquisas adicionais priorizando o trato genital. Os resultados deste estudo contribuem para o desenvolvimento de medidas de controle aprimoradas para doenças crônicas do gado e fornecem orientações valiosas para investigações futuras.



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Molecular epidemiology of *Leptospira* spp. serogroup Sejroe associated with chronic bovine leptospirosis

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ABSTRACT

Pathogenic bacteria *Leptospira* spp. are commonly associated with bovine leptospirosis, characterized chiefly by chronic and subclinical reproductive disorders. Strains from the Sejroe serogroup play a significant role in these chronic genital infections known as Bovine Genital Leptospirosis (BGL), which notably impact cattle health. This study aims to deepen our understanding of BGL by investigating the genetic diversity, geographical distribution, and specific anatomical sites of infection of the causative agents. Initially, uterine fragments and cervicovaginal mucus were collected from 47 cows and subjected to PCR targeting the *lipL32* gene. Positive samples in *lipL32*-PCR (9 samples) underwent genotyping based on the *secY* gene. Subsequently, sequences were aligned with GenBank entries (108 sequences) and analyzed *in silico*. All nine sequences from this study were identified as *L. interrogans* with an identity >99 % to serogroup Sejroe reference strains (Norma and L53). In the broader analysis, the most prevalent species observed was *L. borgpetersenii*, followed by *L. interrogans* and *L. santarosai*. The haplotype network of *L. interrogans* revealed that haplogroups B and C exclusively included *L. interrogans* strains of genital origin, while haplogroup A encompassed strains from renal sources as well. These findings underscore the significance of the *L. borgpetersenii* genotype Hardjobovis and *L. interrogans* genotype Hardjoprajitno as the predominant circulating strains and highlight the existence of distinct haplogroups of pathogenic leptospires originating from genital sources. We advocate for the use of *secY* as an effective genetic marker for *Leptospira* spp. and stress the necessity for additional research prioritizing the genital tract. The outcomes of this study contribute to the development of improved control measures for chronic cattle diseases and provide valuable guidance for future investigations.

1. Introduction

Bovine leptospirosis is a global concern with significant economic implications in the cattle reproductive sphere, despite its zoonotic potential (Ellis, 2015). The disease is caused by spirochetes belonging to the genus *Leptospira*, which can be classified using genetic or phenotypic methods. Genetically, the genus is currently divided into approximately 64 species, categorized into subclades: pathogenic (P1), intermediate (P2), and saprophytic (S1 and S2) (Vincent et al., 2019). In contrast, the phenotypic classification relies on serovars distinguished by lipopolysaccharides on the outer membrane and grouped into serogroups based on their antigenic similarities (Levett, 2015). While these two

classifications are independent, their combined use is recommended for comprehensive understanding and effective management of leptospirosis (Di Azevedo and Lilenbaum, 2021).

Regarding pathogenicity, bovine leptospirosis can manifest in two distinct forms, influenced by factors related to both the infecting agent and the adaptability of the host. The first type occurs with infection by incidental strains, leading to acute disease characterized by outbreaks of abortions in cattle. The second type is associated with chronic infection by adapted strains, resulting in subtle reproductive signs (Ellis, 2015). Given that chronic carriers primarily transmit the infection among cows, identifying these carriers is crucial for implementing effective control measures (Valenzuela et al., 2020). However, chronic disease is often

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underdiagnosed due to its subclinical nature and the challenges involved in detecting carriers within the genital tract (Martins and Lilenbaum, 2017).

The primary causative agents of bovine leptospirosis are strains from the Sejroe serogroup, which are noted for their adaptability to cattle. Specifically, key genotypes implicated in bovine leptospirosis include Hardjoprajitno (*L. interrogans*) and Hardjobovis (*L. borgpetersenii*) (Ellis, 2015). Experimental studies conducted over the past decades have demonstrated the interaction of these genotypes with bovine hosts (Bercovich et al., 1990; Leonard et al., 1992; Cortese et al., 2007; Rocha et al., 2020; Wilson-Welder et al., 2021), showing that infection by these strains can induce reproductive disease. Hardjoprajitno has been frequently reported in Europe and Brazil (Ellis, 2015; Cosate et al., 2015; Aymée et al., 2023a), while Hardjobovis has been identified as the predominant agent of bovine leptospirosis in North America (Putz et al., 2022). Another emerging agent within the Sejroe serogroup is Guaricura (*L. santarosai*), increasingly reported in Latin America and also associated with reproductive disease (Aymée et al., 2022a; Chinchilla et al., 2023). However, these conclusions are primarily based on leptospiral isolation and serological studies, with limited genetic characterization. Furthermore, most studies focus on diagnosis, while related epidemiological aspects remain sparse or absent.

Serology serves as an indirect diagnostic method based on the host's production of specific antibodies, providing epidemiological insights into circulating serogroups within specific herds or regions (Aymée et al., 2024). While useful for collective diagnosis, serology has limitations, particularly in identifying subclinical infections characterized by low antibody titers in individuals. In contrast, molecular diagnosis via polymerase chain reaction (PCR) offers a direct approach to detect carriers and circulating genotypes. Discrepancies often arise between serological and molecular results, with many seronegative animals testing positive by PCR, highlighting the need for complementary molecular tools alongside serology (Pinna et al., 2018; Di Azevedo and Lilenbaum, 2021).

Due to the host-adapted interaction between Sejroe serogroup leptospires and bovines, the disease typically manifests chronically, characterized by infertility, birth of weak calves, repeated estrus, and pregnancy losses (Aymée et al., 2024). Leptospires can colonize the kidneys or genital tract of cattle for extended periods and are shed in body fluids such as urine, cervicovaginal mucus, and semen. In the environment, these bacteria infect susceptible animals through mucous membranes (Loureiro and Lilenbaum, 2020).

Historically, kidneys have been the primary focus of *Leptospira* spp. colonization studies (Hernández-Rodríguez et al., 2011; Guedes et al., 2021; Saglam et al., 2022). However, recent research has shed light on genital tract infections by Sejroe strains, which have been identified in the uterus (Aymée et al., 2022b; Di Azevedo et al., 2020), follicular fluid (Di Azevedo et al., 2021; Pereira et al., 2022), cervicovaginal mucus (Loureiro et al., 2016; Pimenta et al., 2019; Gonçalves et al., 2021), and semen (Vinodh et al., 2008). Notably, studies in a hamster model demonstrated that Sejroe leptospires exhibit a preference for the uterus over other immunologically privileged sites like the kidneys (Pinto et al., 2020). Chronic genital infection by leptospires is now recognized as Bovine Genital Leptospirosis (BGL) (Loureiro and Lilenbaum, 2020). Recent investigations in South America have linked *Leptospira* spp. serogroup Sejroe to infertility and subtle reproductive signs (Macchi et al., 2024; Oliveira et al., 2021; Libonati et al., 2018), further emphasizing the significance of these strains in chronic bovine leptospirosis.

Many naturally infected cows have been identified carrying Sejroe strains in their genital tracts, with a high incidence of reproductive failures observed among these animals. Despite the availability of a substantial sequence bank for *Leptospira* spp. identified in cattle worldwide, there has been a notable absence of elucidative and comparative studies based on genetically characterized strains. Therefore, the current study aimed to conduct genetic typing of *Leptospira* spp. strains

belonging to serogroup Sejroe, using sequences obtained from isolates and clinical samples, in order to assess the genetic diversity of these strains based on their geographical origin and anatomical site of infection through *in silico* methods. The overarching goal of the research was to provide crucial epidemiological insights into this silent yet neglected disease.

2. Material and methods

To improve the number of sequences of *Leptospira* spp. from genital samples of bovines with a history of reproductive problems, we first performed field sampling, followed by molecular diagnosis and genetic characterization. The study was approved by the Ethical Committee for Animal Use of Federal Fluminense University under protocol number 9527220222.

2.1. Animal specimens, sample collection, and molecular diagnosis

A total of 47 non-pregnant cows from two herds in Southeastern Brazil with poor reproductive profiles were selected based on their history of reproductive failures. Herd #1 was located in Araruama municipality, Rio de Janeiro State, and comprised Tabapuã and Nelore beef cattle aged between 3 and 10 years. The second herd (Herd #2) was situated in Coronel Pacheco municipality, Minas Gerais State, and consisted of Holstein dairy cows aged between 3 and 5 years. Georeferencing was conducted using a portable GPS.

Cows from both herds exhibited a history of chronic subfertility/infertility, embryonic death, and abortions. Each of the 47 cows (41 from Herd #1 and 6 from Herd #2) underwent sampling of uterine fragments (UF) and cervicovaginal mucus (CVM) following procedures recommended by Aymée and colleagues (2021). All 47 UF and 47 CVM samples were processed for DNA extraction using the DNeasy® Blood & Tissue Kit (Qiagen, California, USA) following the manufacturer's instructions.

For the detection of pathogenic *Leptospira* spp., PCR based on the *lipL32* gene was performed as described by Hamond et al. (2014). Each sample set included ultrapure water as a negative control in all reactions, and 10fg of DNA extracted from *Leptospira interrogans* serovar Copenhageni (Fiocruz L1-130) served as a positive control. The PCR products were analyzed by electrophoresis in a 1.5–2 % agarose gel and visualized under ultraviolet light after staining with GelRed®.

2.2. Genetic characterization of pathogenic *Leptospira* spp.

Samples positive in *lipL32*-PCR were submitted for Single Locus Sequence Typing (SLST) based on the *secY* gene, already well described in the literature as an efficient genetic marker of the *Leptospira* genus (Lam et al., 2020). A nested PCR was conducted, with an initial reaction using the primers *secY*_outerF (5'-ATGCCGATCATTTTGGCTTC-3') and *secY*_outerR (5'-CCGTCCCTTAATTTAGACTTCTTC-3'), followed by a second reaction with the amplicons using the primers *secY*_inner_F (5'-CCTCAGACGATTATTCAATGGTTATC-3') and *secY*_inner_R (5'-AGAAGAGAAGTTCCACCGAATG-3') (Grillová et al., 2020). Gel electrophoresis and amplicons' revelation followed the same procedures described above for *lipL32*-PCR.

The amplicons were purified using the PCR Clean-Up System Kit (Promega, USA), following the manufacturer's instructions, and prepared for sequencing. Sequencing reactions were performed using the Big Dye Terminator v Kit. 3.1 Cycle Sequencing (Applied Biosystems, USA) on a 3100 automatic DNA sequencer according to the manufacturer's instructions. The Basic Local Alignment Search Tool (BLAST - NCBI, <https://blast.ncbi.nlm.nih.gov/Blast>) was used to initially identify *Leptospira* species based on nucleotide similarity.

2.3. *In silico* analysis - phylogenetic reconstruction and genetic diversity evaluation

The sequences obtained from the present study were subjected to an *in silico* analysis for phylogenetic reconstruction and genetic diversity evaluation. To enrich molecular epidemiological inferences, *secY* sequences of *Leptospira* spp. available in GenBank were selected globally. Inclusion criteria encompassed: i) sequences identified as belonging to the Sejroe serogroup (either by serological characterization or high sequence identity); ii) sequences exceeding 400 bp in length; iii) sequences associated with published articles.

SeqMan DNASTAR v. 7.0 (Inc., Madison, WI, USA) and BioEdit v. 7.0.1 (Hall, 1999) were employed for sequence editing and analysis. Sequences were aligned using ClustalX v 2.0 software (Larkin et al., 2007), and a maximum likelihood (ML) tree was constructed in MEGA X software (Kumar et al., 2018). The most appropriate DNA substitution model was determined using the Bayesian information criterion. Genetic distances were computed in MEGA X with the selected model based on the dataset. The saprophytic species *Leptospira biflexa* strain Patoc served as an outgroup.

To assess haplotype distribution according to the anatomical site of infection, a haplotype network was constructed using the median-joining inference method (Bandelt et al., 1999) with the population genetics software PopART (Leigh and Bryant, 2015).

3. Results

3.1. Molecular diagnosis and history of herds with reproductive disorders

Of the 47 animals sampled, 32 (68.0 %) tested positive for *lipL32*-PCR, regardless of the analyzed sample. Of these PCR-positive cows, 10 (31.3 %) were positive only in CVM samples, six (18.7 %) were positive only in UF, and 16 (50.0 %) were positive in both CVM and UF. Of the positive animals, chronic infertility and embryonic mortality were the most commonly reported signs observed in 26/47 (55.3 %) and 23/47 (48.9 %) cows, respectively. Abortions were less frequent and were reported in only 6/47 (12.7 %) animals.

3.2. Genetic characterization of pathogenic *Leptospira* spp.

Based on the *secY* gene, it was possible to sequence nine *Leptospira* spp. amplicons (409 bp) from cervicovaginal mucus (n=2) and uterine fragment (n=7) (Table 1). The sequences obtained in this study have been deposited in GenBank under accession numbers OR611992 to OR612000. After Pairwise/ Blast/NCBI comparisons with the GenBank *secY* gene dataset, all sequences were identified as *L. interrogans* with an identity > 99 %. Furthermore, sequences from the present study exhibited high identity (>99 %) with reference strains from the Sejroe serogroup (Norma and L53).

3.3. *In silico* analysis of *Leptospira* spp. sequences from Sejroe serogroup

3.3.1. Sequences from previous studies obtained from GenBank

In addition to the nine sequences from the present study, 108 sequences from GenBank met the inclusion criteria, resulting in a total of 117 sequences for further analysis (Table 1). The majority of sequences belonged to *L. borgpetersenii* genotype Hardjobovis (50.4 %), followed by *L. interrogans* genotype Hardjoprajitno (41 %) and *L. santarosai* serovar Guaricura (8.6 %). These sequences originated from animals in four different countries across three continents: Brazil (69.2 %), Tanzania (12.8 %), the USA (9.4 %), and Uruguay (8.5 %). Regarding the site of collection, the majority of genetically characterized *Leptospira* spp. were of renal origin (n=66, 56.4 %), including 24 from kidneys and 42 from urines, while 43.6 % were from genital sites (n=51), comprising 30 uterine fragments (UF), 10 cervicovaginal mucus (CVM), and 11 follicular fluids. From the obtained sequences, clinical history information

was available for 43.5 % (51/117). Among these cases, chronic subfertility (64.7 %), abortion (33.3 %), and embryonic mortality (41.2 %) were commonly reported. In the remaining cases, cows were described as asymptomatic (21.6 %). Table 2 summarizes all these results.

3.3.2. Phylogenetic reconstruction

The ML phylogenetic tree was constructed using the Tamura 3-parameter (TN92) model, which was identified as the most suitable DNA substitution model. Together with reference sequences, the ML-TN92 tree confirmed the species identification of sequences from the present study (CVM56, CVM62, UT63, UT64, UT67, UT72, UT83, UT90, UT100). These sequences formed a maximum-supported clade (bootstrap = 100 %) with *L. interrogans* sequences, distinct from other common species found in bovines (Fig. 1). Furthermore, sequences from the present study clustered together in a large haplogroup predominantly comprising sequences from the same geographical origin in south and southeast Brazil (Group B in Fig. 1; Table 1). Additionally, two other major haplogroups, A and C, were identified within *L. interrogans*, encompassing 9 and 13 sequences, respectively.

Based on the inclusion criteria, *L. interrogans* and *L. santarosai* were exclusively identified in bovines from Brazil, while *L. borgpetersenii* was also found in the USA, Tanzania, and Uruguay (Fig. 1). Notably, despite originating from different geographic locations, a larger haplogroup consisting of 52 sequences was observed in *L. borgpetersenii*, whereas a smaller haplogroup was exclusively composed of strains from Tanzania (n=9). Moreover, the majority of *L. borgpetersenii* sequences from Brazil originated from the Amazon region (North) (Table 1).

In terms of intraspecific genetic distances (TN92±SD), *L. interrogans* and *L. borgpetersenii* exhibited low diversity with 0.01±0.00 (99 % identity between sequences), whereas *L. santarosai* showed higher intraspecific diversity with a distance of 0.03±0.01 (97 % identity). This diversity in *L. santarosai* resulted in the formation of small haplogroups, each comprising a maximum of three strains (Fig. 1).

Regarding the geographic distribution of Sejroe strains, high intra- and interspecific diversity was observed in Brazil. In contrast, homogeneous populations were noted in other countries such as the USA and Tanzania. Additionally, an exclusive haplogroup was identified in *L. borgpetersenii* from Tanzania (Fig. 1).

3.3.3. Haplotype network of *L. interrogans*

To analyze the distribution of haplotypes based on the site of infection (renal vs. genital), a haplotype network was constructed using *L. interrogans secY* sequences, which included a significant number of genital samples (n=44) in addition to renal samples (n=4). Three main haplogroups (A, B, and C) were identified.

Notably, haplogroups B and C exclusively comprised *L. interrogans* strains originating from genital sources, including cervicovaginal mucus (CVM) and uterine fragments (UF). In contrast, haplogroup A included strains isolated from renal sources and follicular fluid. Specifically, haplogroup B contained sequences from both CVM and UF, whereas haplogroup C consisted solely of sequences from UF (Fig. 2).

4. Discussion

The present study focused on gathering and genetically analyzing leptospiral Sejroe strains from cows to enhance understanding of chronic bovine leptospirosis. In addition to sequences from previous studies, our research contributed nine new sequences from genital samples, enriching the phylogenetic analysis and yielding notable findings.

We observed a high prevalence of PCR-positive animals among the herds studied. Most of the 47 non-pregnant cows exhibited subclinical symptoms such as chronic infertility and embryonic death, with a smaller number experiencing abortions. This aligns with previous reports linking bovine leptospirosis to embryonic death leading to estrus repetitions and abortions in cattle herds (Libonati et al., 2018; Oliveira

Table 1Information regarding *Leptospira* spp. Sejroe serogroup *secY* sequences included in the present study.

Acession number	Strain/Sample ID	Biological sample	<i>Leptospira</i> species	Clinical history	Country	Reference
OR611992	CVM56	CVM	<i>L. interrogans</i>	EM, CS	Brazil - SE	Present study
OR611993	CVM62	CVM	<i>L. interrogans</i>	EM, CS	Brazil - SE	Present study
OR611994	UT63	Uterine Fragment	<i>L. interrogans</i>	EM, CS	Brazil - SE	Present study
OR611995	UT64	Uterine Fragment	<i>L. interrogans</i>	EM, CS	Brazil - SE	Present study
OR611996	UT67	Uterine Fragment	<i>L. interrogans</i>	EM, CS	Brazil - SE	Present study
OR611997	UT72	Uterine Fragment	<i>L. interrogans</i>	EM, CS	Brazil - SE	Present study
OR611998	UT83	Uterine Fragment	<i>L. interrogans</i>	EM, CS	Brazil - SE	Present study
OR611999	UT90	Uterine Fragment	<i>L. interrogans</i>	EM, CS	Brazil - SE	Present study
OR612000	UT100	Uterine Fragment	<i>L. interrogans</i>	EM, CS	Brazil - SE	Present study
OP244772	UT47	Uterine Fragment	<i>L. borgpetersenii</i>	CS, A	Brazil - SE	Aymée et al., 2022b
CP015052	NVSL S 818	Kidney	<i>L. borgpetersenii</i>	Asymptomatic	USA	Llanes et al., 2018
CP015050	NVSL S 1343	Kidney	<i>L. borgpetersenii</i>	Asymptomatic	USA	Llanes et al., 2018
CP084036	TC112	Urine	<i>L. borgpetersenii</i>	NP	USA	Putz et al., 2022
CP084038	TC129	Urine	<i>L. borgpetersenii</i>	NP	USA	Putz et al., 2022
CP084040	TC147	Urine	<i>L. borgpetersenii</i>	NP	USA	Putz et al., 2022
CP084042	TC273	Urine	<i>L. borgpetersenii</i>	NP	USA	Putz et al., 2022
CP021412	203	Kidney	<i>L. borgpetersenii</i>	NP	USA	Alt et al., 2017
CP015045	BK-6	Kidney	<i>L. borgpetersenii</i>	Asymptomatic	USA	Llanes et al., 2018
CP015046	BK-9	Kidney	<i>L. borgpetersenii</i>	Asymptomatic	USA	Llanes et al., 2018
CP015048	BK-30	Kidney	<i>L. borgpetersenii</i>	Asymptomatic	USA	Llanes et al., 2018
CP000350	JB197	Kidney	<i>L. borgpetersenii</i>	NP	USA	Bulach et al., 2006
CP033440	L49	Urine	<i>L. borgpetersenii</i>	A, CS	Brazil - S	Chideroli et al., 2016
MH325388	IP1506001	Urine	<i>L. borgpetersenii</i>	A, CS	Uruguay	Zarantonelli et al., 2018
MH325390	IP1509005	Urine	<i>L. borgpetersenii</i>	A, CS	Uruguay	Zarantonelli et al., 2018
MH325391	IP1509006	Urine	<i>L. borgpetersenii</i>	A, CS	Uruguay	Zarantonelli et al., 2018
MH325396	IP1512013	Urine	<i>L. borgpetersenii</i>	A, CS	Uruguay	Zarantonelli et al., 2018
MH325402	IP1605020	Urine	<i>L. borgpetersenii</i>	A, CS	Uruguay	Zarantonelli et al., 2018
MH325412	IP1704030	Urine	<i>L. borgpetersenii</i>	A, CS	Uruguay	Zarantonelli et al., 2018
MH325413	IP1704031	Urine	<i>L. borgpetersenii</i>	A, CS	Uruguay	Zarantonelli et al., 2018
MH325415	IP1708034	Urine	<i>L. borgpetersenii</i>	A, CS	Uruguay	Zarantonelli et al., 2018
MH325417	IP1708036	Kidney	<i>L. borgpetersenii</i>	A, CS	Uruguay	Zarantonelli et al., 2018
MH325419	IP1709038	Kidney	<i>L. borgpetersenii</i>	A, CS	Uruguay	Zarantonelli et al., 2018
MF955882	C0101	Kidney	<i>L. borgpetersenii</i>	NP	Tanzania	Allan et al., 2018
MF955881	C0097	Kidney	<i>L. borgpetersenii</i>	NP	Tanzania	Allan et al., 2018
MF955880	C0658	Kidney	<i>L. borgpetersenii</i>	NP	Tanzania	Allan et al., 2018
MF955879	C0592	Kidney	<i>L. borgpetersenii</i>	NP	Tanzania	Allan et al., 2018
MF955878	C0561	Kidney	<i>L. borgpetersenii</i>	NP	Tanzania	Allan et al., 2018
MF955877	C0518	Kidney	<i>L. borgpetersenii</i>	NP	Tanzania	Allan et al., 2018
MF955874	C0279	Kidney	<i>L. borgpetersenii</i>	NP	Tanzania	Allan et al., 2018
MF955873	C0273	Kidney	<i>L. borgpetersenii</i>	NP	Tanzania	Allan et al., 2018
MF955869	C0201	Kidney	<i>L. borgpetersenii</i>	NP	Tanzania	Allan et al., 2018
MF955868	C0131	Kidney	<i>L. borgpetersenii</i>	NP	Tanzania	Allan et al., 2018
MF955867	C0118	Kidney	<i>L. borgpetersenii</i>	NP	Tanzania	Allan et al., 2018
MF955866	C0101	Kidney	<i>L. borgpetersenii</i>	NP	Tanzania	Allan et al., 2018
MF955865	C0084	Kidney	<i>L. borgpetersenii</i>	NP	Tanzania	Allan et al., 2018
MF955863	C0051	Kidney	<i>L. borgpetersenii</i>	NP	Tanzania	Allan et al., 2018
MF955862	C0025	Kidney	<i>L. borgpetersenii</i>	NP	Tanzania	Allan et al., 2018
MW714382	M13/20-181	Urine	<i>L. borgpetersenii</i>	NP	Brazil - N	Guedes et al., 2021
MT708164	M02/20-172	Urine	<i>L. borgpetersenii</i>	NP	Brazil - N	Guedes et al., 2020
MT708163	M02/20-166	Urine	<i>L. borgpetersenii</i>	NP	Brazil - N	Guedes et al., 2020
MT708162	M02/20-162	Urine	<i>L. borgpetersenii</i>	NP	Brazil - N	Guedes et al., 2020
MT708161	M02/20-152	Urine	<i>L. borgpetersenii</i>	NP	Brazil - N	Guedes et al., 2020
MT708160	M02/20-143	Urine	<i>L. borgpetersenii</i>	NP	Brazil - N	Guedes et al., 2020
MT708159	M02/20-121	Urine	<i>L. borgpetersenii</i>	NP	Brazil - N	Guedes et al., 2020
MT708158	M02/20-118	Urine	<i>L. borgpetersenii</i>	NP	Brazil - N	Guedes et al., 2020
MT708157	M02/20-116	Urine	<i>L. borgpetersenii</i>	NP	Brazil - N	Guedes et al., 2020
MT708156	M02/20-96	Urine	<i>L. borgpetersenii</i>	NP	Brazil - N	Guedes et al., 2020
MT708155	M02/20-77	Urine	<i>L. borgpetersenii</i>	NP	Brazil - N	Guedes et al., 2020
MT708154	M02/20-68	Urine	<i>L. borgpetersenii</i>	NP	Brazil - N	Guedes et al., 2020
MT708153	M02/20-67	Urine	<i>L. borgpetersenii</i>	NP	Brazil - N	Guedes et al., 2020
MT708152	M02/20-55	Urine	<i>L. borgpetersenii</i>	NP	Brazil - N	Guedes et al., 2020
MT708151	M02/20-53	Urine	<i>L. borgpetersenii</i>	NP	Brazil - N	Guedes et al., 2020
MT708150	M02/20-52	Urine	<i>L. borgpetersenii</i>	NP	Brazil - N	Guedes et al., 2020
MT708149	M02/20-50	Urine	<i>L. borgpetersenii</i>	NP	Brazil - N	Guedes et al., 2020
MT708148	M02/20-44	Urine	<i>L. borgpetersenii</i>	NP	Brazil - N	Guedes et al., 2020
MT708147	M02/20-23	Urine	<i>L. borgpetersenii</i>	NP	Brazil - N	Guedes et al., 2020
MT708146	M02/20-09	Urine	<i>L. borgpetersenii</i>	NP	Brazil - N	Guedes et al., 2020
MT708145	M02/20-06	Urine	<i>L. borgpetersenii</i>	NP	Brazil - N	Guedes et al., 2020
MF187958	BUT36	Uterine Fragment	<i>L. interrogans</i>	NP	Brazil - SE	Pires et al., 2018
MF187959	BUT46	Uterine Fragment	<i>L. interrogans</i>	NP	Brazil - SE	Pires et al., 2018
MT270420	UF4	Uterine Fragment	<i>L. interrogans</i>	NP	Brazil - SE	Di Azevedo et al., 2020
MT270421	UF5	Uterine Fragment	<i>L. interrogans</i>	NP	Brazil - SE	Di Azevedo et al., 2020
MT270422	UF7	Uterine Fragment	<i>L. interrogans</i>	NP	Brazil - SE	Di Azevedo et al., 2020
MT270423	UF9	Uterine Fragment	<i>L. interrogans</i>	NP	Brazil - SE	Di Azevedo et al., 2020

(continued on next page)

Table 1 (continued)

Accession number	Strain/Sample ID	Biological sample	<i>Leptospira</i> species	Clinical history	Country	Reference
MT270424	UF16	Uterine Fragment	<i>L. interrogans</i>	NP	Brazil - SE	Di Azevedo et al., 2020
MT270425	UF19	Uterine Fragment	<i>L. interrogans</i>	NP	Brazil - SE	Di Azevedo et al., 2020
MT270426	UF20	Uterine Fragment	<i>L. interrogans</i>	NP	Brazil - SE	Di Azevedo et al., 2020
MT270427	UF21	Uterine Fragment	<i>L. interrogans</i>	NP	Brazil - SE	Di Azevedo et al., 2020
MT270428	UF24	Uterine Fragment	<i>L. interrogans</i>	NP	Brazil - SE	Di Azevedo et al., 2020
MZ099639	CVM1	CVM	<i>L. interrogans</i>	CS	Brazil - SE	Aymée et al., 2021
MZ099642	UF4	Uterine Fragment	<i>L. interrogans</i>	A, EM	Brazil - SE	Aymée et al., 2021
MT270418	FF13	Follicular fluid	<i>L. interrogans</i>	NP	Brazil - SE	Di Azevedo et al., 2021
MZ773519	Cow037_FF	Follicular fluid	<i>L. interrogans</i>	NP	Brazil - SE	Pereira et al., 2022
MZ773520	Cow002_FF	Follicular fluid	<i>L. interrogans</i>	NP	Brazil - SE	Pereira et al., 2022
MZ773521	Cow128_FF	Follicular fluid	<i>L. interrogans</i>	NP	Brazil - SE	Pereira et al., 2022
MZ773522	Cow095_FF	Follicular fluid	<i>L. interrogans</i>	NP	Brazil - SE	Pereira et al., 2022
MZ773523	Cow038_FF	Follicular fluid	<i>L. interrogans</i>	NP	Brazil - SE	Pereira et al., 2022
MZ773524	Cow183_FF	Follicular fluid	<i>L. interrogans</i>	NP	Brazil - SE	Pereira et al., 2022
MZ773525	Cow185_FF	Follicular fluid	<i>L. interrogans</i>	NP	Brazil - SE	Pereira et al., 2022
MZ773526	Cow109_FF	Follicular fluid	<i>L. interrogans</i>	NP	Brazil - SE	Pereira et al., 2022
MZ773527	Cow114_FF	Follicular fluid	<i>L. interrogans</i>	NP	Brazil - SE	Pereira et al., 2022
OP244760	CVM15	CVM	<i>L. interrogans</i>	EM	Brazil - SE	Aymée et al., 2022b
OP244761	CVM12	CVM	<i>L. interrogans</i>	EM	Brazil - SE	Aymée et al., 2022b
OP244764	UT44	Uterine Fragment	<i>L. interrogans</i>	EM	Brazil - SE	Aymée et al., 2022b
OP244767	UT30	Uterine Fragment	<i>L. interrogans</i>	CS	Brazil - SE	Aymée et al., 2022b
OP244768	UT41	Uterine Fragment	<i>L. interrogans</i>	EM, CS	Brazil - SE	Aymée et al., 2022b
OP244769	UT46	Uterine Fragment	<i>L. interrogans</i>	EM, CS	Brazil - SE	Aymée et al., 2022b
OP244770	UT48	Uterine Fragment	<i>L. interrogans</i>	EM, CS	Brazil - SE	Aymée et al., 2022b
OR338746	UT85	Uterine Fragment	<i>L. interrogans</i>	A, EM	Brazil - SE	Aymée et al., 2023a
OR338747	UT86	Uterine Fragment	<i>L. interrogans</i>	A, EM	Brazil - SE	Aymée et al., 2023a
OR338748	UT87	Uterine Fragment	<i>L. interrogans</i>	EM, CS	Brazil - SE	Aymée et al., 2023a
OR338749	UT89	Uterine Fragment	<i>L. interrogans</i>	EM, CS	Brazil - SE	Aymée et al., 2023a
OR338750	CVM88	CVM	<i>L. interrogans</i>	EM, CS	Brazil - SE	Aymée et al., 2023a
CP012603	norma	Urine	<i>L. interrogans</i>	A	Brazil - SE	Cosate et al., 2015
KY305485	L53	Urine	<i>L. interrogans</i>	A, CS	Brazil - S	Chideroli et al., 2017
MH231559	2015_U349	Urine	<i>L. interrogans</i>	Asymptomatic	Brazil - SE	Pinto et al., 2017
MH231560	2015_U376	Urine	<i>L. interrogans</i>	Asymptomatic	Brazil - SE	Pinto et al., 2017
KP645174	2013_VF52	Vaginal fluid	<i>L. santarosai</i>	NP	Brazil - SE	Loureiro et al., 2016
KP645178	2014_VF66	Vaginal fluid	<i>L. santarosai</i>	NP	Brazil - SE	Loureiro et al., 2016
MT270419	FF18	Follicular fluid	<i>L. santarosai</i>	NP	Brazil - SE	Di Azevedo et al., 2021
OK500434	2015_VF237	Vaginal fluid	<i>L. santarosai</i>	CS	Brazil - SE	Loureiro et al., 2016
OK500433	UT27	Vaginal fluid	<i>L. santarosai</i>	CS	Brazil - SE	Aymée et al., 2022a
OP244771	UT18	Uterine Fragment	<i>L. santarosai</i>	CS	Brazil - SE	Aymée et al., 2022b
KP645175	2014_U76	Urine	<i>L. santarosai</i>	Asymptomatic	Brazil - SE	Loureiro et al., 2016
KP645177	2014_U140	Urine	<i>L. santarosai</i>	Asymptomatic	Brazil - SE	Loureiro et al., 2016
KP263065	U214/2013	Urine	<i>L. santarosai</i>	Asymptomatic	Brazil - SE	Loureiro et al., 2016
KP645176	2014_U81	Urine	<i>L. santarosai</i>	Asymptomatic	Brazil - SE	Loureiro et al., 2016

ID: Identification; CVM: Cervicovaginal mucus; CS: Chronic subfertility; EM: Embryonic mortality; A: Abortion; NP: not provided; ND: not defined; SE: Southeast; N: North; S: South

Table 2

Summarized information regarding *Leptospira* spp. Sejroe serogroup genetically characterized causing chronic bovine leptospirosis.

Species/Serovar	
<i>L. borgpetersenii</i> sv. Hardjo	59/117 (50.4 %)
<i>L. interrogans</i> sv. Hardjo	48/117 (41 %)
<i>L. santarosai</i> sv. Guaricura	10/117 (8.6 %)
Site of investigation	
Renal	66/117 (56.4 %)
Genital	51/117 (43.6 %)
Clinical history*	
Abortion	17/51 (33.3 %)
Chronic subfertility	33/51 (64.7 %)
Embryonic mortality	21/51 (41.2 %)
Asymptomatic	11/51 (21.6 %)
Geographical origin	
Brazil	81/117 (69.2 %)
Tanzania	15/117 (12.8 %)
Uruguay	10/117 (8.5 %)
USA	11/117 (9.4 %)

* When provided

et al., 2021; Aymée et al., 2023b). The frequent occurrence of chronic subfertility/infertility and embryonic deaths underscores the silent nature of leptospirosis in cattle, complicating diagnosis and contributing to its underestimation. Based on these findings, we emphasize the

importance of a diagnostic flowchart proposed for identifying the silent chronic form of animal leptospirosis. This approach involves initial serological screening of herds followed by individual molecular testing using *lipL32*-PCR on genital samples, complemented by *secY* gene sequencing, which provides reliable taxonomic resolution (Di Azevedo and Lilenbaum, 2021).

In this study, we successfully amplified and sequenced nine samples based on the *secY* gene, which represents a smaller number compared to the amplicons obtained from *lipL32*-PCR. This difference can be attributed to the smaller size of the *lipL32* fragment (242 bp) relative to the *secY* amplicon (409 bp), making the amplification of the smaller region more reliable in clinical samples where bacterial DNA quantity and quality may be suboptimal.

All genital *secY* sequences identified in our study were characterized as *L. interrogans* with high identity to Sejroe strains, consistent with previous findings reporting the presence of this pathogenic *Leptospira* spp. in the region (Di Azevedo et al., 2021; Aymée et al., 2021; Pereira et al., 2022). For a broader epidemiological overview, we included 108 sequences from GenBank, combined with our nine sequences, totaling 117 sequences for phylogenetic analysis. The predominant species identified were *L. borgpetersenii* and *L. interrogans*, known as key agents of bovine leptospirosis globally, particularly in countries like the United States and Uruguay, where their presence is frequently detected using molecular methods or culturing of urine and kidney samples

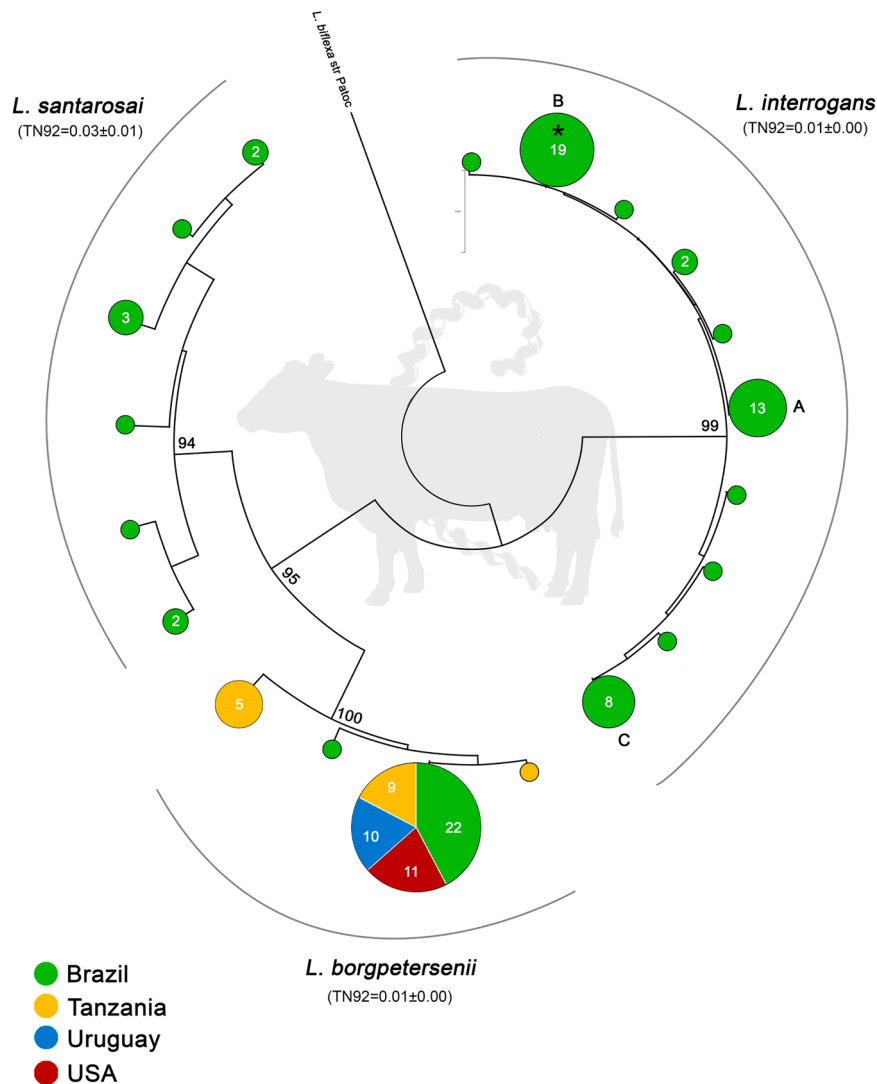


Fig. 1. Maximum likelihood phylogenetic tree inferred from partial *secY* gene sequences of pathogenic *Leptospira* spp. sg. Sejroe from cattle associated with chronic leptospirosis. The size of circles demonstrates the quantity of sequences analyzed and the colors show the origin country. Asterisk (*) indicates haplogroup where sequences from the present study are included. Numbers at nodes are bootstrap values. *Leptospira biflexa* strain Patoc is the outgroup taxa. For each species, estimates of average evolutionary divergence over sequences are shown (Tamura Nei Distance – TN92).

(Zarantonelli et al., 2018; Putz et al., 2022; Hamond et al., 2022). *L. santarosai*, considered an emerging pathogen in the Americas, was also identified, particularly associated with serogroup Sejroe and isolated from both genital and urinary tracts (Aymée et al., 2022a). Therefore, the presence of these three species in association with Sejroe strains in cattle was anticipated.

The phylogenetic analysis revealed multiple distinct clades of *Leptospira* spp. associated with chronic bovine leptospirosis (see Fig. 1). The presence of different bacterial clades causing similar pathogenesis in the same reservoirs suggests mechanisms such as mutations, genetic recombination, or horizontal gene transfer may be at play. The intricate epidemiological cycle of leptospires, involving diverse hosts and ecological niches, likely contributes to their emergence and adaptability under different selective pressures. This pattern of convergent evolution underscores the complex interplay between microbial evolution, host immune responses, and environmental factors (Didelot et al., 2016).

Over the past decades, Brazil has led genetic characterization studies of bovine leptospirosis targeting the *secY* gene, resulting in the country having the largest repository of sequences. However, there remains a notable absence of molecular characterization of *Leptospira* spp. from genital samples in countries outside Brazil, which explains the

predominance of renal samples in global studies. This gap underscores the critical need for new research initiatives focusing on the genital site in diverse geographical regions.

Moreover, many associated studies have not provided detailed reproductive histories of the animals, limiting the ability to draw comprehensive inferences about the disease. The lack of association between infection and clinical signs may stem from studies conducted in slaughterhouses or the subtle manifestations of the disease, which can easily go unnoticed in field settings.

It is important to note that while *L. interrogans* and *L. santarosai* are exclusively reported in Brazil, their absence in other countries cannot be definitively concluded due to the lack of available *secY* sequences meeting inclusion criteria. The genetic studies of leptospires suffer from a lack of standardization in the genetic markers used, which hampers comprehensive molecular epidemiological investigations. To mitigate this issue, we advocate for the widespread adoption of the *secY* gene, recognized for its efficacy in defining genomic species and assessing intraspecific diversity (Lam et al., 2020).

The sequences from our study exhibited high similarity with reference strains from the Sejroe serogroup (Norma and L53), and the ML-TN92 phylogenetic analysis highlighted their close relationship with

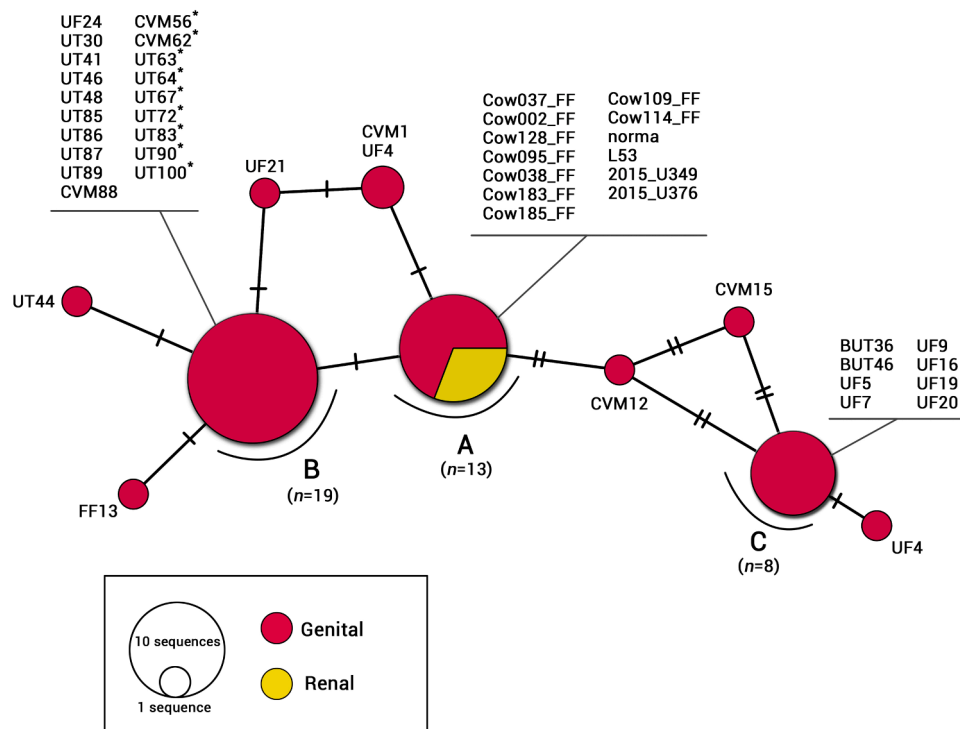


Fig. 2. Haplotype network based on *L. interrogans* *secY* locus (n= 48), constructed according to biological source (genital – pink; renal – yellow). Area of the circle is proportional to number of sequences. The main three haplogroups (A, B and C) are shown. The number of single nucleotide polymorphisms is shown by traits. Sequences from the present study are indicated by asterisk (*).

other strains from the same region (Fig. 1; Table 1). This finding reinforces the epidemiological significance of *L. interrogans* in south-eastern Brazil. Conversely, in northern Brazil, *L. borgpetersenii* predominates, particularly identified from urinary tract samples (Guedes et al., 2020; et al., 2021), with no investigations on genital samples reported.

The genotype Hardjobovis (*L. borgpetersenii*) is known to be more associated with host-host interactions, whereas Hardjoprajitno (*L. interrogans*) is often linked to environmental transmission due to its enhanced environmental resilience. The distinct climatic conditions, rainfall patterns, vegetation types, and interspecies interactions in the Amazon (northern Brazil) versus the Atlantic Forest (southeastern Brazil) biomes likely contribute to these contrasting results. Further studies are imperative to elucidate the influence of biome-specific factors on the distribution and diversity of *Leptospira* spp. genotypes.

The phylogenetic analysis in our study highlighted the genetic similarity among *L. borgpetersenii* strains regardless of their geographical origin, which may be attributed to the species' smaller genome size and potentially reduced genetic diversity (Bulach et al., 2006). This characteristic suggests that *L. borgpetersenii* strains may primarily transmit through direct animal-animal contact rather than persisting in diverse environmental sources.

Our findings suggest that *L. borgpetersenii* genotype Hardjobovis tends to favor colonization of the urinary tract, while *L. interrogans* genotype Hardjoprajitno appears more associated with the genital tract. However, further investigations specifically targeting the genital tract are essential to elucidate why certain strains exhibit a preference for different anatomical sites.

Regarding *L. santarosai*, our study and others have observed high genomic diversity within this species, reflecting its adaptability and emergence in Latin America. It has been isolated from various sources, including urine, vaginal fluid, and uterus, associated particularly with serovar Guaricura (Loureiro et al., 2016). These observations underscore the dynamic nature of *L. santarosai* and its potential impact on bovine health.

A notable outcome of our study is the distinct haplotype distribution observed in the *L. interrogans* phylogenetic network. Haplogroups A, B, and C delineate strains from urinary and genital origins, with B and C exclusively associated with genital strains. These findings underscore significant differences between strains colonizing the genital and renal sites. Although experimentally demonstrated (Pinto et al., 2020), the complex host-pathogen relationship observed between cattle and Sejroe strains remains unclear, and scientists have yet to elucidate why some strains seem to have a predilection for urinary or genital colonization in cows. More experimental and genomic studies are needed, and more novel outcomes may offer important insights to address this gap in understanding.

The study underscores the genital site as a crucial reservoir for leptospiral infection in cattle, particularly associated with Bovine Genital Leptospirosis (BGL). It emphasizes the urgent need for clinical veterinarians and researchers to prioritize the use of genital samples like uterine fragments and cervical, vaginal mucus for accurate diagnosis, aligning with the diagnostic protocol proposed by Aymée et al. (2023b).

Moreover, current vaccines against leptospirosis are based on international reference strains primarily isolated from the renal tract. Given the distinct epidemiological role of genital strains in reproductive failures, there is uncertainty about the efficacy of these vaccines against BGL. Therefore, our study advocates for research into vaccination using locally adapted strains, ideally of genital origin, to enhance protection against bovine leptospirosis and mitigate economic losses associated with reproductive failures.

5. Conclusion

Herein, we have presented a groundbreaking molecular epidemiological perspective on the primary causative agents of chronic bovine leptospirosis, affirming that *L. borgpetersenii* genotype Hardjobovis and *L. interrogans* genotype Hardjoprajitno are predominant circulating strains. Importantly, our study has demonstrated that pathogenic leptospires originating from genital sources cluster within an exclusive

haplogroup, distinctly separate from strains isolated from renal sites. This novel finding provides valuable insights into the distribution and circulation patterns of pathogenic *Leptospira* spp. associated specifically with chronic bovine leptospirosis. Furthermore, our results underscore the potential benefits of developing vaccines composed of locally adapted strains. Given the distinct epidemiological roles of genital strains in bovine reproductive health, such vaccines could play a pivotal role in enhancing prophylactic measures against bovine leptospirosis. Continued research in this direction holds promise for improving disease management and livestock health outcomes.

CRedit authorship contribution statement

Ana Luiza dos Santos Baptista Borges: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Luiza Aymée:** Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Filipe Aníbal Carvalho-Costa:** Writing – review & editing, Resources, Project administration, Investigation, Funding acquisition, Conceptualization. **Walter Lilenbaum:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Maria Isabel Nogueira Di Azevedo:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

None of the authors has any other financial or personal relationships that could inappropriately influence or bias the content of the paper.

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ARTIGO 3: Caracterização genética e potencial zoonótico de *Leptospira interrogans* identificada em pequenos mamíferos não voadores da Mata Atlântica Sudeste, Brasil

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A leptospirose é uma zoonose de importância global para a saúde pública, causada por bactérias do gênero *Leptospira*. Pequenos mamíferos não voadores são importantes reservatórios do patógeno. A Mata Atlântica Brasileira é um *hotspot* de biodiversidade localizado em uma área densamente povoada e sujeita a intensa degradação. Embora documentada por meio de inquéritos sorológicos e da detecção de DNA de leptospiros em pequenos mamíferos selvagens, nenhum estudo realizou a caracterização genética da bactéria na região. O presente estudo teve como objetivo avaliar a diversidade genética de leptospiros patogênicos identificadas em pequenos mamíferos não voadores na Mata Atlântica Sudeste e realizar inferências genéticas intraespecíficas com outros hospedeiros. A área estudada incluiu cinco diferentes unidades de conservação. O diagnóstico molecular foi realizado com base no gene *lipL32*. O método de tipificação SLST foi aplicado com base no gene *secY*. No total, 56% das amostras foram positivas na *lipL32*-PCR e identificadas como *L. interrogans*, com alta identidade genética entre elas, distribuídas em quatro haplótipos principais. O maior haplótipo também incluiu sequências de referência de humanos, cães e ratos urbanos, todos pertencentes ao sorogrupo Icterohaemorrhagiae. Nossos resultados reforçam o papel de pequenos mamíferos como importantes portadores de *L. interrogans* e destacam a Mata Atlântica como um ambiente significativo para a circulação e disseminação de espiroquetas com potencial zoonótico.



Article

Genetic Characterization and Zoonotic Potential of *Leptospira interrogans* Identified in Small Non-Flying Mammals from Southeastern Atlantic Forest, Brazil

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Abstract: Leptospirosis is a zoonotic disease of global public health importance caused by bacteria of the genus *Leptospira*. Small non-flying mammals are important reservoirs of the pathogen. The Brazilian Atlantic Forest is a biodiversity hotspot located in a densely populated area and subject to intense degradation. Although documented through serosurveys and the detection of leptospiral DNA in wild small mammals, no study has performed a genetic characterization of the bacteria in the region. The present study aimed to evaluate the genetic diversity of pathogenic leptospires identified in small non-flying mammals in the Southeast Atlantic Forest and to perform intraspecific genetic inferences with other hosts. The studied area included five different conservation units. Molecular diagnosis was performed based on the *lipL32* gene. The SLST typing method was applied based on the *secY* gene. In total, 56% of samples were *lipL32*-PCR-positive and identified as *L. interrogans*, with a high genetic identity among them, distributed in four main haplogroups. The largest haplogroup also included reference sequences from humans, dogs, and urban rats, all belonging to the Icterohaemorrhagiae serogroup. Our results reinforce the role of small mammals as important carriers of *L. interrogans* and highlight the Atlantic Forest as a significant environment for the circulation and dissemination of spirochetes with zoonotic potential.

Keywords: leptospirosis; zoonosis; one health; *Akodon*; *secY*



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1. Introduction

Leptospirosis is a globally neglected, reemerging zoonotic disease of global public health importance caused by pathogenic spirochetal bacteria of the genus *Leptospira* that circulates among a wide spectrum of hosts, including humans and domestic and wild animals [1]. Leptospires can survive in the environment, particularly in moist conditions, and can be transmitted through different routes: (i) direct contact with the urine of an infected animal; (ii) indirect contact with contaminated water, soil, or feed; and (iii) sexual and vertical transmission in some species (e.g., bovine) [2].

Historically, the etiology of leptospiral infections can be divided into those determined by host-adapted or incidental strains [3]. Adapted strains are observed in the reservoir

animal host, resulting from a long process of host–pathogen coevolution, leading to asymptomatic or inapparent disease or mild and subclinical symptoms of a chronic nature [3]. These maintenance hosts are infected asymptomatic animals that shed viable leptospires in their fluids, acting simultaneously as hosts and reservoirs. They play an important role in the epidemiology of leptospirosis, harboring and shedding leptospires in the environment for long periods, becoming important sources of leptospiral infection for non-adapted hosts [4]. Examples of adapted serovars are Bratislava in horses, Pomona in pigs, Hardjo in ruminants, Canicola in dogs, and Icterohaemorrhagiae in rodents [5]. On the other hand, incidental serovars infect hosts other than the maintenance ones, leading to an apparent acute, often severe illness [4]. Importantly, the *L. interrogans* serogroup Icterohaemorrhagiae is the most associated with acute human disease [6], highlighting the incidental nature of infection in people.

The clinical signs of leptospirosis in humans and animals are diverse and can affect multiple organ systems. In humans, the disease can present with a wide spectrum of symptoms, ranging from mild, flu-like symptoms to severe, life-threatening conditions such as Weil's disease, characterized by jaundice, renal failure, and hemorrhagic manifestations [7]. The bacteria enter through abrasions or cuts in the skin and mucous membranes, including the conjunctiva [8]. Macerated skin, resulting from prolonged exposure to water, is also a suspected route of infection [9].

Small non-flying mammals, such as rodents and marsupials, play a crucial role in the epidemiology of leptospirosis and act as reservoirs of *Leptospira* spp. [8]. These animals shed the bacteria in their urine, contaminating the environment, particularly water sources. This can lead to human infection through direct or indirect contact with contaminated water or soil, representing a constant risk, especially in urban and peri-urban environments [10].

Due to its high biodiversity and species endemism in a small remaining area, the Brazilian Atlantic Forest is among the world's 36 biodiversity hotspots [11]. It harbors a wide range of mammalian hosts, which provide favorable biotic conditions for the bacterial cycle. In addition, it is located in a tropical region and presents ideal environmental abiotic conditions, such as a hot and humid climate, that favor the survival of leptospires [12]. Several wild and domestic animals live in the region, including 72% of the Brazilian population [13]. Leptospirosis is endemic in this region, with an average of 3810 cases reported annually and with higher numbers in urban areas, including the Rio de Janeiro state, located in the southeastern region of the Atlantic Forest, where the disease remains a significant public health concern, with periodic outbreaks, particularly during the rainy season [14]. Poverty and urbanization, among other environmental and socioeconomic factors, influence morbidity and mortality rates [15].

The few studies that have documented the occurrence of *Leptospira* spp. in small mammals in the Atlantic Forest [16–19] were based on serosurveys, which present limited value since reservoirs may not be detected by serology [20]. Previous studies conducted by our group in the region have identified the presence of pathogenic leptospiral DNA in wild marsupials and rodents based on conventional PCR [21,22]. However, to date, no study has performed a genetic characterization of the bacteria.

Therefore, the present study aimed to analyze the genetic diversity of pathogenic leptospires identified in small non-flying mammals in the Atlantic Forest of southeastern Brazil, comparing the identified haplotypes with those described in human and domestic animals to assess the risk of zoonotic transmission and provide new epidemiological inferences.

2. Materials and Methods

All experimental procedures were approved by the Ethics Committee on Animal Use of the Oswaldo Cruz Institute (CEUA-IOC, Comissão de Ética no Uso de Animais,

Instituto Oswaldo Cruz, protocol number P-70/13-2, license number LW-39/14). Fieldwork was in full compliance with the federal license issued by the Instituto Brasileiro do Meio Ambiente e dos Recursos Naturais Renováveis of the Brazilian Ministry of the Environment (IBAMA/SISBIO, license number 13373-1).

2.1. Animal Capture and Study Areas

The capture and taxonomic identification procedures of the small non-flying mammals are shown in [21,22]. Briefly, specimens were captured with Sherman® and Tomahawk® live traps set in sampling transects, 4–10 in each study area, each with 20 traps, evenly spaced 10–20 m apart. The traps were verified and rebaited daily during seven to ten consecutive sampling nights. In total, 265 kidney samples were obtained, one for each host, including the orders Didelphimorphia ($n = 21$) and Rodentia ($n = 244$).

The study area included five different conservation units within the Atlantic Forest in the Rio de Janeiro state, Brazil: (A) Serra dos Órgãos National Park/Casimiro de Abreu; (B) Itatiaia National Park/Itatiaia; (C) Area of Relevant Ecological Interest Itapebussus/Macaé; (D) Sapucaia Power Plant/Campos dos Goytacazes; and (E) Poço das Antas Reserve/Silva Jardim (Figure 1). Areas A, B, and E are strictly protected regions with minimal human interference, while areas C and D are zones with more pronounced anthropogenic activities due to industrial, agricultural, or urban development.

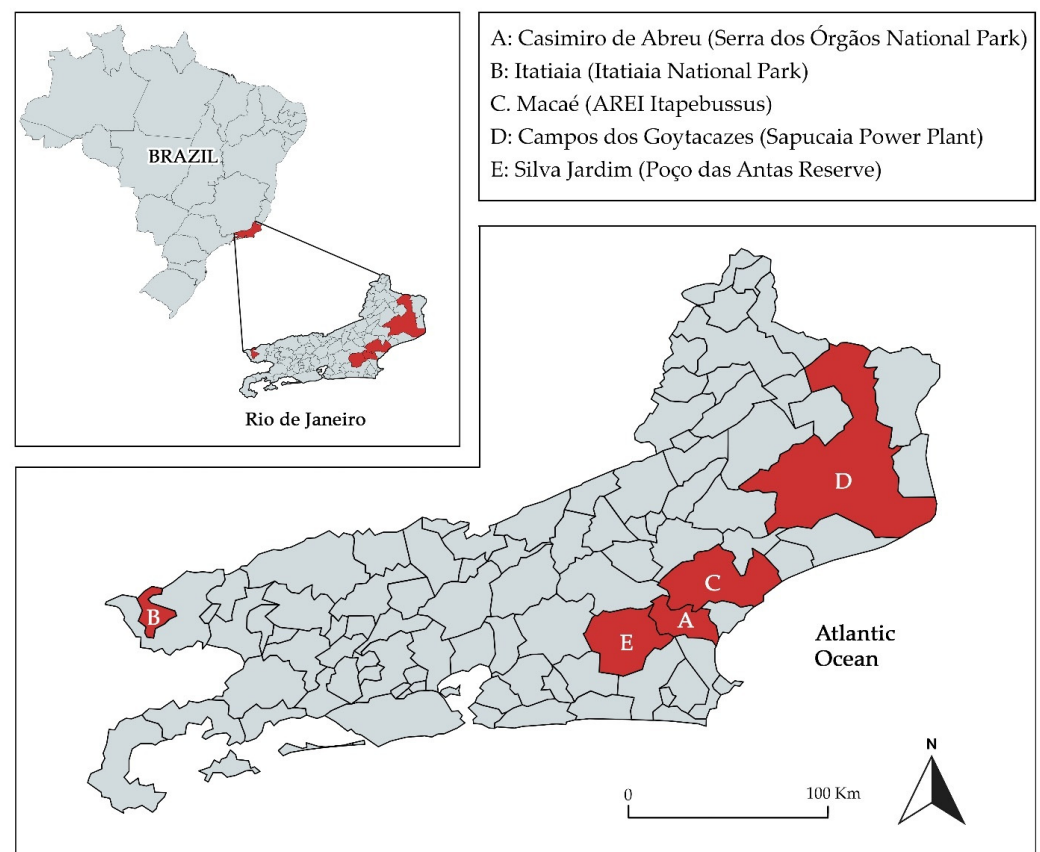


Figure 1. Map of the State of Rio de Janeiro, Brazil, showing the collection sites (municipality/conservation area) of small non-flying mammals (red). The Atlantic Forest is the biome found along the coast of the Atlantic Ocean.

Samples from areas A ($n = 15$), B ($n = 77$) and C ($n = 34$) had been previously subjected to amplification based on the *lipL32* gene. Of these, 63/126 (50%) were positive [21,22]. Samples from areas D ($n = 52$) and E ($n = 87$) were processed in the present study, totaling 139 samples. These were subjected to DNA extraction and molecular diagnosis.

2.2. Molecular Diagnosis

All unprocessed small mammal kidney samples ($n = 139$) were subjected to DNA extraction using a DNeasy® Blood & Tissue Kit (Qiagen, Hilden, Germany following the manufacturer's instructions. For detecting pathogenic *Leptospira* spp., *lipL32* gene-based PCR was performed using primers and conditions described by Hamond et al. (2014) [23]. Each sample set included ultrapure water as a negative control in all reactions, and 10fg of DNA extracted from *Leptospira interrogans* serovar Copenhageni (Fiocruz L1–130) served as a positive control. PCR products were analyzed by 2% agarose gel electrophoresis and visualized under ultraviolet light after staining with GelRed®.

2.3. Genetic Characterization of Pathogenic *Leptospira* spp.

All samples positive for the *lipL32* gene were subjected to a nested PCR based on the *secY* gene, a genetic marker with good taxonomic resolution, whose use is well established for the genetic characterization of leptospires [24]. Reactions were performed using primers and conditions described in Grillová et al., 2020 [25], with the enzyme Platinum Taq DNA Polymerase kit (Invitrogen, Carlsbad, CA, USA). The amplicons were purified with a PCR Clean-Up System Kit (Promega, Madison, WI, USA) according to the manufacturer's instructions and intended for sequencing. The reactions were performed in both directions using a Big Dye Terminator v. 3.1 Cycle Sequencing Kit (Applied Biosystems, Foster City, CA, USA) on a 3100 automated DNA sequencer according to the manufacturer's instructions.

2.4. Phylogenetic Analyses

For molecular epidemiology purposes, we used the Single Locus Sequencing Typing (SLST) typing method. For this, sequences of the *secY* gene were selected from GenBank and subjected to phylogenetic analyses with the sequences obtained in the present study. Very short sequences (<400 bp) were excluded to avoid bias. After aligning the sequences in the ClustalX v 2.0 software [26], a maximum likelihood (ML) tree was constructed in MEGA X software v 11.0.13 [27] using the best-fitting DNA substitution model determined by the Bayesian information criterion. Genetic distances were calculated in MEGA X with the model previously determined by the analysis of the data set.

2.5. Assessment of Genetic Diversity and Determination of Haplotype Circulation Patterns

The identification of unique haplotypes was performed after screening in the DAMBE software v 7.3.32 [28]. To assess the intraspecific genetic diversity of *Leptospira* spp. populations and evaluate the dynamics of haplotype circulation between different hosts and regions, haplotype networks were constructed from the aligned sequences and scripts in nexus format in the PopART population genetics software using the median-joining inference method [29].

3. Results

3.1. Molecular Identification of Pathogenic *Leptospira* spp.

Of the 139 samples processed and submitted to molecular diagnosis based on the *lipL32* PCR, 78 (56%) were positive. Those 78 plus the 63 previously positive samples [21,22] were subjected to a nested PCR based on the *secY* genetic marker, totaling 141 DNA samples. It was possible to obtain strong amplicons, suitable for sequencing, from 93 samples.

3.2. Genetic Characterization of Pathogenic *Leptospira* spp. Identified

Based on the *secY* gene sequencing, all samples were identified as *L. interrogans* with >99% identity with reference sequences after searching against the BLAST/NCBI platform (accessed 15 January 2025). The majority of hosts were from the order Rodentia ($n = 83$),

with few from the order Didelphimorphia ($n = 10$), totaling 10 different genera. From these, *Akodon* and *Mus* had the highest number of positive animals with generated sequences ($n = 23$ and $n = 21$, respectively) (Table 1). The distribution of genetically characterized samples according to location and hosts is shown in Supplementary Table S1.

Table 1. Haplogroup distribution of *Leptospira interrogans* identified in the present study according to collection site and host.

Collection Site	Host	Haplogroup				
		#1	#2	#3	#4	ND
(A) Serra dos Órgãos NP ($n = 8$)	<i>Didelphis aurita</i> ($n = 1$)	1				
	<i>Akodon montensis</i> ($n = 3$)	1			1	1
	<i>Oligoryzomys nigripis</i> ($n = 3$)	1			1	1
	<i>Trinomys marmosa</i> ($n = 1$)	1				
(B) Itatiaia NP ($n = 34$)	<i>Monodelphis</i> sp. ($n = 4$)	2		2		
	<i>Akodon</i> sp. ($n = 8$)	6			1	1
	<i>Bucepattersonius</i> sp. ($n = 10$)	8		1	1	
	<i>Delomys</i> sp. ($n = 6$)	3		1		2
	<i>Oligoryzomys</i> sp. ($n = 2$)	2				
	<i>Oxymycterus</i> sp. ($n = 4$)	3				1
(C) AREI Itapebussus ($n = 10$)	<i>Marmosa paraguayana</i> ($n = 5$)	1	4			
	<i>Akodon cursor</i> ($n = 3$)		3			
	<i>Mus musculus</i> ($n = 2$)		2			
(D) Sapucaia Power Plant * ($n = 15$)	<i>Mus musculus</i> ($n = 15$)	14		1		
(E) Poço das Antas Reserve * ($n = 26$)	<i>Akodon</i> sp. ($n = 9$)	9				
	<i>Mus musculus</i> ($n = 4$)	4				
	<i>Necomys lasiurus</i> ($n = 13$)	12				1
Total		68	9	5	4	7

AREI: Area of Relevant Ecological Interest; NP: National Park; ND: not defined. Samples from sites indicated with an asterisk (*) were extracted and amplified based on the *lipL32* gene in the present study. The other samples were only amplified and sequenced for the *secY* gene, as they had already been previously amplified for the *lipL32* gene in previous studies [21,22].

3.3. Phylogenetic Analyses and Haplotype Distribution

The Tamura Nei model (TN92) was the best DNA substitution model determined. Based on this, a high genetic identity was found among the sequences from the present study (TN92 = 0.01 ± 0.01 ; 99% identity). The phylogenetic analyses and haplotype network demonstrated the presence of four main haplogroups, Hap#1 ($n = 68$), Hap#2 ($n = 9$), Hap#3 ($n = 5$), and Hap#4 ($n = 4$), included in two main clades, with Hap#1, Hap#2, and Hap#3 included in clade 1 and Hap#4 included in clade 2 (Figure 2). This clade included only the sequences from Serra dos Órgãos National Park and Itatiaia National Park, while clade 1 included sequences from all studied sites. Hap#2 was formed exclusively by sequences from AREI Itapebussus. Seven haplotypes were identified as unique, not forming haplogroups (Table 1, Figure 2). There were no specific haplogroups formed according to hosts (Figure 2).

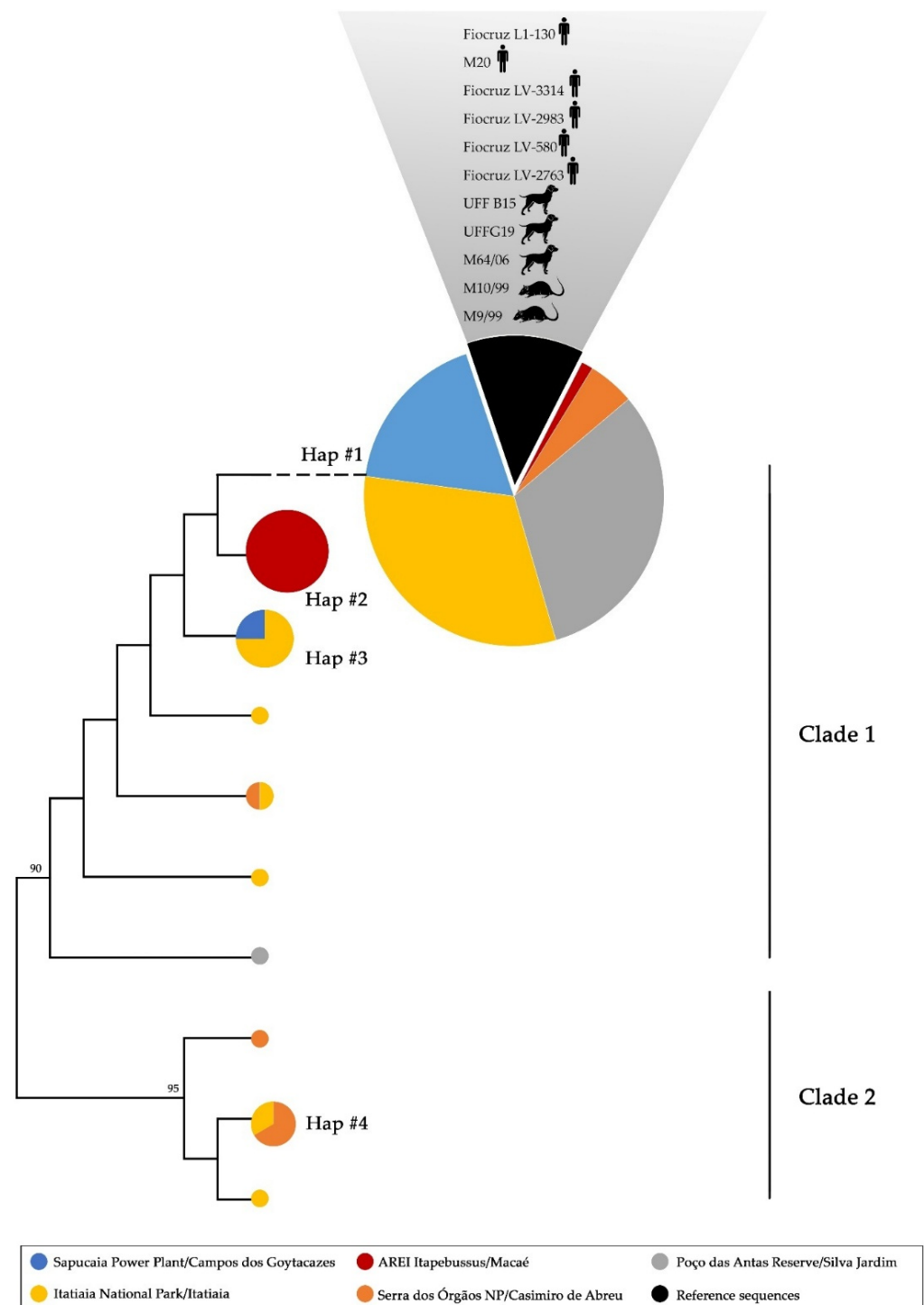


Figure 2. Maximum likelihood (ML) phylogenetic tree and haplogroup distribution inferred from partial *secY* sequences of *L. interrogans*. The area of the circle is proportional to the number of sequences. Colors indicate the place of origin of the sequence (legend). Reference sequences obtained from GenBank were also included, identified in hosts (identified by a vector) and also from the Atlantic Forest (represented in black). AREI: Area of Relevant Ecological Interest; NP: National Park.

Hap#1 included the vast majority of sequences in the present study, including those from rodents and marsupials from the five regions studied. Furthermore, they presented 100% identity with sequences identified in humans, dogs, and urban rats, all from the Icterohaemorrhagiae serogroup from the same geographical area (Figure 2).

4. Discussion

Herein we performed, for the first time, the genetic characterization of pathogenic leptospires identified in 15 species of small non-flying wild mammals from conservation units of the Brazilian Atlantic Forest. A high occurrence was found (56%), which agrees with studies carried out in the same region [21,22]. Another study conducted in small wild rodents from the southeast Brazilian Atlantic Forest, but in rural areas, revealed a lower prevalence (21.3%) and host diversity (five species) based on qPCR diagnosis [30]. This is not surprising, since conservation areas often support greater diversity and abundance of reservoir hosts, greater environmental persistence of the bacteria, low human impact and less use of control measures, greater interactions between species and trophic networks, and less competition with synanthropic species [31]. A study also conducted in conservation areas of the Brazilian Amazon biome revealed a high prevalence of leptospiral DNA in small mammals (44.7%) from 16 different species [32], reinforcing that conservation areas, with their rich biodiversity, can harbor multiple natural reservoirs for *Leptospira*.

Most of the hosts were from the order Rodentia ($n = 83$), with emphasis on the genera *Akodon* ($n = 23$) and *Mus* ($n = 21$). The genus *Akodon*, which includes neotropical wild rodents, has been identified as a major carrier of pathogenic *Leptospira* in the Atlantic Forest region of Brazil. Studies have shown that a substantial proportion of *Akodon* spp. are PCR-positive for pathogenic *Leptospira*, highlighting their role as reservoirs in this environment [21,22]. This suggests that *Akodon* rodents contribute to the maintenance and transmission of leptospires in areas where they are prevalent, although their impact may be more pronounced in rural or forested areas rather than urban settings like Rio de Janeiro. Studies on *Akodon* and other sigmodontine rodents in other South American regions, such as the Paraná Delta in Argentina, further support their involvement in the transmission of *Leptospira*. The species *Akodon azarae* was identified as a host for *Leptospira interrogans* serovar Icterohaemorrhagiae, suggesting that these rodents can carry and potentially transmit zoonotic *Leptospira* strains across their geographic range, which includes parts of Brazil [33]. In contrast, the role of *Mus musculus* in transmitting *Leptospira* spp. varies by region. While they can be relevant carriers in certain areas, such as urban settlements in Nairobi, Kenya [34], and rural settings in Puerto Rico [35], their role seems to be less pronounced in urban areas in Brazil, like in Rio de Janeiro, where *Rattus* species are the major reservoirs [36]. Other rodents identified in the present study included *Oligoryzomys nigripis*, *Trinomys marmosa*, *Brucepattersonius* sp., *Delomys altimontanus*, *Delomys dorsalis*, *Oxymycterus* sp., and *Necomys lasiurus*, previously identified as hosts of *Leptospira* in the region [21,22,30].

Regarding the marsupials, *Didelphis aurita*, *Monodelphis* sp., and *Marmosa paraguayana* were herein identified as hosts of *L. interrogans*. Other studies have investigated marsupials, particularly opossums, as hosts of *Leptospira* spp. In a study performed in Southern Brazil, the white-eared opossum (*Didelphis albiventris*) was identified as a reservoir of pathogenic *Leptospira* spp. [37]. Another study examined a broader range of wildlife in Brazil, including opossums, and identified *L. santarosai* as the prevalent species, with a higher risk of infection observed in adult and male opossums when compared to their younger and female counterparts [19]. Additionally, a study conducted in Rio Grande do Sul, Brazil, detected pathogenic *Leptospira* spp. DNA in various wild mammals, including the white-eared opossum. The species *L. borgpetersenii* and *L. interrogans* were identified [38]. These studies collectively highlight the significance of marsupials, particularly opossums, as hosts of *Leptospira* in Brazil, with potential contributions to the maintenance of the bacterial transmission cycle.

Regarding the phylogenetic analyses based on the SLST method applied, it was possible to observe a high genetic identity between the sequences from the present study

(TN92 = 0.01 ± 0.01 ; 99% identity) (Figure 2). This can be explained by the geographic proximity between the studied areas, which present shared environmental features such as similar climatic and ecological conditions (e.g., temperature, humidity, and water bodies). Moreover, some *Leptospira* spp. strains exhibit genetic stability over time, especially if they are well adapted to their host and/or environment [39], which appears to be the case for rodent and marsupial hosts in this region of the Atlantic Forest. Limited selective pressure in certain environments may reduce the rate of genetic divergence [40]. On the other hand, it is important to note that, while in the Atlantic Forest, a low genetic diversity was observed among *Leptospira* spp., a study conducted in the Western Amazon Forest identified three different species, with high intraspecific genetic variation [32]. This greater genetic diversity of *Leptospira* in the Amazon region is probably related to the high biodiversity of hosts and less fragmentation of the habitat. The Atlantic Forest, with its high fragmentation and smaller number of reservoir species, may be subject to a genetic bottleneck effect, limiting the diversity of bacterial lineages [41].

Regarding intraspecific analysis, our results showed the presence of four main haplogroups, not related to hosts or geographic locations, except for Hap#2, which was formed exclusively by sequences from AREI Itapebussus (Figure 2). Importantly, Hap#1 included the largest number of sequences in the present study ($n = 68$) that also comprised *L. interrogans* reference sequences from humans, dogs, and urban rats from the Icterohaemorrhagiae serogroup. It is well known that the clonal subpopulation Fiocruz L1-130 of *L. interrogans* serogroup Icterohaemorrhagiae is widely distributed in Brazil and Latin America in different hosts with a major zoonotic role [42,43]. The data from the present study demonstrate that small wild mammals from the Atlantic Forest are also reservoirs of this strain, which seems to be ubiquitous in that biome. This outcome raises an important public health concern: *L. interrogans* from the Icterohaemorrhagiae serogroup is the major agent of leptospirosis in humans [8], and rodents in forested areas, particularly in regions like the Atlantic Forest, can act as a direct or indirect link to human infection.

The Atlantic Forest is one of the most biodiverse biomes in the world but is facing severe degradation due to urbanization, agriculture, and logging, with much of its original cover already destroyed [44]. It is the most deforested forest in Brazil, and currently, only 12.4% of the original forest remains [45]. This degradation has a direct impact on local ecology and public health, especially on the transmission of zoonotic diseases such as leptospirosis, which may occur due to different factors: (i) habitat fragmentation and increased human–wildlife interaction; (ii) disruption of ecosystem balance (disturbed trophic interactions and bacterial persistence in altered environments); (iii) increased exposure to contaminated water sources; (iv) changes in biodiversity that may lead to shift in host species; and (v) climate change, which can promote favorable conditions for the spread of the bacteria [46].

5. Conclusions

The present study revealed a high genetic identity among pathogenic leptospires circulating in small non-flying mammals from the southeastern Atlantic Forest biome, highlighting the presence of a larger haplogroup that includes a highly virulent strain also detected in humans, dogs, and urban rats (*L. interrogans* serogroup Icterohaemorrhagiae, strain Fiocruz L1-130), and evidenced the zoonotic potential of the identified leptospires. Our results reinforce the role of small mammals, such as rodents and marsupials, as important carriers of *Leptospira interrogans* and draw attention to the Atlantic Forest as an important environment for the circulation and dissemination of the spirochaete. Detecting virulent strains in animals of different species that inhabit forests and peri-urban areas warns of the imminent risk to public health, especially in regions where interactions

between humans and wildlife are frequent. This study highlights the urgent need for epidemiological surveillance and control actions in southeast Brazil, aiming to mitigate the transmission of leptospirosis and prevent future outbreaks, especially in highly populated areas from the Atlantic Forest submitted to intense fragmentation and anthropic actions. In this context, a One Health approach is fundamental, integrating human, animal, and environmental health to effectively manage and mitigate the impact of zoonotic diseases such as leptospirosis.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/tropicalmed10030062/s1>, Supplementary Table S1: Complementary information regarding the origin and hosts of pathogenic leptospires from the Brazilian Atlantic Forest genetically characterized in the present study based on the *secY* gene.

Author Contributions: Conceptualization, M.I.N.D.A. and W.L.; Methodology, A.C.d.R.S., C.E. and A.S.V.; Software, M.I.N.D.A.; Validation, M.I.N.D.A., F.A.C.-C., A.S.V. and W.L.; Formal analysis, M.I.N.D.A., A.C.d.R.S., C.E. and A.S.V.; Investigation, C.E., F.A.C.-C. and W.L.; Resources, F.A.C.-C. and W.L.; Data curation, M.I.N.D.A., A.C.d.R.S. and A.S.V.; Writing—original draft, M.I.N.D.A., A.C.d.R.S. and C.E.; Writing—review & editing, M.I.N.D.A., F.A.C.-C., A.S.V. and W.L.; Visualization, A.S.V.; Supervision, W.L.; Project administration, W.L.; Funding acquisition, F.A.C.-C. and W.L. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement: The data that support the findings of this study are openly available in GenBank under accession numbers PQ999001–PQ999093.

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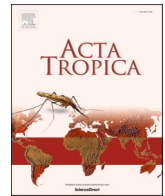
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ARTIGO 4: Distribuição ecológica e associações bioma-hospedeiro de *Leptospira* patogênica no Brasil: uma perspectiva de Saúde Única em uma área tropical

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Espécies patogênicas do gênero *Leptospira* causam uma zoonose de distribuição global com ciclos de transmissão complexos envolvendo múltiplos hospedeiros e ambientes. A vasta diversidade ecológica do Brasil e a alta circulação de *Leptospira* spp. exigem abordagens integrativas para compreender a diversidade genética e a distribuição do patógeno. Compilamos um extenso conjunto de dados moleculares de *Leptospira* spp. patogênicas no Brasil, abrangendo sequências ($n=825$) de diversos marcadores genéticos e dados de sequenciamento de genoma completo (WGS). Um panorama molecular foi elaborado com base em metadados associados, incluindo espécie hospedeira, origem geográfica e tipo de fonte (clínica, animal, ambiental). Os registros abrangem quatro biomas brasileiros, com a maior diversidade e número de sequências provenientes da Mata Atlântica (69,7%) e da Amazônia (22,2%), com ênfase em grandes centros urbanos. *Leptospira interrogans* foi predominante (74,9%), seguida por *L. santarosai* (9,8%), *L. borgpetersenii* (7,2%), *L. kirschneri* (3,9%) e *L. noguchii* (3,9%). Em relação aos sorogrupos, Icterohaemorrhagiae foi o mais frequente (44,9%). Os hospedeiros incluíram 78 espécies de 12 ordens, entre grandes animais terrestres domésticos (36%), animais terrestres silvestres (21%), humanos (15%), animais de estimação (13,8%), roedores urbanos (7,6%) e mamíferos marinhos (1,3%). Sequências de leptospirosas patogênicas ambientais também foram identificadas (5,3%). Esta análise molecular integrativa revela a extensa diversidade, amplitude ecológica e dinâmica complexa de transmissão de leptospirosas patogênicas nos biomas Brasileiros. Os resultados reforçam a importância de uma perspectiva de Saúde Única e fornecem uma base sólida para a vigilância molecular padronizada e intervenções direcionadas para mitigar a carga da leptospirose em regiões tropicais.



Ecological range and host–biome associations of pathogenic *Leptospira* in Brazil: A One Health perspective from a tropical area

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ABSTRACT

Pathogenic *Leptospira* spp. cause a globally distributed zoonosis with complex transmission cycles involving multiple hosts and environments. Brazil's vast ecological diversity and high *Leptospira* spp. circulation demand integrative approaches to understand the pathogen's genetic diversity and distribution. We compiled an extensive molecular dataset of pathogenic *Leptospira* spp. in Brazil, encompassing sequences ($n = 825$) from several genetic markers and WGS data. A molecular landscape was provided based on associated metadata, including host species, geographic origin, and source type (clinical, animal, environmental). Records spanned four Brazilian biomes, with the highest diversity and number of sequences from the Atlantic Forest (69.7 %) and Amazon (22.2 %), with emphasis on large urban centers. *Leptospira interrogans* predominated (74.9 %), followed by *L. santarosai* (9.8 %), *L. borgpetersenii* (7.2 %), *L. kirschneri* (3.9 %), and *L. noguchii* (3.9 %). Regarding serogroups, Icterohaemorrhagiae was the most frequent (44.9 %). Hosts encompassed 78 species from 12 orders, including domestic large land animals (36 %), wild land animals (21 %), humans (15 %), pets (13.8 %), urban rodents (7.6 %), and marine mammals (1.3 %). Importantly, sequences from environmental pathogenic leptospires were also identified (5.3 %). This integrative molecular analysis reveals the extensive diversity, ecological breadth, and complex transmission dynamics of pathogenic *Leptospira* in those biomes. The findings underscore the importance of a One Health perspective and provide a robust foundation for standardized molecular surveillance and targeted interventions to mitigate leptospirosis burden in tropical regions.

1. Introduction

Leptospirosis is a widespread zoonotic disease caused by spirochetes of the genus *Leptospira*, which includes both free-living and pathogenic species (Giraud-Gatineau et al., 2024). Traditionally classified by serology into >300 serovars across over 30 serogroups (Giraud-Gatineau et al., 2025), *Leptospira* spp. are also taxonomically divided into 64 species by genomic criteria, grouped into saprophytic ($n = 26$), intermediate ($n = 21$), and pathogenic ($n = 17$) clades (Vincent et al., 2019). The disconnect between serological and molecular classification systems underscores the need for integrative approaches in epidemiological surveillance.

Pathogenic *Leptospira* infect a wide range of vertebrate hosts, with over 160 mammal species identified as reservoirs or incidental hosts (Cilia et al., 2020, 2021). The bacteria are mainly excreted in urine and disseminated through contaminated water and soil. Reservoir hosts, such as rodents and livestock, can excrete viable bacteria for extended

periods, maintaining environmental contamination and enabling transmission to incidental hosts, including humans (Putz and Nally, 2020). Transmission dynamics are influenced by ecological factors such as flooding, soil characteristics, and reservoir host abundance (Soni et al., 2024), highlighting the intrinsic links among human, animal, and environmental health that underpin the One Health framework. This integrative perspective is crucial for understanding how ecological, climatic, and social drivers collectively shape the emergence and persistence of leptospirosis (Antoniolli et al., 2024).

Human leptospirosis remains a significant global health concern, with the highest incidence in tropical and subtropical regions, but also reported in temperate areas, often following animal outbreaks (Cataldo et al., 2024). Annually, an estimated one million cases and 60,000 deaths occur worldwide (Costa et al., 2015). Among tropical countries, Brazil stands out as a major hotspot for transmission due to its vast geographical extension, ecological heterogeneity, and intense human–animal–environment interactions (Sohn-Hausner et al., 2023).

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Leptospirosis is endemic and often reaches epidemic proportions during the rainy season, particularly in urban centers with poor sanitation and high rodent infestation (Galan et al., 2021). An average of 3600 cases and 375 deaths are reported, with Southeast and South regions concentrating the highest case burdens (Linhares et al., 2024).

Clinical presentation ranges from mild, self-limiting febrile illness to severe forms such as Weil's disease and severe pulmonary hemorrhagic syndrome, with about 10 % of patients developing life-threatening complications (Nicodemo and Duarte-Neto, 2021). Diagnosis is often hampered by symptom overlap with other febrile illnesses and by limited access to confirmatory laboratory tests such as the microscopic agglutination test (MAT), contributing to underreporting (Ciurariu et al., 2025; Esteves et al., 2025). Prevention includes rodent control, avoidance of potentially contaminated water, personal protective equipment, and, where available, vaccination, although current human vaccines offer short-lived, serovar-specific immunity and are limited to a few countries (Barazzzone et al., 2021; de Oliveira et al., 2022; Sykes et al., 2022a).

In animals, leptospirosis is widespread, affecting species from domestic livestock to wildlife. Dogs commonly develop acute hepatorenal disease (Costa et al., 2022a), while cattle frequently present with chronic reproductive disorders such as Bovine Genital Leptospirosis, a silent economically significant syndrome (Loureiro and Lilenbaum, 2020). Rodents and other small mammals often act as asymptomatic carriers, maintaining environmental contamination (Mohd-Taib et al., 2020). Control in animals requires an integrated approach combining antibiotic therapy, vaccination, and environmental management, with varying efficacy depending on host species, infecting serogroup, and epidemiological context (Martins and Lilenbaum, 2015; Soo et al., 2020). Environmental persistence of the pathogen and the diversity of animal reservoirs make eradication unlikely, underscoring the importance of One Health strategies to reduce transmission at the human-animal-environment interface (Naing et al., 2019). In Brazil, this challenge is magnified by the country's exceptional ecological diversity, encompassing six major biomes (Amazon, Atlantic Forest, Cerrado, Caatinga, Pantanal, and Pampa), each harboring distinct climates, wildlife communities, and degrees of anthropogenic pressure (IBGE, 2019). Such environmental heterogeneity creates unique eco-epidemiological scenarios for *Leptospira* maintenance, adaptation, and transmission.

Although routine surveillance in developing countries still relies

heavily on serological data (Nardoni Marteli et al., 2022), molecular tools have become essential for species-level identification, outbreak investigation, and tracing of transmission routes. Markers such as 16S rRNA, *secY*, *lipL32*, and multilocus sequence typing (MLST) schemes, along with whole-genome sequencing (WGS), have provided taxonomic identification and enabled comparative analyses of circulating strains (Di Azevedo and Lilenbaum, 2021; Sykes et al., 2022b). The increasing availability of these genetic data from public repositories enables large-scale *in silico* studies that integrate genetic, geographic, and ecological information. Despite the availability of hundreds of *Leptospira* sequences from Brazilian isolates and clinical samples, most previous studies have focused on specific hosts, regions, or outbreaks, resulting in fragmented knowledge of the pathogen's molecular diversity and ecological distribution. There is still a lack of nationwide integrative analyses that combine data from multiple hosts, ecosystems, and biomes under a unified molecular and ecological perspective. An integrative approach that bridges data across hosts and biomes is therefore crucial to understand transmission patterns in such a complex tropical landscape. By compiling and analyzing genetic data from humans, domestic animals, wildlife, and environmental sources across all major Brazilian biomes, this study bridges that gap and provides the first comprehensive molecular overview of pathogenic *Leptospira* in Brazil, advancing the understanding of transmission dynamics and host–environment connectivity within a One Health framework in a tropical context.

2. Materials and methods

2.1. Data collection and curation

A comprehensive database was compiled to assess the genetic diversity and circulation patterns of pathogenic *Leptospira* species in Brazil, including human, animal, and environmental records. Nucleotide sequences were retrieved from the NCBI GenBank database using the search string “*Leptospira* AND Brazil” last accessed on 15 July 2025 (GenBank release 257.0). The search covered all available records deposited between 2009 and 2025. Only sequences corresponding to genetically characterized isolates, with an identified *Leptospira* species and associated metadata including host/source and geographical location, were included. Sequences labeled only as “*Leptospira* sp.” were excluded because they lack reliable species-level identification, which is essential for accurate phylogenetic and epidemiological inferences

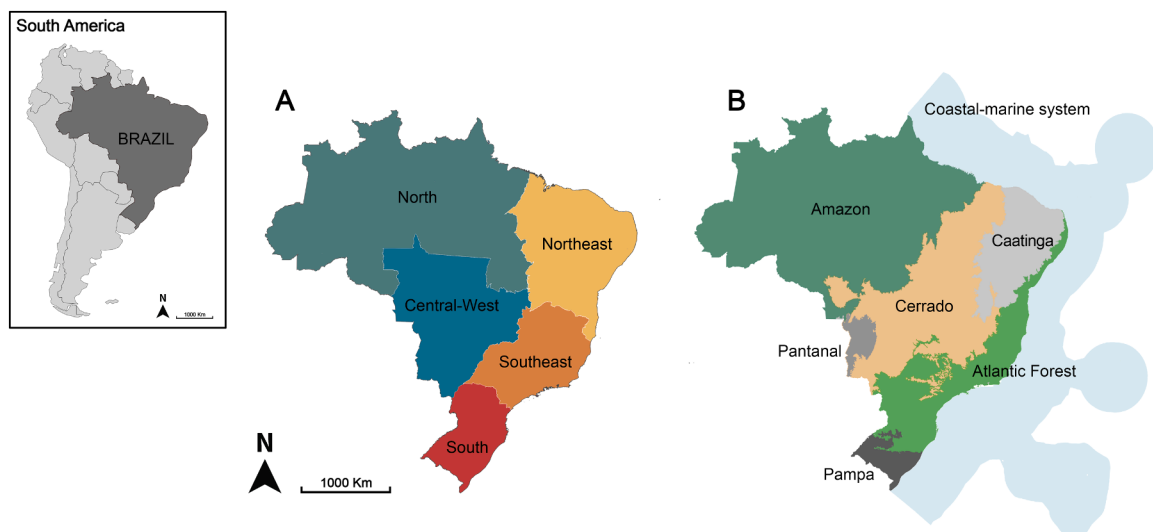


Fig. 1. Geographic division of Brazil used for spatial analyses. (A) Official Brazilian macro-regions. (B) Major Brazilian biomes and the Coastal-Marine System (CMS). The location of Brazil within South America is shown in the inset. Shapefiles for both maps were obtained from the Brazilian Institute of Geography and Statistics (IBGE) geographic data repository.

(Riley, 2018). Although some of these records may contain partial metadata, their exclusion minimizes potential misclassification and ensures consistency in downstream analyses. We acknowledge that this criterion may slightly reduce the representation of poorly characterized or newly described taxa, but it strengthens the reliability of species-level comparisons across datasets. Only pathogenic species were included, specifically, members of the P1 clade as defined by Vincent et al. (2019). Intermediate and saprophytic species were excluded.

Each eligible record was manually reviewed and annotated with the following attributes: GenBank accession number, *Leptospira* species, strain or sample identification (ID), genetic marker applied, serovar/serogroup (when available), host species (taxonomic and ecological categories) or environmental source, geographical location (city, geographic region, and biome), and associated reference. Metadata referring to host identity, sampling location, and serovar information were cross-checked with the original publications cited in GenBank whenever available, to ensure accuracy and consistency. Records with incomplete, ambiguous, or conflicting metadata were retained only if the core attributes (species, genetic marker, and host or source) could be confidently verified; otherwise, they were excluded from the dataset. The data was organized in an Excel spreadsheet.

2.2. Host and geographical categorization

To enable ecological and epidemiological comparisons, besides taxonomic categorization of host species, they were classified into seven broad categories: "Domestic large land animal", "Wild land animal", "Pet", "Human", "Marine mammal", "Urban rodent", and "Environment". These categories were based on ecological and epidemiological criteria widely used in zoonotic surveillance frameworks (e.g., OIE and CDC guidelines) but were adapted to reflect the host diversity represented in the dataset. Similarly, geographical locations were also assigned to one of the five major Brazilian biomes, defined according to the Brazilian Institute of Geography and Statistics (IBGE): "Amazon", "Atlantic Forest", "Pampa", "Cerrado", and "Caatinga", based on the original collection site. The Atlantic Ocean was also included as a separate category (coastal-marine system – CMS).

The Amazon is a humid rainforest with >2000 mm annual rainfall and exceptional biodiversity (Flores et al., 2024). The Atlantic Forest along the eastern coast is highly fragmented and rich in endemic species (Colombo and Joly, 2010). The Caatinga is a semi-arid region with xerophytic vegetation and seasonal droughts (da Silva and Lacher, 2020). The Cerrado, a tropical savanna, exhibits a grassland-woodland mosaic and marked dry season (Colli et al., 2020). The Pampa, in southern Brazil, consists of temperate grasslands (Roesch et al., 2009), while the Pantanal is a vast wetland with seasonal flooding (Wantzen et al., 2024). The CMS includes estuaries, mangroves, and oceanic waters linking terrestrial and marine ecosystems (Álvares-Da-Silva et al., 2022). Fig. 1 shows the geographical divisions of Brazil and its biomes.

2.3. Data analysis and host-species inventory

Descriptive statistics were used to summarize the number and proportion of *Leptospira* species, serogroups, host categories, and biomes represented in the dataset. The data compiled in spreadsheets were analyzed using the R programming environment (version 4.5.1; R Core Team, 2024). The dataset was imported using the `readxl` package. The final table, including absolute counts and percentages, was exported using the `writexl` package for further tabular presentation. No inferential or quantitative statistical analyses (e.g., species richness or diversity indices) were performed, as the study was designed to provide a descriptive and integrative overview of molecular records rather than to test specific hypotheses.

Additionally, a host-species inventory table was manually created to document all known hosts of pathogenic *Leptospira* in Brazil based on genetically confirmed records. This matrix included the taxonomic

Table 1

Summary of pathogenic *Leptospira* records genetically identified in Brazil, compiled from 825 sequences retrieved from public databases. Percentages refer to the proportion of records within each category relative to the total dataset.

Variable	Category	n (%)
Species	<i>L. interrogans</i>	618 (74.9 %)
	<i>L. santarosai</i>	81 (9.8 %)
	<i>L. borgpetersenii</i>	59 (7.2 %)
	<i>L. noguchii</i>	32 (3.9 %)
	<i>L. kirschneri</i>	32 (3.9 %)
	<i>L. yasudae</i>	2 (0.2 %)
	<i>L. stimsonii</i>	1 (0.1 %)
Genetic marker	<i>secY</i>	546 (66.2 %)
	16S rRNA	120 (14.6 %)
	WGS	99 (12 %)
	<i>lipL32</i>	38 (4.6 %)
	MLST	17 (2 %)
	<i>gyrB</i>	4 (0.5 %)
	<i>pfbK</i>	1 (0.1 %)
Serogroup*	Icterohaemorrhagiae	140 (44.9 %)
	Sejroe	95 (30.5 %)
	Pomona	17 (5.5 %)
	Grippotyphosa	14 (4.6 %)
	Autumnalis	12 (3.8 %)
	Australis	7 (2.2 %)
	Canicola	7 (2.2 %)
	Tarassovi	6 (1.9 %)
	Panama	5 (1.6 %)
	Shermani	2 (0.6 %)
	Bataviae	3 (1 %)
	Pyrogenes	2 (0.6 %)
	Mini	1 (0.3 %)
	Sarmin	1 (0.3 %)
	Artiodactyla	293 (37.5 %)
Hosts (taxonomic order) [†]	Rodentia	156 (20 %)
	Carnivora	129 (16.5 %)
	Primates	126 (16.1 %)
	Didelphimorphia	24 (3 %)
	Chiroptera	20 (2.6 %)
	Americheylidia	11 (1.4 %)
	Perissodactyla	9 (1.2 %)
	Crocodylia	5 (0.6 %)
	Lagomorpha	4 (0.5 %)
	Squamata	3 (0.4 %)
	Pilosa	1 (0.1 %)
	Domestic large land animal	297 (36 %)
	Wildland animal	173 (21 %)
	Human	125 (15 %)
	Pet	114 (13.8 %)
Host/Source (ecological context)	Urban rodent	64 (7.6 %)
	Environment	44 (5.3 %)
	Marine mammal	11 (1.3 %)
Geographical region	Southeast	369 (44.7 %)
	Northeast	211 (25.6 %)
	North	183 (22.2 %)
	South	62 (7.5 %)
	Atlantic Forest	575 (69.7 %)
Biome	Amazon	183 (22.2 %)
	Pampa	48 (5.8 %)
	Atlantic Ocean	11 (1.3 %)
	Cerrado	8 (1 %)
	Rural	303 (36.7 %)
Landscape	Urban	287 (34.8 %)
	Sylvatic	140 (17 %)
	Peri-urban	95 (11.5 %)

*Percentages for serogroups were calculated only from sequences with available serogroup information ($n = 312$)

[†] Percentages for hosts were calculated only from sequences from animals ($n = 781$), excluding those of environmental origin.

classes, orders, and species (both scientific and common names), the *Leptospira* species detected in each host, and the biome(s) where these interactions were recorded. Scientific names of host species were validated using the NCBI Taxonomy Browser (available at: <https://www.ncbi.nlm.nih.gov/taxonomy>).

Table 2

Hosts so far reported for pathogenic *Leptospira* in Brazil, based exclusively on records supported by genetic characterization (i.e., available nucleotide sequences of any taxonomically informative gene). Host class, order, and species (scientific and common name) are shown. Only leptospires identified at species level were included. The biomes where the bacteria were identified are also shown (represented by acronyms).

	<i>L. interrogans</i>	<i>L. noguchii</i>	<i>L. kirschneri</i>	<i>L. borgpetersenii</i>	<i>L. santarosai</i>
Mammals					
Artiodactyla					
<i>Bos taurus</i> (bovine)	Am, Af, Ce, Pa	Am, Af	Ce, Am	Am, Af, Pa	Am, Af
<i>Bubalus bubalis</i> (water buffalo)	Am			Am	
<i>Capra hircus</i> (domestic goat)		Af			Af
<i>Ovis aries</i> (sheep)	Af				Af
<i>Pontoporia blainvillei</i> (franciscana whale)	Ao				
<i>Rusa unicolor</i> (samba deer)	Af				
<i>Sotalia guianensis</i> (Guyana river dolphin)	Ao				
<i>Steno bredanensis</i> (rough-toothed dolphin)	Ao				
<i>Sus scrofa domestica</i> (pig)	Am, Af, Ce, Pa				Af
Carnivora					
<i>Arctocephalus australis</i> (South American fur seal)	Ao				
<i>Arctocephalus tropicalis</i> (subantarctic fur seal)	Ao				
<i>Canis lupus familiaris</i> (dog)	Am, Af, Pa	Af, Pa	Pa		Af
<i>Conepatus chinga</i> (Molina's hog-nosed skunk)	Pa				
<i>Felis catus</i> (cat)	Pa			Pa	
<i>Leopardus geoffroyi</i> (Geoffroy's cat)	Pa				
<i>Leopardus pardalis</i> (ocelot)	Af				
<i>Leopardus wiedii</i> (margay)	Pa				
<i>Lontra longicaudus</i> (neotropical river otter)				Pa	
<i>Lycalopex gymnocercus</i> (Pampas fox)	Pa				
Chiroptera					
<i>Artibeus lituratus</i> (great fruit-eating bat)	Am				
<i>Artibeus planirostris</i> (flat-faced fruit-eating bat)	Am				
<i>Carollia brevicauda</i> (silky Short-Tailed Bat)	Am				
<i>Desmodus rotundus</i> (common vampire bat)	Am				
<i>Glossophaga soricina</i> (Pallas's long-tongued bat)	Am				
<i>Lasiurus blossevillii</i> (Western red bat)	Pa				
<i>Molossus currentium</i> (Thomas's mastiff bat)	Pa				
<i>Molossus molossus</i> (Pallas's mastiff bat)	Pa				
<i>Molossus rufus</i> (black Molossus)	Pa				
<i>Myotis levis</i> (vesper bat)	Pa				
<i>Sturmira tilda</i> (tilda yellow-shouldered bat)	Am				
<i>Tadarida brasiliensis</i> (Brazilian free-tailed bat)	Pa				
<i>Uroderma bilobatum</i> (tent-building bat)	Am				
Didelphimorphia					
<i>Didelphis albiventris</i> (white-eared opossum)	Pa			Pa	
<i>Didelphis aurita</i> (big-eared opossum)	Af	Af			
<i>Marmosa demerarae</i> (long-furred woolly mouse opossum)	Am	Am			
<i>Marmosa regina</i> (short-furred woolly mouse opossum)	Am				
<i>Marmosa paraguayana</i> (Tate's woolly mouse opossum)	Af				
<i>Marmosops ocellatus</i> (spectacled slender opossum)	Am				
<i>Marmosops paulensis</i> (Brazilian slender opossum)	Af				
<i>Marmosops</i> sp. (opossum)					Am
<i>Metachirus nudicaudatus</i> (brown four-eyed opossum)					Am
<i>Monodelphis glirina</i> (Amazonian red-sided opossum)	Am				Am

(continued on next page)

Table 2 (continued)

	<i>L. interrogans</i>	<i>L. noguchii</i>	<i>L. kirschneri</i>	<i>L. borgpetersenii</i>	<i>L. santarosai</i>
<i>Monodelphis peruviana</i> (Peruvian short-tailed opossum)					Am
<i>Monodelphis</i> sp. (opossum)	Af				
Lagomorpha					
<i>Lepus europaeus</i> (European hare)	Pa				
<i>Oryctolagus cuniculus</i> (rabbit)	Pa				
Perissodactyla					
<i>Equus caballus</i> (domestic horse)	Af		Af		
Pilosa					
<i>Tamandua tetradactyla</i> (Southern tamandua)	Af				
Primates					
<i>Callithrix jacchus</i> (white ear-tufted marmoset)	Pa				
<i>Homo sapiens</i> (human)	Am, Af, Pa	Pa	Pa		Am
Rodentia					
<i>Akodon cursor</i> (Cursor grass mouse)	Af				
<i>Akodon montensis</i> (wood mouse)	Af				
<i>Akodon</i> sp. (mouse)	Af				
<i>Bucepattersonius</i> sp. (akodont)	Af				
<i>Cavia aperea</i> (Brazilian guinea pig)	Pa				
<i>Delomys altimontanus</i> (montane woodrat)	Af				
<i>Delomys dorsalis</i> (striped Atlantic forest rat)	Af				
<i>Hydrochoerus hydrochaeris</i> (capybara)	Pa		Af	Pa	Am, Af
<i>Mesocricetus auratus</i> (golden hamsters)	Pa				
<i>Mesomys hispidus</i> (spiny tree rat)		Am			
<i>Mus musculus</i> (house mouse)	Af				
<i>Neacomys musseri</i> (Musser's shrew mouse)	Am				
<i>Neacomys spinosus</i> (bristly mouse)	Am				
<i>Nectomys squamipes</i> (South American water rat)					Af
<i>Oligoryzomys microtis</i> (small-eared pygmy rice rat)	Am				
<i>Oligoryzomys nigripes</i> (black-footed pygmy rice rat)	Af				
<i>Oligoryzomys</i> sp. (rat)	Af				
<i>Oxymycterus</i> sp. (hociucudo)	Af			Af	
<i>Proechimys gardneri</i> (Gardner's spiny mouse)	Am	Am			
<i>Proechimys brevicauda</i> (short-tailed Spiny-rat)	Am				
<i>Rattus norvegicus</i> (Norway rat)	Af			Af, Pa	
<i>Rattus rattus</i> (black rat)	Af				
<i>Rhipidomys leucodactylus</i> (white-footed climbing mouse)		Am			
<i>Trinomys</i> sp. (spiny rat)	Af				
Reptiles					
Americhelydia					
<i>Kinosternon scorpioides</i> (scorpion mud turtle)	Am				
Crocodylia					
<i>Caiman latirostris</i> (brown-snouted caiman)	Af				
Squamata					
<i>Boa constrictor</i> (boa)	Af				
<i>Salvator merianae</i> (Tegu lizard)	Af				

Codes: Af: Atlantic Forest; Am: Amazon; Ao: Atlantic Ocean; Ce: Cerrado; Pa: Pampa. References to all data and detailed information are available in Supplementary Material 1.

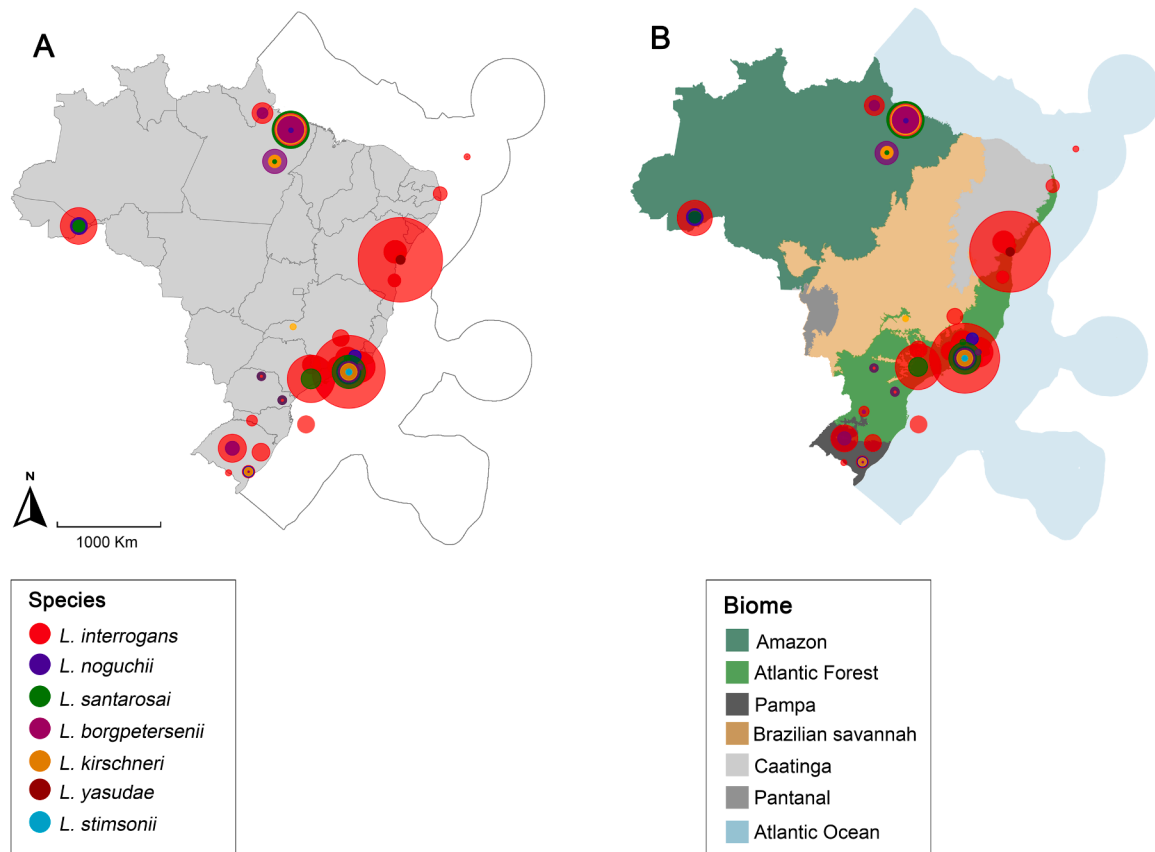


Fig. 2. Geographic distribution of pathogenic *Leptospira* species in Brazil. (A) Distribution by federative units. (B) Distribution by biomes. Colored rings represent different *Leptospira* species confirmed by nucleotide sequence data ($n = 825$). Circle size reflects the number of records per location.

2.4. Geospatial mapping of pathogenic *Leptospira* occurrence data in Brazil

To visualize the spatial distribution of genetically characterized pathogenic *Leptospira* species in Brazil, we generated thematic maps using Python 3.12 (<https://www.python.org/>) in a WSL (Windows Subsystem for Linux) environment. The analysis was performed using the following core packages: pandas (<https://pandas.pydata.org/>) for data manipulation, geopandas (<https://geopandas.org/>) for spatial operations and shapefile handling, shapely (<https://shapely.readthedocs.io/>) for geometric construction of point data, and matplotlib (<https://matplotlib.org/>) for map visualization and figure export.

A curated spreadsheet was compiled, containing three columns: (1) *Leptospira* species name, (2) latitude, and (3) longitude of the municipality where the species was identified. All entries corresponded to genetically confirmed isolates, and geographic coordinates were manually cross-referenced and standardized based on official IBGE data (<https://www.ibge.gov.br/>). When precise coordinates were not provided, the centroid of the corresponding municipality or state was used as a proxy.

Two types of maps were constructed: one overlaid on the political boundaries of Brazilian states, allowing for interpretation in terms of public health jurisdictions, and another overlaid on biome regions, to emphasize the environmental and ecological distribution patterns of the species. For the first map, we used the official shapefile of Brazilian federative units, including CMS, downloaded from the official IBGE geographic data server (FTP repository) (Available at <https://ftp.ibge.gov.br/>).

The second visualization was generated using the official shapefile of Brazilian biomes and CMS, also obtained from IBGE. All geospatial data, including the shapefiles of Brazilian states and biomes, were

standardized to the WGS84 geographic coordinate reference system (EPSG:4326). It expresses locations in decimal degrees of latitude and longitude and ensures compatibility across datasets. In both maps, each pathogenic *Leptospira* species was assigned a unique color, and the size of the circles was scaled proportionally to the number of occurrences using a multiplicative factor ($\text{count} \times 60$) to enhance visual discrimination. To ensure visual consistency, the same circle scaling factor was applied across both maps. As the data were obtained by convenience sampling, the aim of this visualization was not to infer prevalence or abundance, but rather to highlight the ecological range and known distribution of pathogenic *Leptospira* in Brazil, based on genetically validated records.

3. Results

3.1. Dataset composition and overview of pathogenic *Leptospira* circulation

A total of 825 genetically characterized sequences of pathogenic *Leptospira* were included in the final dataset, including those from the genes *secY* ($n = 546$; 66.2 %), 16S rRNA ($n = 120$; 14.6 %), *lipL32* ($n = 38$; 4.6 %), *gyrB* ($n = 4$; 0.5 %), and *pfkB* ($n = 1$; 0.1 %) (Table 1). Additionally, data from MLST ($n = 17$; 2 %) schemes and WGS ($n = 99$, 12 %) were also included. Each record was linked to a specific host or environmental source and a defined geographical location within Brazil. Detailed information is provided in Supplementary Material 1, and general statistical data are shown in Table 1.

The sequences comprised 7 formally identified species, including *L. interrogans* ($n = 618$; 74.9 %), *L. santarosai* ($n = 81$; 9.8 %), *L. borgpetersenii* ($n = 59$; 7.2 %), *L. noguchii* ($n = 32$; 3.9 %), *L. kirschneri* ($n = 32$; 3.9 %), *L. yasudae* ($n = 2$; 0.2 %), and *L. stimsonii* ($n = 1$; 0.1 %).

Regarding serogrouping, only 312/ 825 (37.8 %) of deposits had a defined serovar and/or serogroup. From these, Icterohaemorrhagiae was the most frequent serogroup ($n = 140$; 44.9 %), followed by Sejroe ($n = 95$; 30.5 %), Pomona ($n = 17$; 5.5 %), Grippotyphosa ($n = 14$; 4.6 %), and Autumnalis ($n = 12$; 3.8 %). Other serogroups were identified, but in a few isolates (Table 1; Supplementary Material 1).

Regarding host/source, the majority of sequences were from domestic large land animals ($n = 297$; 36 %), followed by wild land animals ($n = 173$; 21 %) and humans ($n = 125$; 15 %). Pets (cats and dogs) were identified in 114 deposits (13.8 %), while urban rodents and marine mammals were hosts from 64 (7.6 %) and 11 (1.3 %) of pathogenic *Leptospira*, respectively. Samples from environmental sources, like puddle, sewage, and soil, totaled 44 records (5.3 %) (Table 1; Supplementary Material 1).

The Atlantic Forest was the biome with the largest number of pathogenic *Leptospira* records ($n = 575$; 69.7 %), with emphasis on large urban centers, such as Salvador, Rio de Janeiro, and São Paulo, located in the Northeast and Southeast regions, which had the highest number of records (44.7 % and 25.6 %, respectively). The regions North and South of the country totaled 183 (22.2 %) and 62 records (7.5 %), respectively. Amazon, Pampa, and Cerrado totaled 183 (22.2 %), 48 (5.8 %), and 8 (1 %) occurrences, respectively. Eleven marine mammals were identified as hosts on the Southern Brazilian coast. No records were found in the Central-West region, and in the Pantanal and Caatinga biomes. In terms of landscape, most records originated from the rural ($n = 303$; 36.7 %) and urban ($n = 287$; 34.8 %) areas, followed by sylvatic ($n = 140$; 17 %) and peri-urban ($n = 95$; 11.5 %) environments (Table 1; Supplementary Material 1). Overall, the integrated dataset reveals that *L. interrogans* was predominantly detected in urban environments, whereas *L. borgpetersenii*, *L. kirschneri*, and *L. noguchii* were more frequently reported in rural areas. *Leptospira santarosai* showed a more balanced distribution, occurring predominantly in rural and peri-urban settings. No records of *L. noguchii* were found in peri-urban environments, and *L. kirschneri* was not detected in sylvatic areas.

3.2. Host range of *Leptospira* species

In total, the data encompassed isolates/samples from 78 vertebrate host species, including terrestrial, arboreal, aquatic, and synanthropic mammals, as well as a few reptile species. The majority of detections occurred in mammals, encompassing nine different taxonomic orders, including both domestic and wild species. Artiodactyls, rodents, carnivores, and primates were the most common hosts (Table 1, Table 2, Supplementary Material 1).

Among all hosts, *Bos taurus* (domestic cattle) exhibited the highest *Leptospira* species richness, harboring five different species: *L. interrogans*, *L. noguchii*, *L. kirschneri*, *L. borgpetersenii*, and *L. santarosai*. Other hosts with high species diversity included domestic dogs (*Canis lupus familiaris*), associated with four species, and capybaras (*Hydrochoerus hydrochaeris*), with three species (Table 2, Supplementary Material 1).

Several wild mammals were found to carry multiple *Leptospira* species. Notably, marsupials of the genera *Didelphis*, *Marmosa*, and *Monodelphis*, as well as various small rodents such as *Akodon*, *Delomys*, *Oligoryzomys*, and *Proechimys*, were recurrent hosts, primarily associated with *L. interrogans* and *L. santarosai*. Additionally, a notable number of host-pathogen associations involved bat species, which represented a taxonomically and ecologically diverse group within the dataset. At least 14 bat species from multiple genera, such as *Artibeus*, *Carollia*, *Glossophaga*, *Molossus*, *Myotis*, *Desmodus*, and *Tadarida*, were found infected with *Leptospira*, predominantly *L. interrogans*. These species included frugivorous, nectarivorous, and hematophagous bats (Table 2, Supplementary Material 1).

In addition to terrestrial mammals, some marine mammals, such as the Franciscana dolphin (*Pontoporia blainvillei*) and fur seals (*Arctocephalus* spp.), were also found infected with *L. interrogans*. Other occasional detections in non-mammalian hosts included reptiles like the boa,

tegu lizard, and caiman (Table 2, Supplementary material 1).

3.3. Geographic distribution of *Leptospira* species

Pathogenic *Leptospira* species have been genetically reported in 11 states, including four Brazilian biomes and the coastal marine environments, with marked variation in both species' richness and spatial distribution. In addition to the five species previously reported in the hosts, *L. yasudae* and *L. stimsonii* have been reported from environmental sources in Bahia and Rio de Janeiro, respectively, both in the Atlantic Forest. Overall, *L. interrogans* was the most widely distributed species, present in all biomes and across the largest number of states. Other species exhibited more localized patterns, typically restricted to fewer regions (Fig. 2).

The highest concentration of records was observed in the Southeastern and Southern regions of the country, particularly in the states of São Paulo, Rio de Janeiro, and Rio Grande do Sul. These areas exhibited not only the largest number of records but also the highest diversity of *Leptospira* species per location. A high number of *L. interrogans* occurrences was also recorded in the Northeastern state of Bahia, notably around the city of Salvador. In contrast, central regions of the country showed none or fewer records, and the northern region presented more scattered detections (Fig. 2A).

Regarding the biomes, the Atlantic Forest harbored the greatest number of occurrences, with records of all major *Leptospira* species. The Amazon biome also presented a high frequency of detections, particularly of *L. interrogans*, *L. borgpetersenii*, and *L. santarosai*. The Pampa biome showed a remarkable diversity of *Leptospira* species, despite its lower representation compared to the Atlantic Forest and Amazon. The Cerrado had only a few records, while the Pantanal and Caatinga showed no data so far. In addition, several occurrences were located in the marine domain, along the Atlantic Ocean coast, particularly involving *L. interrogans* in marine mammals (Fig. 2B).

Overall, the dataset revealed clear ecological and spatial patterns in the distribution of pathogenic *Leptospira* across Brazil. *Leptospira interrogans* predominated at urban and peri-urban interfaces, particularly in synanthropic and domestic hosts, reflecting its adaptation to anthropized environments and rodent-associated cycles. In contrast, *L. santarosai*, *L. noguchii*, and *L. kirschneri* were more frequent in rural and wildlife contexts, spanning diverse biomes and host groups. Cattle and dogs emerged as major multi-host bridges, connecting urban, rural, and sylvatic transmission networks.

4. Discussion

4.1. Overview of species, hosts, and ecological contexts of *Leptospira* in Brazil

Based on an extensive survey and analysis of genetic and ecological data of *Leptospira* spp. in Brazil, this study compiled the most comprehensive molecular datasets of this pathogen in Brazil to date, encompassing 825 sequences from multiple genetic targets, including *secY*, 16S rRNA, *lipL32*, *rpoB*, MLST loci, and whole genome sequences (Table 1, Supplementary Material 1). The sequences were deposited between 2009 and 2025, reflecting the progressive adoption of molecular tools for leptospirosis surveillance in Brazil in the last decades. The inclusion of diverse markers allowed for broader taxonomic representation and geographic coverage across hosts and environments. This approach reflects both the historical progression of molecular diagnostics in leptospirosis and the growing availability of sequence data from various surveillance and research efforts. On the other hand, this multi-locus dataset highlights an important gap in standardization, which underscores the need for coordinated surveillance and research frameworks adopting molecular targets with both taxonomic and epidemiological relevance, as previously proposed (Di Azevedo and Lilenbaum, 2021). Noteworthy that the data analyzed in this study do

not reflect the actual prevalence of *Leptospira* infection in Brazil, as only records with available species-level genetic sequences were included. Numerous studies based on serological assays or non-sequenced molecular detections were excluded, suggesting that the true geographic and host distribution of pathogenic *Leptospira* in the country is likely much broader than depicted here. Additionally, the dataset exhibits sampling biases inherent to the available molecular records, with a predominance of secY sequences and isolates from the Atlantic Forest region. This uneven representation likely reflects methodological preferences and the concentration of molecular research infrastructure in southeastern Brazil. Such heterogeneity may influence perceived patterns of genetic diversity and should be considered when interpreting ecological and phylogenetic trends.

Despite the known diversity of pathogenic *Leptospira* species (Vincent et al., 2019), the predominance of *L. interrogans* (74.9 %) (Table 1) mirrors its global importance as the most frequent agent of animal and human disease (Rajapakse et al., 2025). In Brazil, it is the most widespread species, including wild, rural and urban scenarios (da Cunha et al., 2016; de Albuquerque et al., 2017a; Jorge et al., 2018; Paz et al., 2019; Rodamilans et al., 2020; Siqueira et al., 2020; Ulsenheimer et al., 2022; Gomes et al., 2022; Freitas et al., 2024; Di Azevedo et al., 2025). Other species, such as *L. santarosai*, *L. borgpetersenii*, *L. noguchii*, and *L. kirschneri*, were also present, although at lower frequencies (3–10 %), suggesting underreporting or reduced molecular surveillance in the hosts or regions where they may circulate more prominently.

Leptospira santarosai has been recognized as an important circulating species in the Americas (Chinchilla et al., 2023). It has been implicated in reproductive disorders in livestock (Di Azevedo et al., 2021; Aymée et al., 2022a), zoonotic transmission in tropical settings (Peláez Sanchez et al., 2016), as well as asymptomatic infections in dogs (Miotto et al., 2016) and wild animals (Moreno et al., 2016; Medeiros et al., 2020; Vieira et al., 2019).

Notably, the vast majority of *L. borgpetersenii* hosts are cattle, and the species has been considered an important agent of bovine genital leptospirosis (Aymée et al., 2024). Evidence of vertical transmission in cattle highlights its complex transmission pathways, which are strongly influenced by host density rather than environmental factors (Bulach et al., 2006). Infections in urban rodents, wild animals, and dogs have also been reported for this species (Moreno et al., 2016; Vieira et al., 2019; Ulsenheimer et al., 2024, 2025).

An outbreak in goats involving reproductive disorders has been associated with *L. noguchii* infection (Aymée et al., 2022b), and the species has also been reported in dogs, cows, and small mammals (Guedes et al., 2021; Loureiro et al., 2020; Medeiros et al., 2020; Sant'Anna et al., 2021). *Leptospira kirschneri* was associated with a human leptospirosis case in southern Brazil, caused by a strain genetically related to a canine isolate from the same region (da Cunha et al., 2016). Other hosts include horses, bovines, and capybaras (Hamond et al., 2016; Jaeger et al., 2019; Guedes et al., 2021). Interestingly, the occurrence of the recently described environmental pathogenic species *L. yasudae* and *L. stimsonii* (Casanovas-Massana et al., 2020), although rare ($n = 3$), reinforces the potential for overlooked diversity within the genus and supports recent taxonomic updates derived from genomic analyses. Similarly, studies conducted in the Amazon region have genetically identified potentially new *Leptospira* species circulating in bats and cattle, although further studies are necessary for their formal description and validation (Guedes et al., 2019; Di Azevedo et al., 2023; Verde et al., 2024).

Regarding serogroup information, <40 % of the sequences were linked to a defined serovar and/or serogroup (Supplementary Table 1). This low proportion reflects not a failure in the deposit process or metadata curation, but rather a methodological limitation related to the need for viable bacterial isolates to perform the Microscopic Agglutination Test (MAT), the current standard for serogroup identification. Since *Leptospira* spp. are notoriously fastidious and slow-growing organisms, bacterial culturing remains a major bottleneck in routine

diagnostics and surveillance. Moreover, the MAT requires a well-equipped laboratory, skilled personnel, and access to a panel of live reference cultures, which are often unavailable in low-resource settings (Ciurariu et al., 2025). These constraints limit the widespread application of serological classification, especially in remote or resource-limited regions. Among the classified records, the predominance of the serogroup Icterohaemorrhagiae (Table 1) is consistent with its well-documented global and national distribution, being the most commonly reported serogroup in both human and canine acute leptospirosis cases (Hagedoorn et al., 2024). As a rodent-adapted serogroup, its high detection rates are typically linked to urban transmission cycles maintained by *R. norvegicus*. However, the occurrence of serogroup Icterohaemorrhagiae detected in the genital tract of cattle and horses represents an unexpected finding, raising important questions about its potential role in genital leptospirosis and sexual transmission, and revealing an underexplored epidemiological interface. Several hypotheses could explain this observation, including cross-species adaptation or spillover from rodent reservoirs, host–pathogen coevolution leading to niche expansion, or even diagnostic artifacts resulting from serological cross-reactivity or environmental contamination. Regardless of the underlying mechanism, this finding underscores the need for targeted molecular and experimental studies to confirm tissue tropism, assess reproductive transmission potential, and elucidate the evolutionary dynamics of Icterohaemorrhagiae in non-traditional hosts. Notably, this contrasts with the expected host–serogroup associations, as Icterohaemorrhagiae is typically linked to acute infections and urban transmission, whereas the predominance of Sejroe-related records remains consistent with its recognized role in livestock infections, particularly in cattle, and its known involvement in chronic, reproductive forms of leptospirosis (Loureiro and Lilenbaum, 2020).

The spatial patterns observed highlight the ecological differentiation among *Leptospira* species. The predominance of *L. interrogans* in urban environments supports its adaptation to synanthropic cycles involving rodents and domestic animals, commonly associated with human infection risk in densely populated areas (Bradley and Lockaby, 2023). In contrast, the higher occurrence of *L. borgpetersenii*, *L. kirschneri*, and *L. noguchii* in rural landscapes reflects their maintenance in livestock and wildlife hosts, consistent with previous reports of host-associated transmission in agricultural settings (Guedes et al., 2019; Soares et al., 2020). The intermediate distribution of *L. santarosai* across rural and peri-urban areas suggests ecological flexibility and potential overlap between domestic and sylvatic transmission networks (Chinchilla et al., 2023). Together, these findings underscore how landscape heterogeneity influences *Leptospira* circulation and host connectivity across Brazil. However, they should be interpreted cautiously, as the dataset results from opportunistic submissions and unequal research efforts across regions and host groups.

4.2. Host range and ecological roles in *Leptospira* transmission

The host distribution in the dataset demonstrates the wide host range of pathogenic *Leptospira* in Brazil, with sequences retrieved from a diverse array of domestic animals (cattle, pigs, goats, sheep, and horses), wildlife (small mammals, bats, large carnivores), pets (dogs and cats), and humans (Table 1). From a taxonomic point of view, 78 host species were identified, including two classes (mammals and reptiles) from 12 distinct orders (Tables 1 and 2). Of these, the most notable were Artiodactyla (ungulate mammals, like ruminants and pigs), Rodentia (rodents), Carnivora (mainly dogs), and Primates (humans) (Tables 1 and 2). This diversity reinforces the classical concept of leptospirosis as a multi-host zoonosis, in which transmission cycles are maintained by a wide variety of species with differing epidemiological roles (Belsare et al., 2022). Similar host diversity has also been reported in studies from other endemic regions, underscoring the global relevance of this ecological plasticity (Dietrich et al., 2018; Ferreira et al., 2020; Motto et al., 2021; Sluydts et al., 2022; Wainaina et al., 2024). From a One

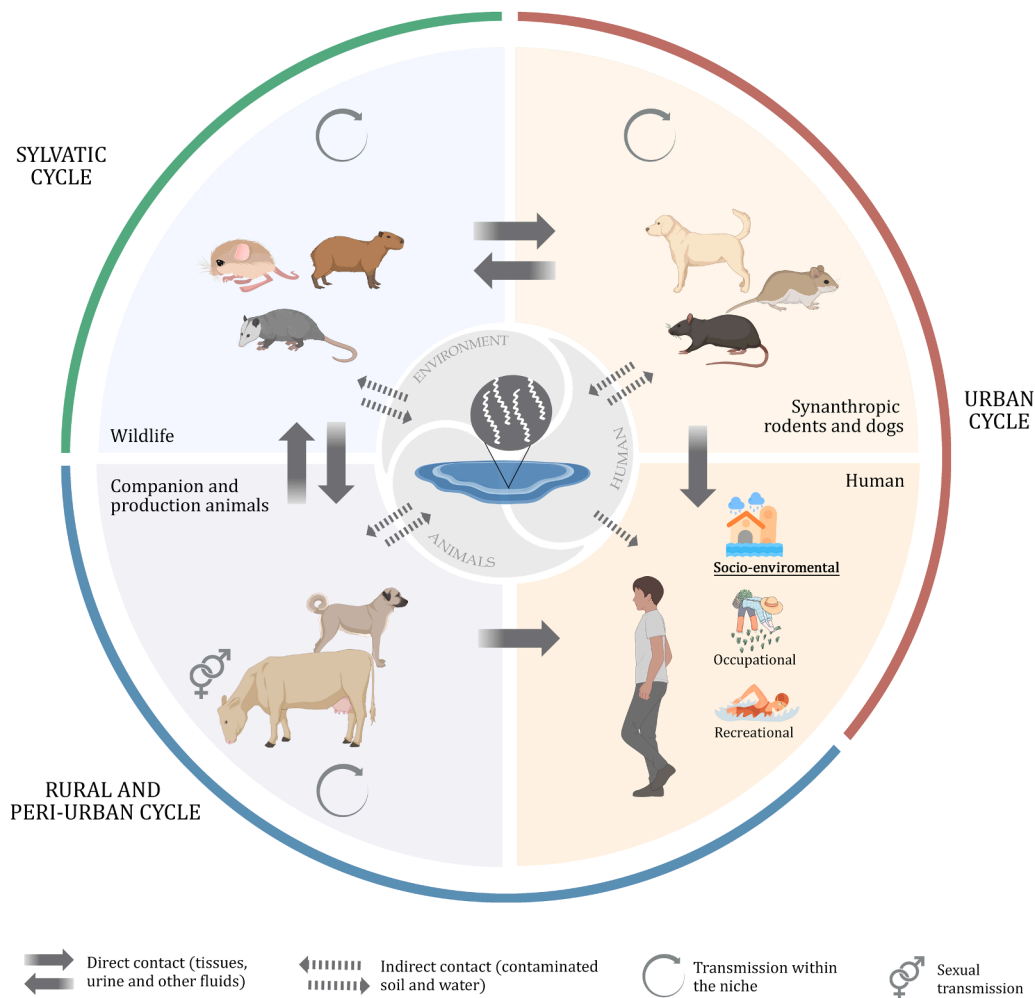


Fig. 3. Epidemiological cycle of *Leptospira* spp. in a tropical area from a One Health perspective. Schematic representation highlighting the interaction between wild animals, synanthropic rodents, companion and production animals, humans, and the environment. The main components of the wild, rural/peri-urban, and urban cycles are shown. Arrows indicate the main transmission routes: direct (contact with tissues, urine, and other fluids), indirect (contact with contaminated soil and water), and sexual (cattle). The contexts of human exposure are shown: socio-environmental (primary), occupational, and recreational. It is important to note that humans are accidental hosts and have no epidemiological significance in the bacterial cycle (notice that the arrows are one-way).

Health perspective, the observed host spectrum reveals potential maintenance and spillover dynamics among different ecological compartments. Cattle and other domestic livestock, along with synanthropic and wild rodents, likely act as primary maintenance reservoirs, sustaining endemic transmission (Martins and Lilenbaum, 2015). Conversely, dogs and humans are more often incidental hosts, reflecting spillover events facilitated by shared environments, water sources, or contaminated soil (Bierque et al., 2020).

Among the host species identified with genetically confirmed *Leptospira* infection in Brazil, a subset appears to represent exclusive or nearly exclusive records for the country, with no equivalent molecular documentation elsewhere. These include forest-dwelling small mammals such as *Bucepattersonius* spp., *Delomys* spp., *Akodon cursor*, *Akodon montensis*, *Monodelphis glirina* and *Trinomys* spp., most of which are endemic to Brazilian biomes (<https://www.gbif.org>) and rarely included in global leptospirosis surveys. This high number of host species exclusively reported in Brazil highlights the country's pivotal role in expanding the global understanding of *Leptospira* ecology, especially in wildlife. Noteworthy, Brazil's exceptionally rich biodiversity and ecological heterogeneity support a broader range of potential reservoir hosts than is typically encountered in temperate or more homogenized environments. These findings reinforce the idea that forest-dwelling small mammals serve as important enzootic reservoirs in sylvatic

transmission cycles, a pattern also seen in parts of South and Central America (Benavidez et al., 2019; Souza Rocha et al., 2020; Silva-Ramos et al., 2025), but less documented in temperate zones. Conversely, many hosts identified in this study, such as cattle, dogs, rats, and horses, have also been reported as leptospiral carriers in other tropical areas, evidencing extensive evidence of their roles as reservoirs beyond Brazil (Orjuela et al., 2022; Andrade-Silveira et al., 2024).

The predominance of sequences from domestic large land animals, particularly livestock such as cattle and pigs, reflects their epidemiological relevance as key reservoirs of pathogenic *Leptospira*. This finding aligns with previous studies indicating high infection rates and chronic carriage in these species (Fernandes et al., 2020; Guedes et al., 2021; Gomes et al., 2022), which not only sustain transmission cycles within animal populations but also pose a direct zoonotic threat to humans through occupational and environmental exposure (Alinaitwe et al., 2019). The substantial proportion of sequences from wild animals highlights the importance of wildlife in maintaining leptospiral diversity and facilitating spillover events, especially in ecotones or areas of habitat fragmentation (Helman et al., 2023; Verde et al., 2024). Conversely, the low number of human-derived sequences likely reflects limited molecular surveillance and underestimates the true burden, given the diversity of animal reservoirs and environmental conditions that favor *Leptospira* transmission in Brazil (Browne et al., 2022).

Likewise, urban rodents represented only 7.6 % of the records, which is inconsistent with their recognized role as key reservoirs in urban leptospirosis cycles (Soni et al., 2024). This underrepresentation likely stems from sampling and reporting biases, as molecular data on rodents are often scarce, localized, or not publicly deposited. It underscores the need for expanded genomic surveillance of rodent populations, especially in urban settings.

Importantly, 114 records (13.8 %) were associated with dogs, underscoring their growing recognition as key players in the epidemiology of leptospirosis. Recent studies have highlighted the role of dogs not only as incidental hosts, but also as potential maintenance hosts in urban and peri-urban areas, where they may act as a bridge between rodents, humans, and other domestic animals (Sant'anna et al., 2017; Santos et al., 2021; Paz et al., 2025). The relevance of canine leptospirosis has been further supported by national seroprevalence studies, which report consistent exposure to serogroups commonly associated with disease in humans and dogs, particularly Icterohaemorrhagiae, even in asymptomatic animals (Pinheiro et al., 2022; Ricardo et al., 2024).

Notably, the dataset also revealed host–pathogen associations in bats, with at least 14 species found infected. These included frugivorous (*Artibeus*, *Carollia*), nectarivorous (*Glossophaga*), insectivorous (*Tadarida*, *Molossus*, *Myotis*), and hematophagous (*Desmodus*) taxa. The predominance of *L. interrogans* suggests widespread exposure in this group, regardless of feeding habits. On the other hand, a high diversity of *Leptospira* species was found in bats from Colombia, in the Magdalena-Caribbean tropical humid zonobiome (Monroy et al., 2021). Recent studies in Brazil have highlighted bats as emerging reservoirs of *Leptospira*, particularly in the periurban and forest-edge environments, with high infection rates in Amazonian and Atlantic Forest regions (Di Azevedo et al., 2023; Verde et al., 2024). Their mobility and capacity to occupy both natural and anthropized habitats make them epidemiologically relevant for spillover and dispersal events, especially in light of global concerns around bat-borne zoonoses.

Although not an exclusive species of Brazil, records of active infection in capybaras by *Multiple species of Leptospira* are, to date, documented mainly in Brazil. These largest living rodents are highly adapted to semi-aquatic environments, emerging as an epidemiologically significant host in the country (da Silva et al., 2023; Ulsenheimer et al., 2024). In this study, they were found carrying four distinct *Leptospira* species (*L. interrogans*, *L. kirschneri*, *L. borgpetersenii*, and *L. santarosai*) across multiple biomes, reinforcing their role in both sylvatic and synanthropic transmission cycles. Their gregarious behavior, frequent use of shared water bodies, and increasing presence in urban, periurban, and agricultural landscapes facilitate environmental contamination and cross-species transmission, particularly with livestock and humans. Unlike similar rodents in other regions, capybaras demonstrate a consistent association with leptospiral shedding, and may act as amplifier hosts, sustaining high bacterial loads and contributing to the environmental persistence of the pathogen (de Albuquerque et al., 2017b).

The identification of *Leptospira* in marine mammals ($n = 11$) highlights an emerging and understudied component of the epidemiological cycle, particularly in coastal and estuarine ecosystems. These findings align with growing evidence that aquatic mammals, including pinnipeds, cetaceans, and manatees, can be exposed to and infected by pathogenic *Leptospira*, likely through contact with contaminated freshwater runoff, sewage discharge, or infected prey (Torres et al., 2025). In this context, marine mammals may serve not only as incidental hosts but also as effective sentinels of environmental contamination and ecosystem health. Their infections can signal broader ecological risks, including the potential spillover to humans and domestic animals in coastal communities where human–wildlife–environment interfaces are intensifying.

Taken together, these patterns highlight distinct functional roles among host groups in *Leptospira* transmission networks. Livestock

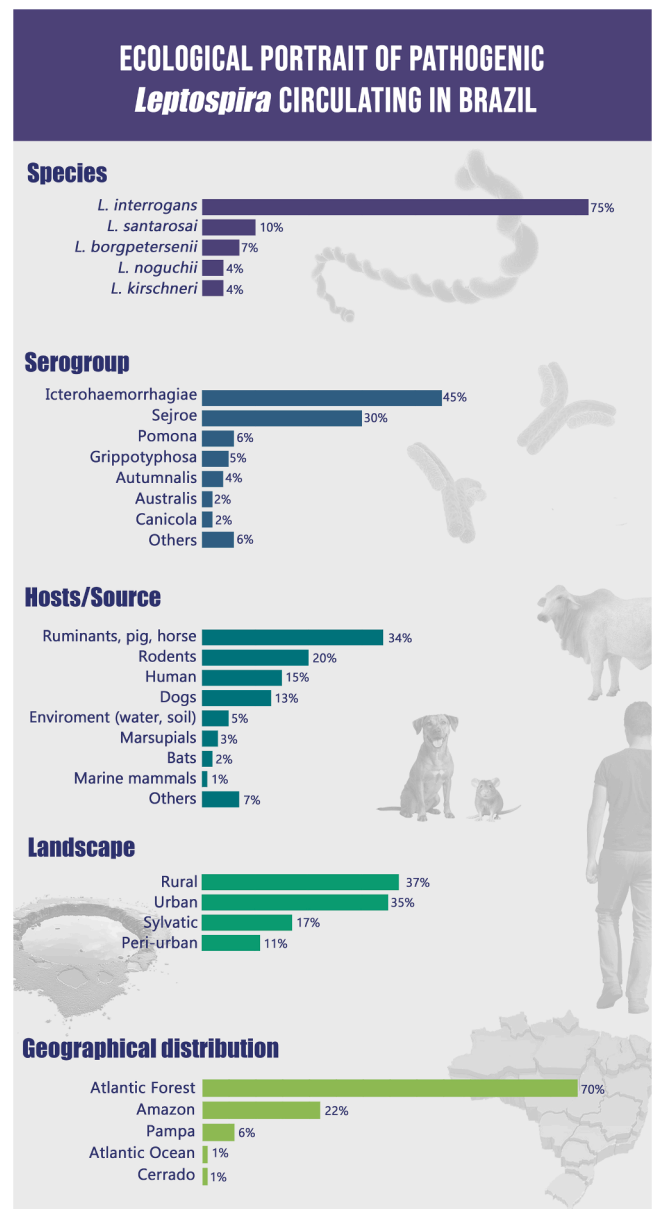


Fig. 4. Infographic summarizing the main characteristics of pathogenic *Leptospira* circulating in Brazil based on genetic, ecological, and geographic data compiled in this study. The figure integrates information already presented in the tables and text, providing a visual overview for didactic purposes and improved interpretation of the results. These data are limited to *Leptospira* spp. with available genetic sequences in public databases and provide only a partial snapshot of the bacterium's impact and characteristics in Brazil.

species such as cattle and pigs primarily sustain enzootic cycles in rural environments, while synanthropic and companion animals, particularly dogs and rodents, act as bridge hosts facilitating pathogen spillover to humans. Recognizing these transmission interfaces is essential for prioritizing control and surveillance strategies under the One Health framework.

4.3. Geographic distribution and biome-specific patterns of *Leptospira* circulation

The spatial analysis revealed marked heterogeneity in the distribution of pathogenic *Leptospira* species across Brazil, highlighting both regional hotspots and ecological gradients (Table 1, Table 2, Fig. 2).

While *L. interrogans* demonstrated a broad geographic range, detected in all major biomes and landscapes, coastal environments, and across 11 states, other species exhibited more localized distributions, suggesting ecological or surveillance-related constraints. The Southeast and South regions, particularly the states of São Paulo, Rio de Janeiro, and Rio Grande do Sul, presented the highest number of records and the greatest species richness per location, mainly in urban and rural areas (Table 1, Fig. 2A). It likely reflects a combination of factors, including greater research infrastructure, historical investment in surveillance, urban population density, and environmental conditions conducive to transmission, such as frequent flooding and rodent infestation (Nardoni Marteli et al., 2022; Michelon et al., 2025). These regions are also sites of major zoonotic outbreaks, often linked to *L. interrogans* serogroup Icterohaemorrhagiae, particularly in urban areas with inadequate sanitary conditions (Costa et al., 2022b; Teles et al., 2023; Linhares et al., 2024). Interestingly, Salvador, in the Northeast region, also showed a high number of *L. interrogans* detections. This corroborates epidemiological data indicating recurrent outbreaks in this region and suggests that *Leptospira* transmission is not confined to southern urban centers, but also affects tropical coastal cities, where socioenvironmental vulnerabilities persist (Nabity et al., 2020; Khalil et al., 2021; de Oliveira et al., 2024).

From an ecological perspective, the Atlantic Forest biome stood out as a major hotspot, with occurrences of all major pathogenic species, including the rarely reported *L. yasudae* and *L. stimsonii* (Fig. 2B). The high diversity in this biome likely results from its complex mosaic of urban, rural, and forested environments, enabling overlapping transmission cycles among domestic animals, wildlife, and humans (Martins and Lilenbaum, 2015). Likewise, the Amazon biome also harbored a high frequency of *Leptospira* species, particularly *L. interrogans*, *L. santarosai*, and *L. borgpetersenii*, reflecting both its biodiversity and the growing body of studies focused on sylvatic hosts in the region (Guedes et al., 2019; Medeiros et al., 2020; de Souza Rocha et al., 2022; Gomes et al., 2022; Di Azevedo et al., 2023; Verde et al., 2024). Of note, the Pampa biome, despite a lower number of records, showed a considerable species diversity. This may reflect unique ecological dynamics in southern grassland ecosystems, including interactions between livestock and wild mammals (Michel and Overbeck, 2024). By contrast, central and western regions of the country, including the Cerrado, Caatinga, and Pantanal biomes, were notably underrepresented or absent in the dataset. This likely reflects surveillance gaps rather than the true absence of circulation. Despite the absence of deposited genetic sequences, *Leptospira* sp. has been previously molecularly and/or serologically identified in animals and humans from these regions (Vieira et al., 2016; Galan et al., 2023; Gomes et al., 2024; de Lisboa Padilha et al., 2025). These findings underscore a geographic and methodological bias in available molecular data and emphasize the need to expand and standardize surveillance efforts in understudied regions.

4.4. Environmental reservoirs and transmission cycles

Beyond hosts and geographic distribution, our dataset also revealed the environmental component of *Leptospira* spp. transmission as a relevant yet underexplored dimension (Table 1; Supplementary Material 1). The identification of 44 sequences from environmental sources, including puddles, sewage, soil, and freshwater, underscores the ability of pathogenic *Leptospira* to persist in aquatic and semi-aquatic environments. Importantly, pathogenic leptospires are capable of forming biofilms on abiotic surfaces and sediments, a trait that enhances persistence by buffering cells against desiccation, UV exposure, nutrient limitation, and antimicrobial stresses (Ristow et al., 2008; Meganathan et al., 2022). Field investigations in Brazilian urban settings with poor sanitation, particularly Salvador (Bahia), have repeatedly detected pathogenic *Leptospira* in open sewers, floodwater, and street puddles (Casanovas-Massana et al., 2020, 2022), supporting the role of environmental matrices as maintenance and spillover interfaces for

rat-borne leptospirosis where biofilm-friendly conditions (moisture, organic load, slightly alkaline pH) prevail (Bierque et al., 2020; Davignon et al., 2023). Together, these experimental and field data indicate that environmental reservoirs, potentially structured as biofilms, play a crucial role in indirect transmission, especially in flood-prone urban areas and agricultural settings, where human and animal exposure increases during rainfall or water accumulation. This emphasizes the importance of environmental management and infrastructure improvements, such as sanitation, drainage, and flood-control measures, to reduce environmental contamination and human exposure in endemic areas.

To better illustrate the complex epidemiological scenario of leptospirosis in Brazil, we developed a conceptual diagram (Fig. 3) representing the main transmission routes and host interfaces. The diagram highlights the coexistence of distinct but interconnected cycles: (i) the urban cycle, dominated by synanthropic rodents and dogs; (ii) the rural and peri-urban cycle, involving livestock, companion animals, and humans in shared environments; and (iii) the wild cycle, where transmission occurs among native wildlife species. All these cycles interact dynamically through direct and indirect transmission routes, including contact with contaminated environments and, in some cases, sexual transmission. This multifaceted system reinforces the importance of integrative approaches that consider ecological, socio-environmental, and behavioral factors in leptospirosis surveillance and control.

To provide a visual synthesis of the main findings of the present study, we developed an infographic summarizing the key genetic, ecological, and geographic characteristics of pathogenic *Leptospira* species circulating in Brazil, as detailed in the tables and results presented above (Fig. 4).

5. Conclusions

This study provides the first nationwide, integrative molecular overview of pathogenic *Leptospira* circulating across Brazilian biomes, combining genetic, ecological, and epidemiological data from humans, domestic animals, wildlife, and environmental sources. We revealed a broad taxonomic and ecological host range, encompassing terrestrial, arboreal, aquatic, synanthropic, and even non-mammalian taxa. While the heterogeneous use of molecular markers and uneven geographic coverage reveal important gaps in surveillance, the dataset and analytical framework presented here provide a valuable foundation for standardized molecular surveillance and for guiding targeted interventions to reduce the burden of leptospirosis in tropical countries. It is important to highlight that this marker heterogeneity may limit direct species comparability and phylogenetic resolution and future studies should aim to harmonize marker selection by adopting standardized genes, such as *secY* and genome-wide approaches, which would enhance consistency and enable more robust evolutionary and epidemiological inferences across regions. The inclusion of environmental isolates further reinforces the One Health perspective, underscoring the role of multiple reservoirs and exposure pathways in leptospirosis epidemiology, particularly under tropical conditions. Strengthening this integrative approach will require coordinated surveillance and policy actions across human, animal, and environmental health sectors, fostering data sharing and early detection strategies to effectively mitigate leptospirosis transmission.

CRedit authorship contribution statement

Maria Isabel Nogueira Di Azevedo: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Walter Lilenbaum:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.actatropica.2025.107895](https://doi.org/10.1016/j.actatropica.2025.107895).

Data availability

Data will be made available on request.

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ARTIGO 5: Diversidade genética e expansão clonal de *Leptospira* patogênica no Brasil: um panorama multi-hospedeiro e multi-regional

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A leptospirose é uma zoonose de distribuição global, de grande relevância para a saúde pública e veterinária, causada por espécies patogênicas do gênero *Leptospira*. O Brasil representa um local crítico de transmissão devido à sua diversidade ecológica e às complexas interfaces hospedeiro-ambiente. Este estudo explorou a diversidade e a estrutura genética de espécies patogênicas de *Leptospira* spp. circulantes no Brasil por meio de uma análise de tipificação de sequência de locus único (SLST) baseada no gene *secY*. Um total de 531 sequências foram obtidas do GenBank e submetidas a análises filogenéticas e de diversidade de haplótipos. A reconstrução por máxima verossimilhança revelou clados fortemente suportados para sete espécies, sendo *L. interrogans* a mais prevalente e amplamente distribuída entre hospedeiros e regiões. Essa espécie apresentou evidências de expansão clonal, com um haplótipo dominante ($n = 242$) compartilhado por humanos, animais domésticos e animais silvestres. Em contraste, *L. santarosai* e *L. noguchii* exibiram alta diversidade haplotípica e estruturas de rede reticuladas, refletindo maior variabilidade evolutiva. As espécies *L. kirschneri* e *L. borgpetersenii* apresentaram variação haplotípica reduzida, sendo a última associada principalmente ao gado, o que condiz com seu perfil de adaptação ao hospedeiro. Redes de haplótipos baseadas em hospedeiros e biomas revelaram tanto a ampla adaptabilidade ecológica de certas linhagens quanto a presença exclusiva de haplótipos restritos a ambientes específicos, como os encontrados em mamíferos marinhos do Oceano Atlântico. Análises de distância genética confirmaram a forte resolução taxonômica do gene *secY*, que distinguiu efetivamente espécies intimamente relacionadas, ao mesmo tempo que capturou a diversidade intraespecífica. Esses achados fornecem uma visão molecular abrangente de leptospirosas patogênicas no Brasil, destacando a conectividade ecológica entre hospedeiros e biomas, bem como a dinâmica evolutiva contrastante entre as espécies. Além de descrever padrões genéticos, nossas análises enfatizam processos evolutivos, conectividade hospedeiro-ambiente e as implicações para a abordagem "Uma Só Saúde". Essa estrutura integrativa fortalece a base para estratégias de vigilância e controle em outras regiões endêmicas do mundo.



Article

Genetic Diversity and Clonal Expansion of Pathogenic *Leptospira* in Brazil: A Multi-Host and Multi-Regional Panorama

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Abstract

Leptospirosis is a globally distributed zoonosis of major public health and veterinary relevance, caused by pathogenic species of the genus *Leptospira*. Brazil is a hotspot for transmission due to its ecological diversity and complex host–environment interfaces. This study explored the genetic diversity and structure of circulating pathogenic *Leptospira* spp. in Brazil through a single-locus sequence typing (SLST) analysis based on the *secY* gene. A total of 531 sequences were retrieved from GenBank and subjected to phylogenetic and haplotype diversity analyses. Maximum likelihood reconstruction revealed strongly supported clades for seven species, with *L. interrogans* being the most prevalent and broadly distributed across hosts and regions. This species showed evidence of clonal expansion, with a dominant haplotype ($n = 242$) shared by humans, domestic animals, and wildlife. In contrast, *L. santarosai* and *L. noguchii* exhibited high haplotypic diversity and reticulated network structures, reflecting greater evolutionary variability. The species *L. kirschneri* and *L. borgpetersenii* displayed reduced haplotypic variation, the latter mainly associated with cattle, consistent with its host-adapted profile. Host- and biome-based haplotype networks revealed both the broad ecological adaptability of certain lineages and the exclusive presence of haplotypes restricted to specific environments, such as those found in marine mammals from the Atlantic Ocean. Genetic distance analyses confirmed the strong taxonomic resolution of the gene *secY*, which effectively distinguished closely related species while capturing intraspecific diversity. These findings provide a comprehensive molecular overview of pathogenic *Leptospira* in Brazil, highlighting ecological connectivity across hosts and biomes, as well as the contrasting evolutionary dynamics among species. Beyond describing genetic patterns, our analyses emphasize evolutionary processes, host–environment connectivity, and the implications for One Health. This integrative framework strengthens the basis for surveillance and control strategies in other endemic regions in the world.

Keywords: molecular epidemiology; phylogenetics; haplotype network; zoonosis; one health



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1. Introduction

Leptospirosis is a widespread zoonotic disease caused by pathogenic species of the genus *Leptospira*, which circulate among multiple animal hosts and persist in diverse ecological settings [1]. More than 160 mammalian species, along with several reptile and amphibian species, have been identified as hosts or reservoirs of the bacteria. These

animals shed the pathogen primarily through urine, releasing it into the environment and contaminating soil and water sources [2]. The primary route of human infection is indirect, occurring when mucous membranes (eyes, mouth, nose) or abraded skin come into contact with water, soil, or food contaminated by the urine of infected animals. Risk is heightened in humid environments, during flooding, or in waterlogged soils, which enhance the survival and dissemination of leptospires, particularly in urban settings with inadequate sanitation and high rodent densities [3]. Intense rainfall events can further mobilize leptospires from soil into surface waters, thereby amplifying human exposure [4]. Given its complex transmission network involving humans, animals, and the environment, leptospirosis exemplifies a quintessential One Health challenge, demanding integrated surveillance and control strategies across sectors [5].

Globally, leptospirosis represents one of the most widespread zoonoses of bacterial origin, with an estimated one million human cases annually and mortality exceeding 60,000 deaths per year. The disease burden is especially high in tropical countries of South America and Southeast Asia, where climatic conditions favor environmental persistence and host density is elevated [6]. Brazil represents one of the most important global hotspots for leptospiral transmission due to its ecological heterogeneity and complex human–animal–environment interfaces. The country encompasses six major biomes, Amazon, Atlantic Forest, Cerrado, Caatinga, Pantanal, and Pampa, each harboring distinct climatic regimes, wildlife communities, and anthropogenic pressures. This environmental mosaic provides ideal conditions for the coexistence of multiple *Leptospira* species maintained by both wild and domestic hosts [7]. In the country, human leptospirosis is endemic and often reaches epidemic levels during the rainy season, particularly in capital cities and metropolitan areas. Outbreaks are closely linked to flooding, inadequate waste management, and high rodent infestation, which together increase exposure in low-income communities. In rural settings, occupational contact with livestock, contaminated water and soil represent major risk factors [8]. Urban incidence is highest in the North and Northeast regions, whereas rural cases predominate in the Southeast and South [9].

Leptospira remains a taxonomically complex and genetically diverse genus, encompassing more than 60 species that display a continuum of pathogenicity and ecological adaptation [10]. Although several local studies have characterized *Leptospira* spp. from specific hosts or regions in Brazil, comprehensive nationwide assessments integrating phylogenetic, ecological, and host data are still scarce. Most available information is fragmented or limited to regional case studies, lacking a unified comparative framework across biomes. Such integrative molecular analyses are essential to elucidate transmission pathways, assess genetic variability within and between species, and understand the ecological connectivity of circulating lineages. In this context, single-locus sequence typing (SLST) has proven to be a particularly valuable approach, especially in situations where bacterial isolation is unfeasible and only clinical samples are available. This technique relies on a single, highly informative gene that provides sufficient resolution for both genetic and phylogenetic analyses and has already been successfully applied to other bacterial groups [11,12]. The *secY* gene has emerged as one of the most informative loci for molecular detection and classification of pathogenic *Leptospira*. This housekeeping gene, which encodes a component of the protein translocation channel, combines high interspecific variability with sufficient conservation to allow amplification from a wide range of clinical and environmental samples. Its sequence polymorphism enables both species-level identification and intraspecific differentiation, while maintaining strong phylogenetic congruence with multilocus typing and whole-genome approaches. Furthermore, the gene's broad availability in public databases facilitates large-scale comparative analyses and epidemiological surveillance [13,14].

In this study, we investigated the phylogenetic diversity and haplotype structure of pathogenic *Leptospira* spp. in Brazil by analyzing sequences derived from human and animal sources. By combining phylogenetic reconstruction, haplotype network inference, and genetic distance analyses, we aimed to characterize species-level diversity, host and biome associations, and evolutionary patterns of the main pathogenic taxa circulating in the country. This integrative approach provides new insights into the molecular epidemiology of *Leptospira* in a megadiverse tropical country and establishes a framework for One Health surveillance and control strategies.

2. Materials and Methods

2.1. Data Collection and Curation

Nucleotide sequences were obtained from the NCBI GenBank repository using the query “*Leptospira* AND Brazil AND *secY*” (accessed on 15 July 2025). We included only sequences with confirmed species-level identification and available metadata regarding host/source and geographic origin. Entries described solely as “*Leptospira* sp.” without species assignment were discarded, since accurate species identification is essential for molecular epidemiology analyses [15]. For this study, only pathogenic taxa belonging to the P1 clade, as proposed [10], were retained, while intermediate and saprophytic species were excluded. Each selected record was manually checked and annotated with key attributes: GenBank accession, species name, strain or sample identifier, host species, location of identification, and bibliographic reference. The curated dataset was organized in an Excel spreadsheet.

To allow broader ecological and epidemiological interpretation, host species were grouped into seven general categories: “Domestic large land animals”, “Wild land animals”, “Pets”, “Humans”, “Marine mammals” and “Urban rodents”. Likewise, sampling sites were assigned to one of the five main Brazilian biomes defined by the Brazilian Institute of Geography and Statistics (IBGE): Amazon, Atlantic Rain Forest, Pampa, Cerrado (Savannah), and Caatinga (semi-arid), according to the original collection site. The Atlantic Ocean was treated as a separate category (coastal-marine system, CMS).

2.2. Phylogenetic Analysis

Only *secY* sequences longer than 400 base pairs were retained to ensure reliable alignments and robust phylogenetic inference, as shorter fragments may reduce tree resolution and inflate homoplasy. For genome-derived records, *secY* sequences were extracted using the NCBI Genome Annotation tool, based on the corresponding GenBank or BioProject accession numbers. All command-line analyses were performed using the WSL terminal environment to ensure reproducibility and compatibility with bioinformatics tools. All *secY* sequences corresponded to a partial fragment of the gene, covering approximately positions 250–655 of the reference strain *L. interrogans* Fiocruz L1-130 (GenBank accession AE016823). After alignment and trimming, the final analyzed fragment spanned 405 base pairs (bp).

Sequences were aligned using MAFFT v7.505 [16] with default settings. Phylogenetic reconstruction was performed using IQ-TREE v2.2.6 [17], applying the Tamura–Nei (TN93) substitution model, as it was the best-fitting DNA substitution model determined by the Bayesian information criterion, with 1,000 ultrafast bootstrap replicates to assess branch support. A bootstrap value of $\geq 70\%$ was considered indicative of strong support. The maximum likelihood (ML) tree generated was subsequently imported into MEGA X v11.0.13 [18] for visualization and editing. Minor adjustments to tree layout, taxon labels, and branch display were made to improve clarity and produce the final version used for interpretation and figure export.

2.3. Haplotype Networks

To further explore genetic patterns, median-joining haplotype networks were constructed using PopArt v1.7 using the median-joining inference method [19], based on the aligned *secY* dataset for each *Leptospira* species. Two separate networks were generated: one grouping haplotypes by host category and another by biome (as previously defined). Annotation was performed via a custom NEXUS input file to assign colors and groupings.

For single-locus analyses such as this, identical *secY* sequences were grouped into unique haplotypes, defined as sets of identical nucleotide sequences within the alignment. The term “haplotype” is used here in its phylogenetic context to describe sequence clusters derived from a single genetic locus, rather than the multilocus “sequence types (STs)” used in conventional MLST schemes.

2.4. Estimation of Genetic Distances

To provide additional insights into the genetic cohesiveness of each species and the degree of divergence between taxa occurring in Brazil, intraspecific and interspecific genetic distances were calculated using MEGA X, based on the aligned *secY* sequences. Mean pairwise distances within and between species were calculated under the TN93 model. Standard error (SD) estimates were obtained by a bootstrap procedure (1000 replicates).

3. Results

3.1. Phylogenetic Relationships

A total of 531 sequences met the criteria and were included in the genetic analysis. Selected sequences and associated information are shown in Supplementary Material Table S1. The alignment spanned 405 bp, and the ML-TN93 tree recovered well-supported monophyletic clades for each recognized species, with bootstrap values exceeding 90% for most nodes. Notably, *L. interrogans* showed the highest number of total sequences and the greatest representation across multiple hosts and regions (Figure 1 and Supplementary Material Table S1). Moreover, it exhibited a remarkable clonal expansion represented by a predominant haplotype within the dataset comprising 242 sequences, including the reference strain Fiocruz L1-130. Other frequent haplotypes within *L. interrogans* also contained 21 and 11 sequences.

Leptospira noguchii, *L. kirschneri*, and *L. borgpetersenii* showed less diversity of haplotypes, with a predominance of one or two groups among 4 and 29 sequences, while *L. santarosai* presented several haplotypes formed by a maximum of 5 sequences, as well as presenting many unique sequences. The recently described species *L. yasudae* and *L. stimsonii* were represented by fewer sequences, but still formed distinct clusters, clearly indicating their genetic divergence from previously described species (Figure 1).

3.2. Haplotype Network Analysis by Host and Biome

The haplotype network based on *L. interrogans* sequences (Figure 2) revealed a total of 19 haplotypes, with one central haplotype comprising the majority of sequences ($n = 242$). This predominant haplotype (Fiocruz L1-130-like group) included representatives from several host categories, being practically equally distributed among humans, wild animals, domestic large land animals, and dogs, with few urban rodents. Other larger haplotypes, including about 20 sequences, are mostly from bovine origin. Small haplotypes, genetically close to the larger ones, were formed by sequences from marine mammals, dogs, and wild animals (Figure 2A).

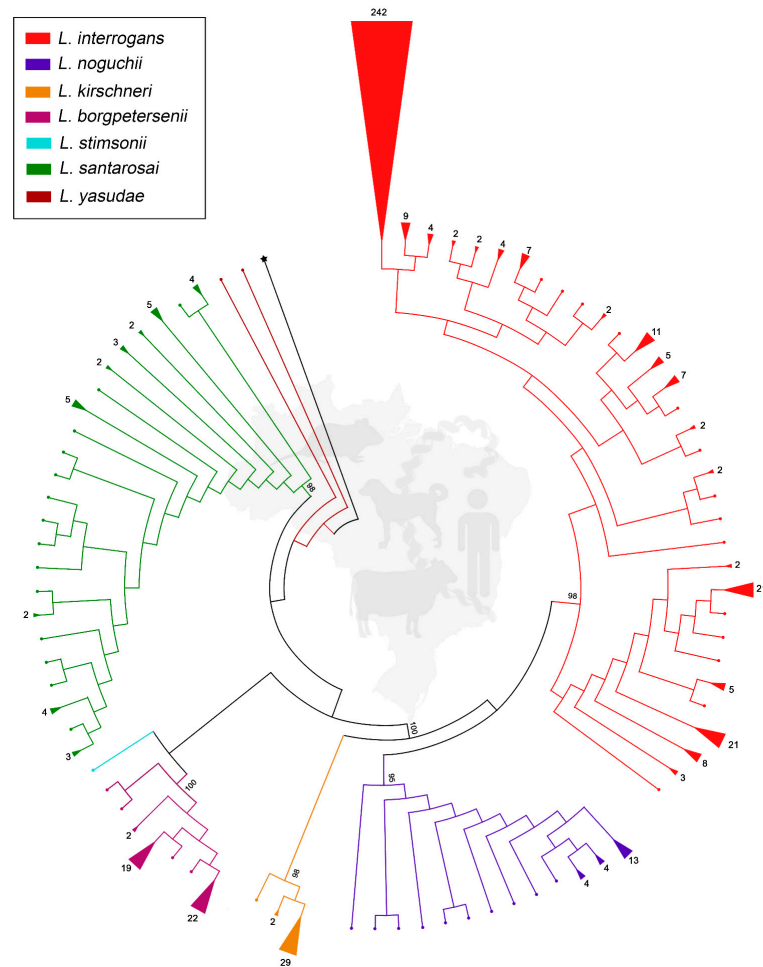


Figure 1. Maximum likelihood phylogenetic tree inferred from partial *secY* gene sequences ($n = 531$) of pathogenic *Leptospira* spp. identified in humans and animals in Brazil. Species identity is represented by colors, and triangle size is proportional to the number of identical sequences (number of sequences is shown). Small dots represent unique sequences. Numbers at nodes are bootstrap values. *Leptospira biflexa* strain Patoc is the outgroup taxon (represented by star).

In terms of geographic distribution (Figure 2B), the central haplotype encompassed sequences with absolute predominance of Atlantic Forest, with few from Amazon and Pampa. Several haplotypes were exclusive to the Atlantic Forest and Amazon. Sequences from the Atlantic Ocean formed three exclusive haplotypes, not shared with other biomes, such as that presented in the host network. The few sequences from Cerrado were identical to the sequences from Atlantic Forest and Amazon.

The sequences comprising each haplotype for the five species are detailed in Supplementary Material Table S2.

For *L. noguchii*, the haplotype network exhibited a high degree of haplotypic diversity, with several intermediate nodes and no single dominant haplotype. The sequences were broadly distributed among multiple haplotypes, many of which were shared across host categories. Domestic large land animals represented the majority of hosts, followed by wild animals, humans, and dogs (Figure 3A). In the biome-specific network, haplotypes were spread across three biomes, with notable abundance in the Atlantic Forest. Several haplotypes were exclusive to a single biome, while two were shared across more regions (Figure 3B).

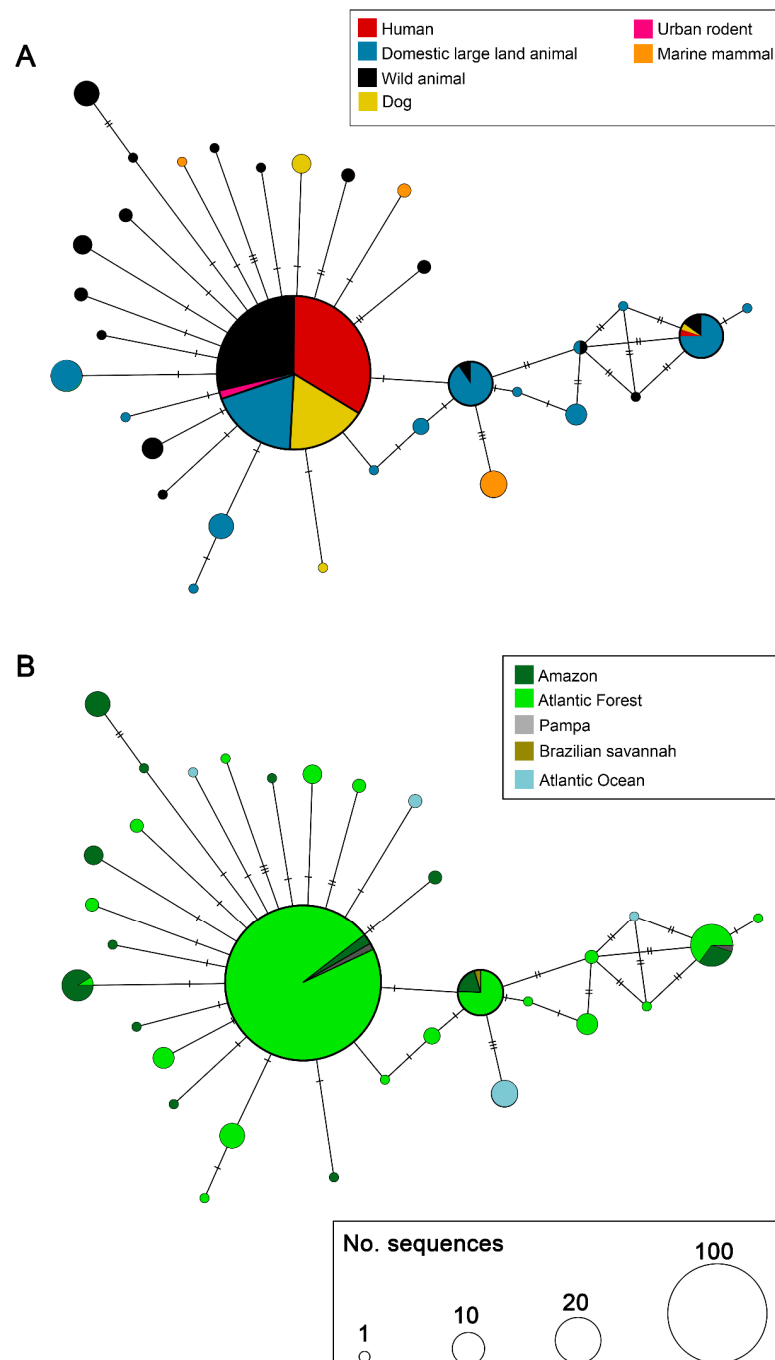


Figure 2. Haplotype network of *Leptospira interrogans* based on *secY* gene sequences. **(A)** Distribution of haplotypes according to host category. **(B)** Distribution of haplotypes according to the Brazilian biome. Each circle represents a unique haplotype, and its size is proportional to the number of sequences. Colors within circles indicate the host category **(A)** or biome **(B)**, as specified in the legends. Hatch marks between circles represent the number of mutational steps between haplotypes.

For *L. kirschneri*, the network revealed low haplotypic diversity, with a predominant central haplotype encompassing most sequences. This main haplotype included sequences from domestic, large land and wild animals. A few peripheral haplotypes were connected by a few mutational steps (Figure 3C). In the biome-based network, sequences were mainly associated with the Atlantic Forest, Amazon, and Cerrado, with a small number linked to the Pampa biome, which formed an exclusive haplotype (Figure 3D).

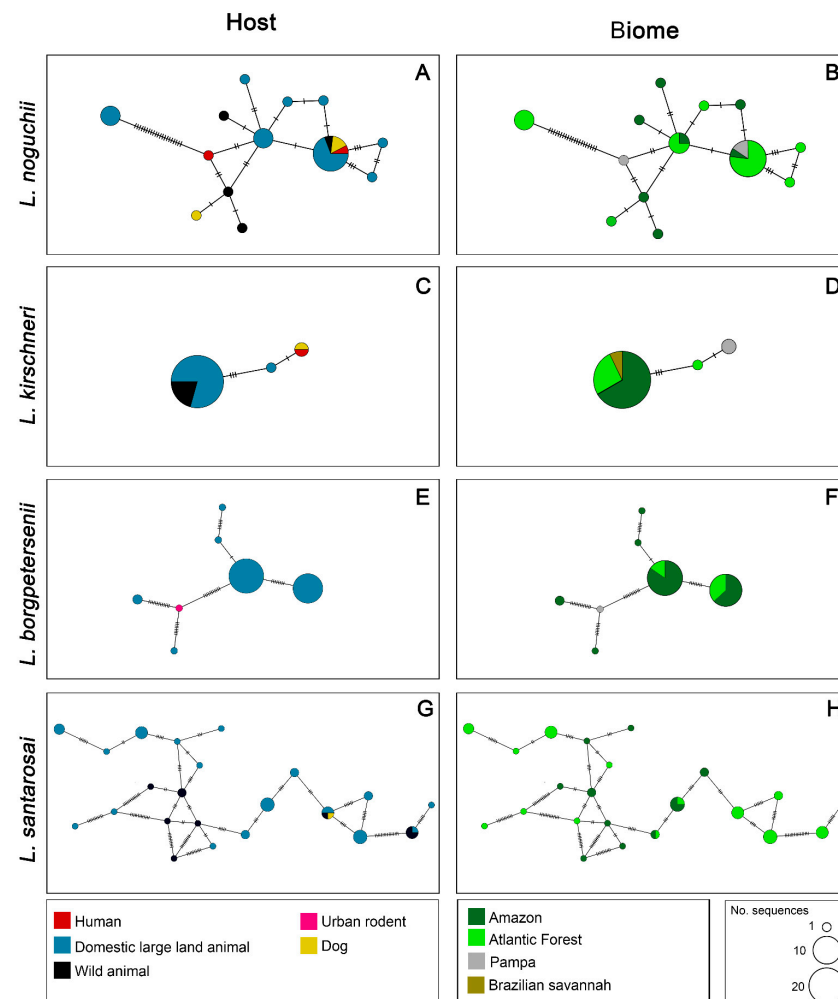


Figure 3. Haplotype networks of four pathogenic *Leptospira* species circulating in Brazil based on *secY* gene sequences. Each row represents one species: *L. noguchii*, *L. kirschneri*, *L. borgpetersenii*, and *L. santarosai*, respectively. The left column shows the distribution of haplotypes by host category (A,C,E,G), and the right column shows the distribution by biome (B,D,F,H). Each circle represents a unique haplotype, and its size is proportional to the number of sequences. Colors within circles correspond to the host categories (left) or biomes (right) as indicated in the legends. Hatch marks between circles represent the number of mutational steps separating haplotypes.

Leptospira borgpetersenii also showed low haplotypic diversity, with a few dominant haplotypes. The host haplotype network was formed almost exclusively by sequences originating from cattle, with a single haplotype composed of an urban rodent sequence (Figure 3E). In the biome-based network, sequences were predominantly from the Amazon, followed by the Atlantic Forest. The urban rodent sequence was from the Pampa biome (Figure 3F). Overall, the network structure was simpler compared to other species, with fewer mutational steps and a more clustered organization.

The network for *L. santarosai* revealed the highest haplotypic diversity among the species analyzed, with numerous haplotypes distributed along a reticulated and complex structure. This network structure was characterized by multiple internal connections, and no single haplotype was dominant. Host distribution showed dominance of domestic large land animals and wild animals, with smaller contributions from dogs (Figure 3G). The biome-based network was similarly diverse, with sequences from only two biomes, the Atlantic Forest and Amazon, with only two shared between regions (Figure 3H).

3.3. Genetic Distances Within and Between *Leptospira* Species

Genetic distances (TN93) based on 531 nucleotide sequences of the *secY* gene revealed low to moderate intraspecific variability across the five pathogenic *Leptospira* species circulating in Brazil (bold values along the diagonal in Table 1). The lowest intraspecific distance was observed for *L. kirschneri* (0.002), followed by *L. noguchii* (0.017) and *L. interrogans* (0.004), while *L. santarosai* exhibited the highest intraspecific divergence (0.034). Regarding interspecific comparisons (below the diagonal in Table 1), the smallest genetic distances were found between *L. interrogans*, *L. noguchii*, and *L. kirschneri* (ranging from 0.02 to 0.03), suggesting closer evolutionary relationships among these taxa. In contrast, *L. borgpetersenii* and *L. santarosai* showed greater divergence in relation to the other species, with distances reaching up to 0.24 when compared to *L. noguchii*. Standard deviations were generally low, indicating consistency in pairwise comparisons (above diagonal in italics in Table 1).

Table 1. Interspecific genetic distance (below the diagonal) between the five pathogenic *Leptospira* species circulating in Brazil based on 534 nucleotide sequences of the *secY* gene. Standard deviation is shown above the diagonal in italics. Intraspecific genetic distances are in bold along the diagonal.

	<i>L. interrogans</i>	<i>L. noguchii</i>	<i>L. kirschneri</i>	<i>L. borgpetersenii</i>	<i>L. santarosai</i>
<i>L. interrogans</i>	0.004	0.02	0.02	0.03	0.03
<i>L. noguchii</i>	0.11	0.017	0.02	0.03	0.02
<i>L. kirschneri</i>	0.11	0.11	0.002	0.03	0.02
<i>L. borgpetersenii</i>	0.23	0.21	0.21	0.015	0.02
<i>L. santarosai</i>	0.23	0.24	0.20	0.11	0.034

4. Discussion

To explore the genetic structure of circulating *Leptospira* spp. in Brazil, 531 *secY* gene sequences were selected, a marker well established for both taxonomic and epidemiological resolution [13,14]. This extensive dataset, encompassing sequences from multiple hosts and biomes, provided a robust basis for investigating lineage diversification and geographic structuring at a national scale. Herein, the *secY* locus has been consistently shown to balance interspecific conservation with sufficient intraspecific variability, allowing the reconstruction of reliable phylogenetic relationships while detecting microevolutionary signals within populations. By combining *secY*-based phylogenetic reconstruction, haplotype network analysis, and genetic distance estimation, we were able to delineate distinct evolutionary clusters corresponding to recognized species and to visualize the degree of haplotypic connectivity among hosts and regions.

The maximum likelihood phylogeny recovered strongly supported monophyletic clades for each species, with *L. interrogans* showing the largest number of sequences and broadest distribution across hosts and regions. This dominance underscores the remarkable ecological versatility of *L. interrogans*, a species capable of maintaining stable transmission cycles in both urban and rural settings, facilitated by its adaptability to a wide range of mammalian hosts [20]. A striking feature was the evidence of clonal expansion, with a single haplotype encompassing 242 sequences, including the reference strain Fiocruz L1-130, detected in humans, domestic animals, and wildlife, and spanning multiple biomes. Such extensive representation across hosts and regions indicates that this clone has achieved ecological dominance, likely supported by its capacity to persist in both maintenance and incidental hosts. The presence of identical sequences in geographically distant areas also suggests recent or ongoing gene flow among populations, possibly mediated by the movement of livestock, companion animals, and synanthropic rodents. The success of this

lineage reflects both low genetic variability and wide ecological adaptability, consistent with its role as the major cause of human outbreaks in Brazil, especially in urban centers with high rodent infestation [21]. This evolutionary success likely results from a combination of host plasticity and environmental resilience, allowing the bacteria to survive under fluctuating conditions of humidity, temperature, and salinity. The lineage's widespread distribution also suggests a long-term evolutionary stability, possibly linked to efficient maintenance within rodent reservoirs and recurrent spillovers to other mammals. The persistence of this haplotype across hosts and landscapes highlights its epidemiological significance: clonal stability may facilitate efficient transmission, while reduced antigenic diversity could simultaneously favor vaccine targeting and amplify risks of large-scale outbreaks if this lineage continues to expand unchecked. From an epidemiological standpoint, this pattern illustrates the dual nature of clonal expansion: it enhances predictability for diagnostic and vaccine development yet increases vulnerability to epidemic resurgence when environmental or climatic conditions become favorable for transmission. Continuous genomic monitoring of this dominant haplotype could therefore serve as an early-warning indicator for urban leptospirosis outbreaks. A clonal subpopulation of *L. interrogans* was previously reported during leptospirosis outbreaks in Brazil [22]. Similarly, clonal expansion of *L. interrogans* was identified as the predominant driver of leptospirosis cases in the Indian archipelagos [23]. These findings, observed across independent epidemiological settings, highlight a possible broader global trend in which a few highly adapted *L. interrogans* lineages sustain the majority of human infections. Such convergence across continents suggests evolutionary optimization toward transmission efficiency rather than diversification. However, further studies employing standardized genetic characterization approaches are needed to confirm this trend and elucidate its evolutionary and epidemiological implications.

By contrast, *L. kirschneri* and *L. borgpetersenii* presented simpler network topologies, with limited haplotypic variation and few mutational steps between clusters. The predominance of a small number of haplotypes may reflect host adaptation and reduced environmental survival, particularly in the case of *L. borgpetersenii*, which has undergone genome reduction and relies more on direct host-to-host transmission [24,25]. Recent genomic studies reinforce this restricted variability, showing that *L. borgpetersenii* lineages circulating in cattle often cluster into very few haplotypes or STs, highlighting clonal expansion and limited environmental persistence [26,27]. Similarly, *L. kirschneri* isolates frequently display low genetic variability and well-structured clusters, consistent with local host adaptation and geographically restricted lineages [28]. This restricted diversity is consistent with their frequent association with cattle and other livestock, reinforcing the role of domestic animals in maintaining adapted *Leptospira* lineages and shaping transmission cycles [29].

Haplotype networks stratified by host and biome provided further resolution of phylogeographic patterns. For *L. interrogans*, the dominant haplotype was widely shared among humans, cattle, dogs, and wildlife, and was particularly prevalent in the Atlantic Forest biome. Such patterns corroborate previous findings of clonally dominant lineages circulating in small mammals in this biome [30] and illustrate strong ecological connectivity across hosts and environments. Shared haplotypes spanning different Brazilian biomes emphasize lineage stability at continental scales, suggesting successful dispersal mechanisms possibly mediated by synanthropic reservoirs such as rodents. Similar patterns have been documented elsewhere: in urban areas of Guangzhou, China, the coexistence of *L. interrogans*, *L. borgpetersenii*, and *L. kirschneri* in commensal rodents was reported, highlighting their role as long-term carriers and effective vehicles for dissemination across ecological boundaries [31]. At the genomic level, Chinchilla and colleagues (2023) analyzed

914 genomes of *L. santarosai* from the Americas and revealed both region-specific clonal groups and evidence of historical lineage movements across distant areas, pointing to global-scale dispersal dynamics. Together, these findings suggest that haplotype sharing in Brazil may not be an isolated phenomenon but rather part of a broader ecological and evolutionary pattern in which synanthropic reservoirs facilitate the persistence and geographic expansion of pathogenic *Leptospira* worldwide.

Exclusive haplotypes identified in marine mammals from the Atlantic Ocean expand the ecological boundaries of leptospiral circulation. Although the genetic distances separating these lineages were minimal (one to three substitutions), their exclusivity suggests ecological isolation, with limited connectivity to terrestrial reservoirs. Nevertheless, the zoonotic potential of this environment cannot be disregarded, as marine–terrestrial interfaces may represent overlooked transmission routes [32]. These findings reinforce the need for frameworks that encompass both terrestrial and aquatic ecosystems.

The analysis of genetic distances supported the phylogenetic and haplotype network results. Intraspecific distances were lowest for *L. kirschneri* (0.002) and *L. interrogans* (0.004), consistent with clonal expansion and host-adapted transmission. By contrast, *L. santarosai* showed the highest intraspecific divergence (0.034), reinforcing its evolutionary variability. Interspecific distances indicated closer genetic relatedness among *L. interrogans*, *L. noguchii*, and *L. kirschneri*, while *L. borgpetersenii* and *L. santarosai* were more distinct, a finding consistent with whole-genome and cgMLST analyses [10,33,34]. Importantly, the absence of overlap between maximum intraspecific and minimum interspecific distances underscores the strong taxonomic resolution of *secY*, supporting its use as a robust marker for species delimitation and intraspecific diversity assessment.

From a One Health perspective, the genetic connectivity observed among *Leptospira* lineages circulating in humans, domestic animals, and wildlife underscores the importance of coordinated surveillance across sectors. Molecular approaches such as *secY*-based typing provide a practical and cost-effective tool to monitor pathogen flow between hosts and environments, identify potential spillover interfaces, and guide preventive measures adapted to local ecological contexts. By bridging molecular epidemiology with ecological data, this study contributes to the integrated understanding required for sustainable leptospirosis control in endemic tropical regions.

Despite these advances, some limitations should be considered. The reliance on a single-locus marker, although informative, precludes the detection of recombination events and fine-scale genomic adaptations. Whole-genome approaches will be essential to uncover hidden diversity and clarify evolutionary processes such as horizontal gene transfer. Furthermore, sequence availability is uneven across hosts and regions, potentially biasing haplotype frequencies and underrepresenting circulation in poorly sampled biomes such as the Caatinga and Pantanal. Because the dataset analyzed here was entirely derived from publicly available *secY* sequences, it reflects the representativeness of deposited records rather than the true prevalence of genotypes in nature. The apparent predominance of certain haplotypes may therefore result from uneven research efforts, regional sampling preferences, or repeated sequencing of epidemiologically important isolates. Consequently, the patterns described in this study should be interpreted as indicators of genetic diversity and connectivity within the available dataset, not as estimates of infection frequency or epidemiological dominance. Finally, future studies combining Brazilian and international datasets will be essential to determine whether the country harbors unique endemic clades or mainly reflects the global diversity structure of pathogenic *Leptospira*. In this context, formal nomenclature of *secY*-based haplotypes should be pursued to facilitate comparability among research groups and ensure consistent lineage recognition across regions worldwide.

Addressing these gaps through coordinated sampling efforts will be critical for a truly comprehensive picture of *Leptospira* diversity in Brazil and in the world.

5. Conclusions

This study provides a comprehensive molecular overview of the genetic diversity of pathogenic *Leptospira* circulating in Brazil. Phylogenetic analyses based on the *secY* gene revealed well-supported clades for the main pathogenic species, while haplotype network analyses uncovered contrasting evolutionary patterns, ranging from clonal expansion in *L. interrogans* to extensive haplotypic diversity in *L. santarosai* and *L. noguchii*. Host and biome-based networks highlighted both the ecological adaptability of dominant lineages and the existence of exclusive haplotypes in specific environments, such as marine mammals in the Atlantic Ocean. Genetic distance estimates confirmed the strong taxonomic resolution of *secY*, distinguishing closely related species while capturing intraspecific diversity. Together, these findings confirm the complex interplay of host, environment, and pathogens shaping leptospiral diversity. By bridging phylogenetics with ecological and epidemiological contexts, this work contributes to a deeper understanding of leptospiral transmission dynamics and provides a valuable framework for One Health surveillance and control strategies in endemic regions.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/microorganisms13112512/s1>, Table S1: Complete curated dataset of the 531 pathogenic *Leptospira* sequences included in this study. For each record, the table compiles genetic, geographic, and host-related metadata retrieved from GenBank; Table S2: List of sequences included in each haplotype for the five pathogenic *Leptospira* species analyzed in this study. One species per worksheet is shown. For each species, haplotypes were defined based on *secY* gene sequences (>400 bp) through haplotype network analysis on PopArt software.

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DISCUSSÃO

O presente estudo apresenta um panorama inédito sobre a diversidade genética de leptospiros patogênicos circulantes no Brasil, integrando dados de múltiplos hospedeiros e regiões. Essa abordagem permitiu caracterizar as estirpes em nível genético, inferir padrões de circulação e rotas de transmissão, além de elucidar o potencial zoonótico das linhagens identificadas.

A alta identidade genética entre estirpes isoladas de cães e humanos confirma a estreita interface zoonótica entre essas populações, sugerindo que os cães atuam não apenas como sentinelas, mas também como amplificadores da infecção em áreas urbanas e periurbanas, mesmo quando assintomáticos (Sant'Anna et al., 2017; Sant'Anna et al., 2021). Em diferentes regiões da América do Sul, a soroprevalência canina varia de 18% a 20%, sendo mais elevada em animais com acesso à rua, oriundos de áreas de vulnerabilidade social e com maior exposição a ambientes contaminados, presença de roedores e contato com água ou solo úmido (Esteves et al., 2023; Ricardo et al., 2024; Griebisch et al., 2025). Cães errantes, filhotes, machos e idosos também apresentam risco aumentado (Griebisch et al., 2025). Em canis, surtos podem ocorrer mesmo em animais vacinados, devido à circulação de sorogrupos não contemplados nas vacinas comerciais, reforçando a necessidade de vigilância contínua (Miotto et al., 2018; Balboni et al., 2022).

A transmissão direta de cães para humanos é possível, sobretudo pelo contato com urina de animais infectados, havendo relatos de casos humanos associados a surtos caninos (Azócar-Aedo e Monti, 2022; Balboni et al., 2022; Guzmán et al., 2023; Ricardo et al., 2024), inclusive em regiões de baixo risco, como Wyoming, EUA (Waranius et al., 2024), onde o aumento da vacinação canina foi associado à redução do risco para humanos expostos. Em regiões tropicais, cães utilizados em atividades de caça apresentam alta soroprevalência e podem atuar como sentinelas para os humanos envolvidos nessas práticas (Orr et al., 2022), reforçando seu papel como elo entre ambientes silvestres e domésticos. Esses achados reforçam o papel importante dos cães na epidemiologia da leptospirose e destacam a importância de incluir esses animais

em programas integrados de vigilância e controle, especialmente em comunidades vulneráveis com alta circulação de roedores.

Nos bovinos, observou-se epidemiologia distinta dos cães, marcada pela predominância de *L. borgpetersenii* e *L. interrogans* do sorogrupo Sejroe, sobretudo em infecções genitais crônicas. A detecção de haplótipos exclusivos do trato reprodutivo, não compartilhados com estirpes renais, sugere adaptação dessas leptospiros ao ambiente genital bovino. Foi recentemente demonstrado que estirpes de *L. santarosai* isoladas de trato genital apresentam adaptações genômicas específicas que favorecem colonização e persistência nesse nicho (Di Azevedo et al., 2025). Evidências moleculares confirmam a presença de leptospiros em tecidos genitais mesmo na ausência de excreção urinária detectável (Aymée et al., 2021), consolidando o conceito de leptospirose genital bovina e expondo as limitações dos métodos sorológicos tradicionais para detectar infecções silenciosas. Essa forma subclínica e crônica representa desafio significativo à saúde animal e pública, pois facilita a disseminação silenciosa no rebanho e no ambiente (Loureiro e Lilenbaum, 2020; Aymée et al., 2024; Barnabé et al., 2024). A detecção de leptospiros em fluidos genitais e sêmen (Maiolino et al., 2021; Pedrosa et al., 2024) aponta para uma rota de transmissão venérea ainda subexplorada. Tal via pode comprometer a eficiência reprodutiva e ampliar o risco de exposição humana, sobretudo entre trabalhadores rurais e inseminadores, que manipulam secreções contendo leptospiros viáveis (Alinaitwe et al., 2019; Mai et al., 2021). Espécies do sorogrupo Sejroe já foram isoladas de casos humanos (Romero e Yasuda, 2006; Kakita et al., 2020), evidenciando seu potencial zoonótico e a necessidade de vigilância sanitária que inclua testes moleculares e vacinação com cepas locais. A ausência de obrigatoriedade de notificação da leptospirose animal em países como o Brasil contribui para subnotificação e lacunas críticas na vigilância integrada, contrariando os princípios da abordagem Uma Só Saúde.

Já a análise de pequenos mamíferos silvestres na Mata Atlântica revelou ampla circulação de *L. interrogans* com haplótipos idênticos à estirpe Fiocruz L1-130, indicando intensa atividade zoonótica mesmo em ecossistemas preservados. Essa homogeneidade genética reflete a plasticidade ecológica de

clones epidêmicos capazes de se adaptar a diferentes nichos e hospedeiros. A positividade observada (56%) está em consonância com achados prévios na região (Vieira et al., 2019a, 2019b). Em áreas rurais do mesmo bioma, foram registradas menores prevalências (21,3%) e menor diversidade de hospedeiros (Silva et al., 2023), o que é esperado, uma vez que as áreas de conservação tendem a sustentar maior diversidade e abundância de reservatórios, maior persistência ambiental da bactéria e redes tróficas mais estáveis, além de apresentarem uso reduzido de medidas de controle e menor competição com espécies sinantrópicas (Bradley e Lockaby, 2023). Por outro lado, a degradação da Mata Atlântica, que atualmente mantém apenas 12,4% de sua cobertura original (Ribeiro et al., 2023), intensifica o contato entre humanos, animais domésticos e silvestres, elevando o risco de transmissão de doenças zoonóticas. Fatores como fragmentação de habitat, alterações tróficas, aumento da exposição a águas contaminadas e mudanças climáticas contribuem para a persistência e disseminação de leptospiros (Pfenning-Butterworth et al., 2024).

De uma forma geral, a compilação de 825 sequências obtidas entre 2009 e 2025 de origem humana e animais domésticos, sinantrópicos e silvestres constitui o conjunto molecular mais abrangente de *Leptospira* spp. no Brasil, evidenciando o avanço das ferramentas genéticas na vigilância do patógeno. Embora a diversidade de espécies patogênicas seja conhecida (Pyskun e Richter, 2025), a predominância de *L. interrogans* (74,9%) reforça seu papel como principal agente infeccioso global (Rajapakse et al., 2025) e no Brasil (Albuquerque et al., 2017a; Jorge et al., 2018; Paz et al., 2019; Siqueira et al., 2020; Gomes et al., 2022; Freitas et al., 2024). Outras espécies, como *L. santarosai*, *L. borgpetersenii*, *L. noguchii* e *L. kirschneri* foram detectadas com menor frequência (3–10%), o que pode refletir não apenas subnotificação ou desigualdade na vigilância molecular entre regiões, mas também diferenças ecológicas e adaptativas, uma vez que essas espécies tendem a apresentar nichos mais restritos, menor capacidade de persistência ambiental ou menor eficiência de transmissão em comparação a *L. interrogans*.

A predominância de *L. interrogans* em ambientes urbanos confirma sua adaptação a ciclos sinantrópicos envolvendo roedores e cães (Bradley e

Lockaby, 2023), enquanto *L. borgpetersenii*, *L. kirschneri* e *L. noguchii* são mais frequentes em áreas rurais, associadas a hospedeiros domésticos e silvestres (Guedes et al., 2019; Soares et al., 2020). *Leptospira santarosai*, distribuída entre zonas rurais e periurbanas, apresenta flexibilidade ecológica e potencial de sobreposição entre ciclos silvestres e domésticos (Chinchilla et al., 2023). Essa heterogeneidade reflete o papel da paisagem na circulação das leptospiros, embora os dados devam ser interpretados com cautela, pois resultam de esforços amostrais desiguais entre regiões e hospedeiros.

Do ponto de vista taxonômico, foram identificadas 78 espécies hospedeiras, incluindo duas classes (mamíferos e répteis) de 12 ordens distintas. Destas, as mais notáveis foram Artiodactyla (mamíferos ungulados, como ruminantes e suínos), Rodentia (roedores), Carnivora (principalmente cães) e Primates (humanos). Essa diversidade reforça o conceito clássico de leptospirose como uma zoonose multihospedeira, na qual os ciclos de transmissão são mantidos por uma ampla variedade de espécies com diferentes funções epidemiológicas (Belsare et al., 2022).

Leptospira interrogans apresentou ampla distribuição geográfica, ocorrendo em todos os biomas e paisagens. As regiões Sudeste e Sul concentraram o maior número de registros e riqueza de espécies, provavelmente devido à maior infraestrutura de pesquisa e às condições ambientais favoráveis à transmissão, como inundações frequentes e infestação por roedores (Nardoni Marteli et al., 2022; Michelon et al., 2025). A Mata Atlântica destacou-se como o principal *hotspot* de *Leptospira*, seguida pela Amazônia, enquanto o Cerrado, a Caatinga e Pantanal permanecem sub-representados, apesar das evidências de circulação previamente documentadas por detecções moleculares e sorológicas em animais e humanos (Vieira et al., 2016; Galan et al., 2023; Lisboa Padilha et al., 2025).

A presença de 44 sequências ambientais (poças, esgoto, solo, água doce) evidencia a importância das matrizes ambientais na persistência e transmissão de *Leptospira*. A capacidade de formar biofilmes em superfícies abióticas confere resistência e sobrevivência prolongada (Ristow et al., 2008; Meganathan et al., 2022). Estudos brasileiros em áreas urbanas com saneamento precário

confirmam a detecção recorrente de *Leptospira* patogênicas em esgoto e águas pluviais (Casanovas-Massana et al., 2020, 2022), reforçando o papel do ambiente como interface de manutenção e transbordamento (Bierque et al., 2020; Davignon et al., 2023).

A análise de 531 sequências do gene *secY* revelou cinco espécies patogênicas principais e forte expansão clonal de *L. interrogans*, com haplótipo dominante incluindo à estirpe Fiocruz L1-130. Essa linhagem, associada a surtos urbanos e infestação por roedores (Jaeger et al., 2018), apresentou baixa variabilidade genética e alta adaptabilidade ecológica. *Leptospira borgpetersenii* e *L. kirschneri* exibiram baixa diversidade haplotípica e restrição geográfica, compatíveis com adaptação ao hospedeiro e menor sobrevivência ambiental (Bulach et al., 2006; Picardeau et al., 2008; Aymée et al., 2022; Benvin et al., 2025).

As redes haplotípicas mostraram haplótipos compartilhados entre diversos hospedeiros, com destaque para humanos, cães, bovinos e mamíferos silvestres, com maior ocorrência na Mata Atlântica. O compartilhamento entre biomas indica estabilidade genética e dispersão mediada por reservatórios sinantrópicos, padrão também observado em roedores urbanos na Ásia (Shao et al., 2022). Haplótipos exclusivos detectados em mamíferos marinhos do Atlântico, embora geneticamente próximos aos terrestres (1–3 substituições), indicam isolamento ecológico e possíveis rotas marinho-terrestres de transmissão negligenciadas, o que reforça a importância de abordagens que integrem ecossistemas terrestres e aquáticos (Torres et al., 2025).

A análise das distâncias genéticas confirmou os padrões observados na filogenia e nas redes haplotípicas. As menores divergências intraespecíficas foram registradas para *L. kirschneri* (0.002) e *L. interrogans* (0.004), refletindo expansão clonal e adaptação ao hospedeiro, enquanto *L. santarosai* apresentou maior variabilidade (0.034). As distâncias interespecíficas revelaram maior proximidade entre *L. interrogans*, *L. noguchii* e *L. kirschneri*, e maior distinção de *L. borgpetersenii* e *L. santarosai*, em concordância com análises genômicas e de cgMLST (Vincent et al., 2019; Xu et al., 2016; Guglielmini et al., 201). A ausência

de sobreposição entre distâncias intra e interespecíficas reforça a alta resolução taxonômica do gene *secY* como marcador para delimitação de espécies.

A conectividade genética observada entre linhagens de *Leptospira* em humanos, animais domésticos e silvestres reforça a necessidade de vigilância integrada entre setores. Abordagens moleculares baseadas no gene *secY* são ferramentas eficazes e de baixo custo para monitorar o fluxo do patógeno, identificar interfaces de transbordamento e orientar medidas preventivas adaptadas aos contextos ecológicos locais. O uso de um único marcador, entretanto, limita a detecção de recombinações e adaptações genômicas. Estudos amplos baseados em sequenciamento de genoma completo são essenciais para revelar diversidade oculta e processos evolutivos, como transferência horizontal de genes. Além disso, a desigualdade amostral entre hospedeiros e regiões pode enviesar frequências haplotípicas, especialmente em biomas pouco amostrados. Como o conjunto analisado baseou-se em registros públicos, os padrões refletem a representatividade dos dados depositados, e não necessariamente a prevalência real. Futuros estudos integrando bancos brasileiros e internacionais serão fundamentais para verificar se o país abriga clados endêmicos ou reflete a estrutura global de diversidade de *Leptospira*. A adoção de uma nomenclatura formal para haplótipos baseados em *secY* será igualmente crucial para padronizar comparações entre grupos de pesquisa e regiões.

CONCLUSÕES

- As análises *in silico* de sequências de *Leptospira* spp. associadas à infecção canina na América Latina evidenciaram baixa variabilidade genética e ampla distribuição geográfica de linhagens clonais, reforçando o papel dos cães como hospedeiros sentinelas na vigilância da leptospirose.
- A detecção molecular em amostras reprodutivas de bovinos do Sudeste brasileiro confirmou a circulação de *Leptospira* spp. patogênicas associadas à forma crônica da leptospirose bovina, destacando a

relevância do trato genital como sítio de manutenção e persistência do patógeno.

- A caracterização genética de *Leptospira* spp. em pequenos mamíferos da Mata Atlântica revelou elevada diversidade de espécies e haplótipos, evidenciando o papel desses reservatórios silvestres na manutenção ecológica do agente e seu potencial zoonótico.
- A integração de sequências provenientes de diferentes genes, hospedeiros e biomas permitiu identificar padrões ecoepidemiológicos consistentes de circulação de *Leptospira* spp. patogênicas no Brasil, sob a perspectiva Uma Só Saúde.
- As análises filogenéticas e de diversidade genética indicaram heterogeneidade evolutiva entre as espécies circulantes, com evidências de expansão clonal em algumas linhagens e maior estruturação genética em outras, refletindo processos de adaptação a hospedeiros e ambientes distintos.

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